

Intel's Core-based 2S server processors

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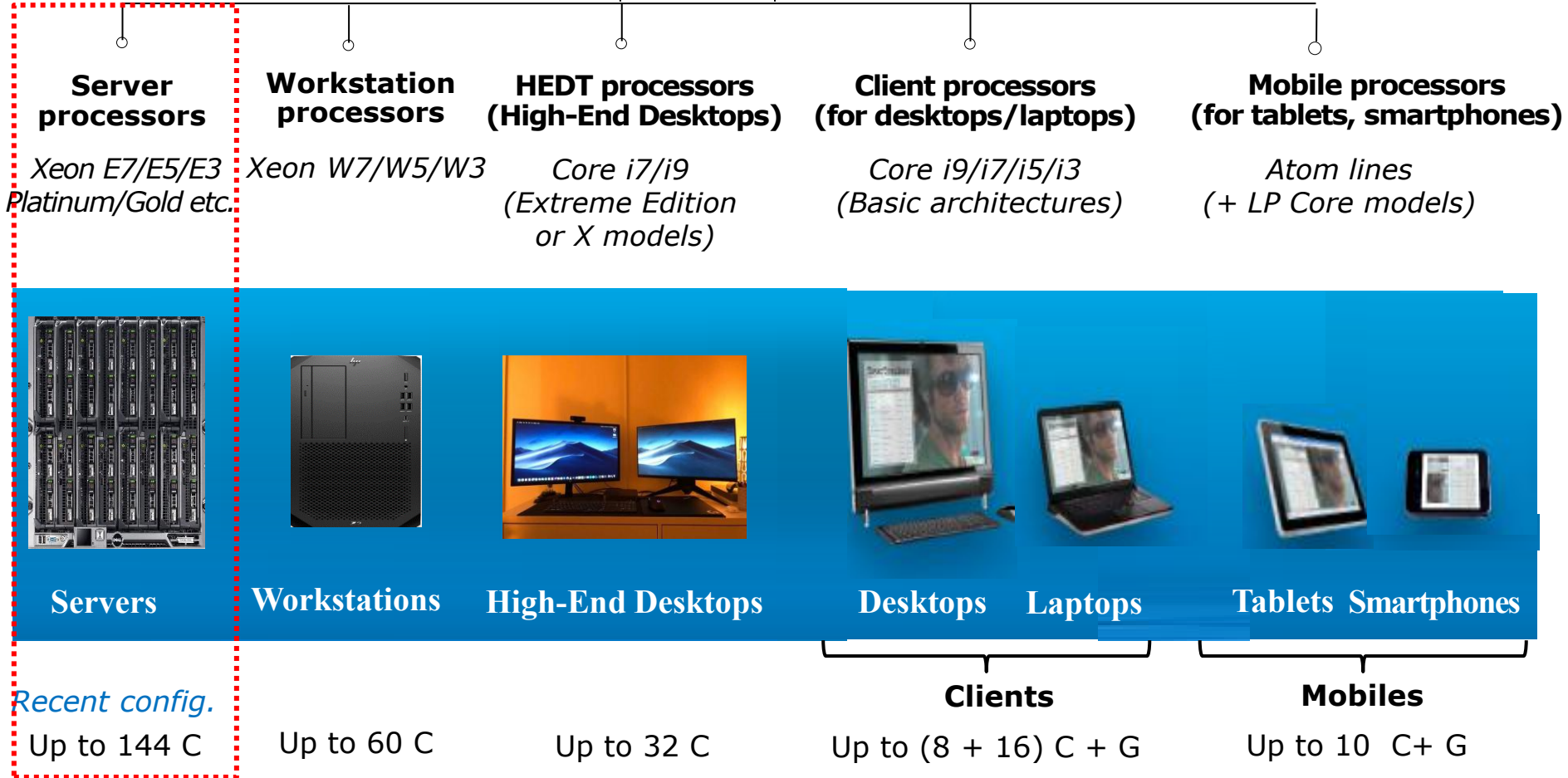
Intel's Core-based 2S server processors

- 1. Introduction to Intel's Core-based 2S server processors
- 2. Exploring Intel's 2S server processors
- 3. Case examples for 2S server boards
- 4. References

Intel's Core-based 2S server processors (1)

Intel's Core-based 2S server processors -1

Recent major processor categories of Intel



Note that both Intel and AMD designate **laptop processors** as **mobile processors**.

Intel's Core-based 2S server processors -2

After an introduction, in this Section we are concerned with the following questions:

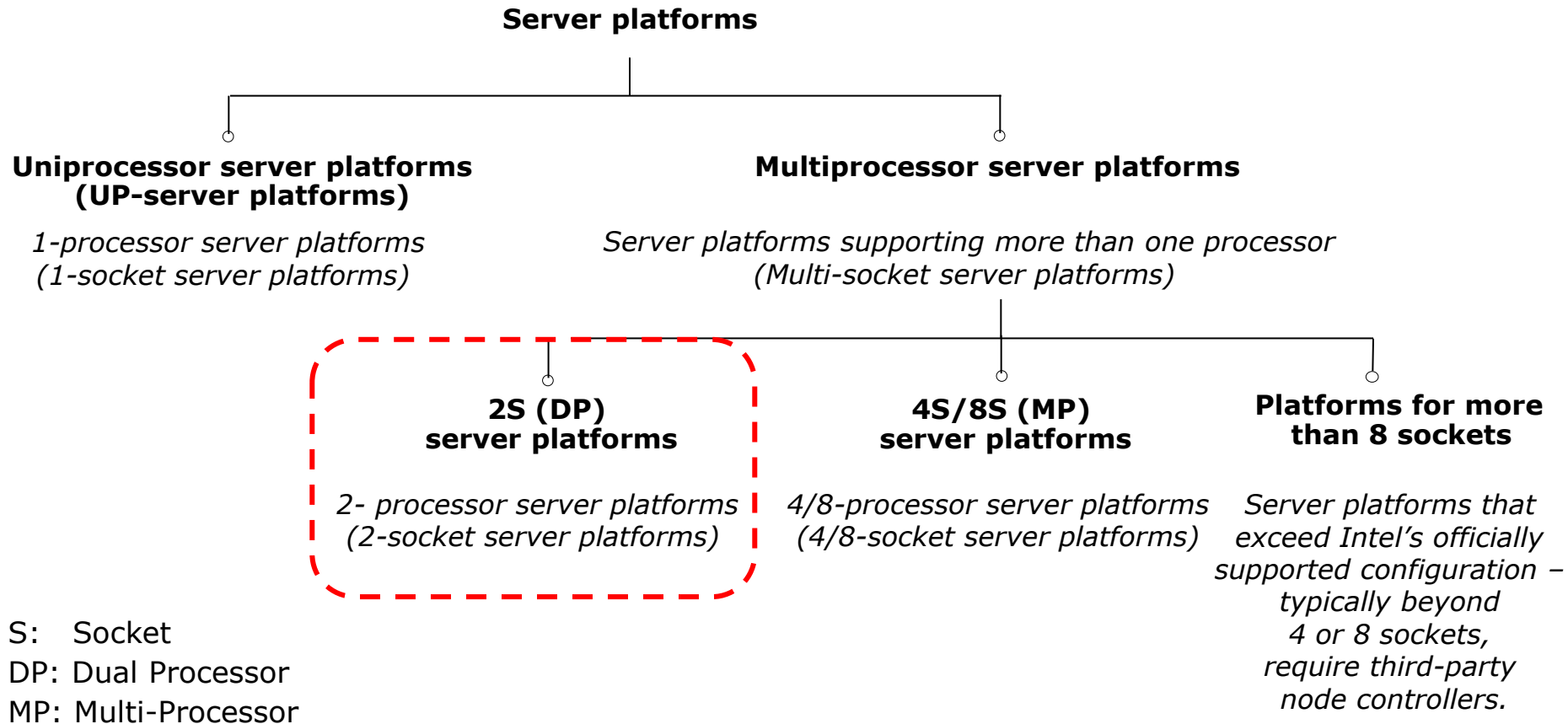
- Are Intel's server processor designs **monolithic or multi-die**?
- **How large** are Intel's 2S **server dies**, and how many transistors do they include?
- **At what rate could core counts and memory speeds be increased** in Intel's high-end Core-based 2S server processors?
- What are the implications **if memory rates rise slower than core counts** over server generations?
- What **power consumption** have Intel's 2S server processors (in terms of TDP) and how they are **socketed**?

1. Introduction to Intel's Core-based 2S server processors

1. Introduction to Intel's Core-based 2S server processors (1)

1. Introduction to Intel's Core-based 2S server processors

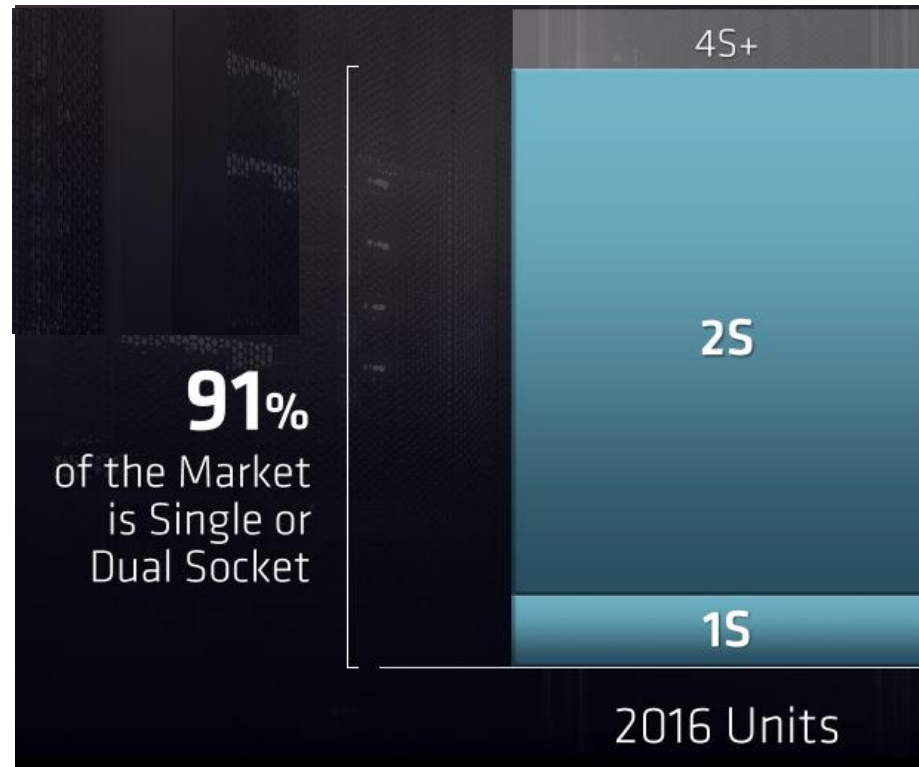
Server platforms can be categorized **easily** based on the number of sockets they include.



Note that in the context of servers, the terms "processor" and "socket" are often used interchangeably. This Section focuses exclusively on **2S server processors**, which **dominate the server market**, as illustrated in the following Figure.

1. Introduction to Intel's Core-based 2S server processors (2)

2016 market share of 1S/2S/4S servers (in number of units sold) [116]



As of 2016, 2S servers dominated the server market, accounting for approximately 80% of the total shipment.

A similar dominance is expected in recent years as well.

1. Introduction to Intel's Core-based 2S server processors (3)

Evolution of Intel's 64-bit multicore 2S server processor according to their basic layout

- In 2004, Intel made a substantial change in its server processor designs by transitioning from 32-bit to a 64-bit architecture.

This upgrade enabled support for significantly larger memory capacities, and improved overall system performance.

- At that time, Intel's server processors were still based on single-core designs.

However, shortly thereafter, in 2005, the company introduced its first dual-core server processors, however, implemented on dual dies, marking the beginning of the multi-core era.

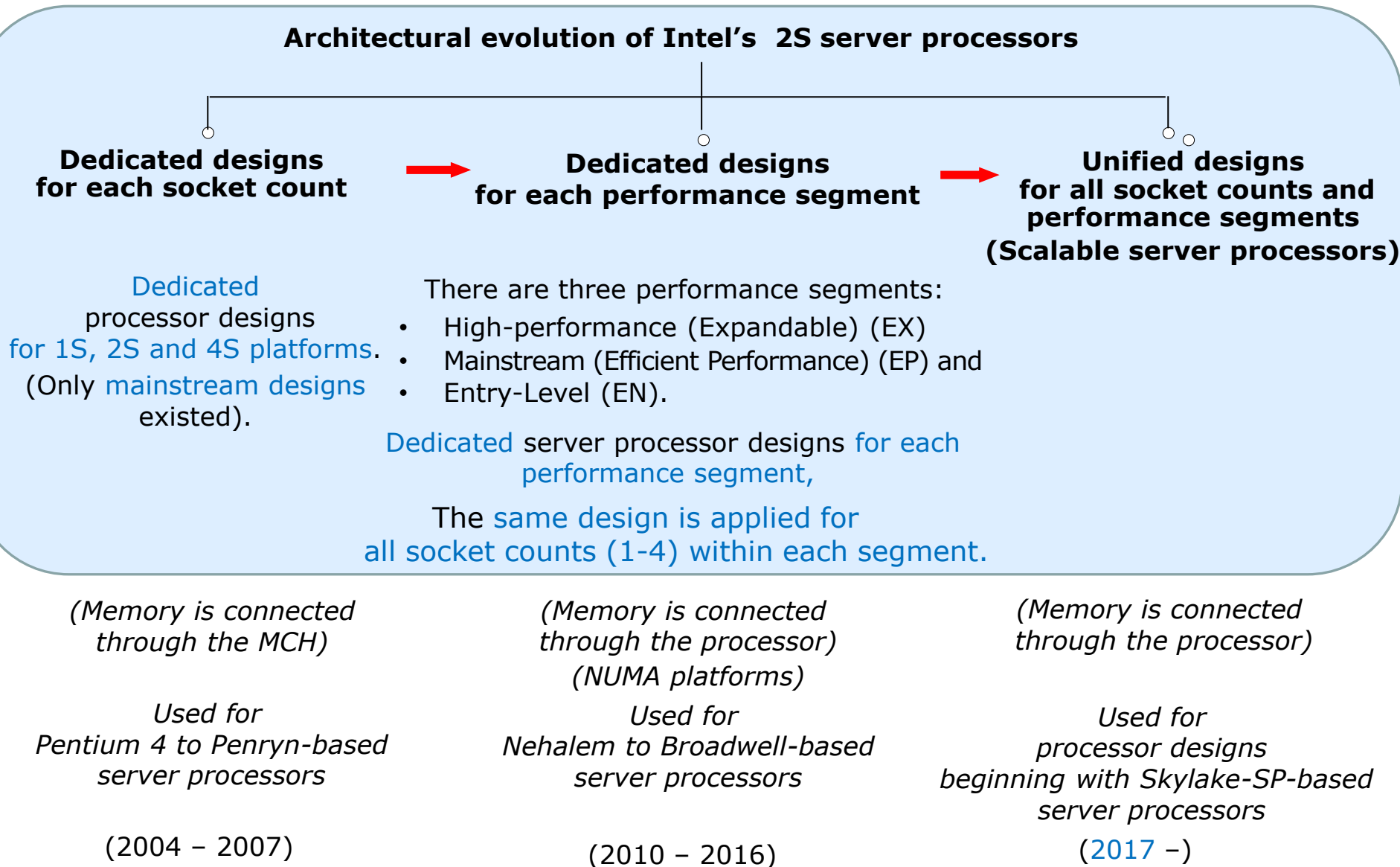
- Since the introduction of 64-bit multi-core server processors, Intel's server processor designs have evolved tremendously across multiple dimensions.
- In the following overview, we'll focus specifically on 2S server processors, as 2S processors are the most common server type.

Among the many aspects of this evolution, one crucial aspect is how their basic architecture has evolved over time.

- We obtain a clear understanding of this progression when we'll break it down into three distinct phases, as shown in the following Figure.

1. Introduction to Intel's Core-based 2S server processors (4)

Architectural evolution of Intel's 2S server processors -1 (beginning with 64-bit multicore designs)



1. Introduction to Intel's Core-based 2S server processors (5)

The 1st phase of the architectural evolution of Intel's 2S server processors

- In the 1st phase of the evolution, the company designed dedicated server processor architectures tailored to each socket count, namely 1S, 2S, and 4S servers.
- Designs for different socket counts typically incorporate different processor types, employed varying memory connection technologies, and supported distinct MCHs.
- These processors were designed as mainstream solutions, with no further segmentation with entry-level or high-end variants.
- This design approach characterized Intel's server processors from the Pentium 4 through the Penryn-based server processor generations, spanning the years 2004 to 2007.
- During this period, memory was typically connected through the MCH, as will be explained in the following.

1. Introduction to Intel's Core-based 2S server processors (6)

Alternative methods for connecting memory to a server platform -1

Alternative methods for connecting memory to a platform

Memory is connected through the MCH

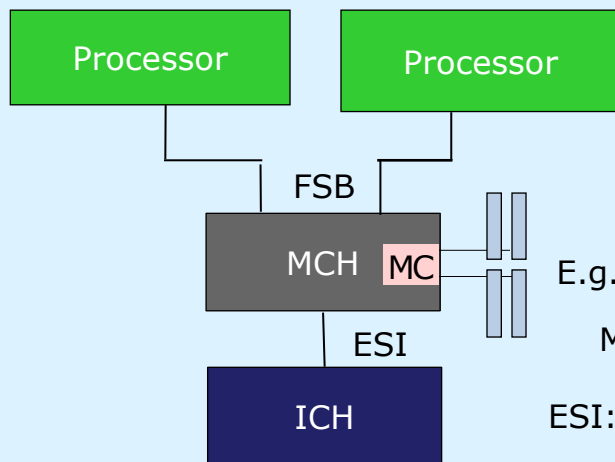
The MC is implemented on the MCH die.
Memory is connected via the MCH and the FSB.

All processors access memory
with the same latency.

Known as **UMA platforms**.

Multiprocessors (Multi socket systems)
with **Uniform Memory Access (UMA)**

Examples



E.g. DDR2-533
E.g. DDR3-1333
MCH: Memory Control Hub
FSB: Front Side Bus
ESI: Enterprise System Interface
ICH: I/O Control Hub

1st phase of server processors

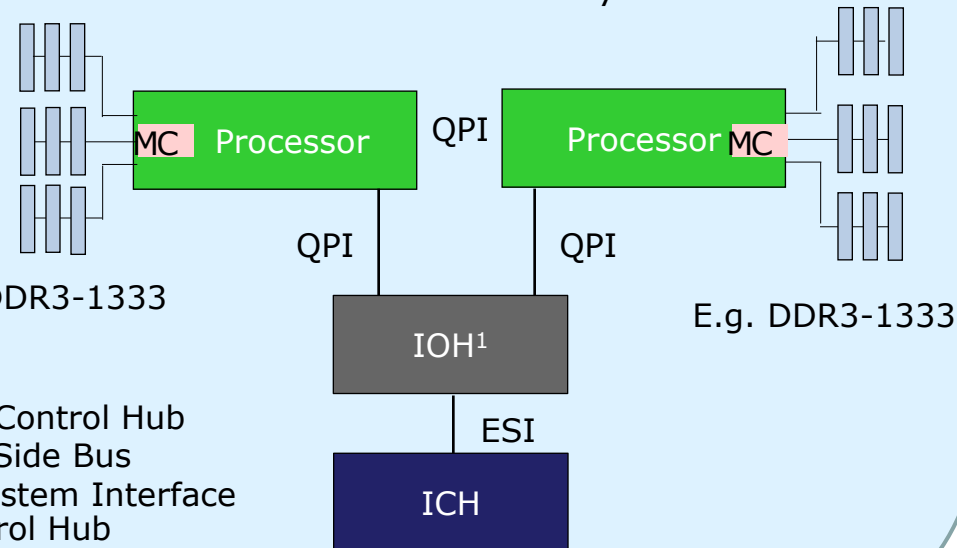
Memory is connected through the processor

The MCs are implemented on the processor die.
Memory is directly connected to the processors
through memory channels.

Processors access memory
with different latencies.

Known as **NUMA platforms**.

Multiprocessors (Multi socket systems)
with **Non-Uniform Memory Access**



From the 2nd phase of server processors onward

1. Introduction to Intel's Core-based 2S server processors (7)

Alternative methods for connecting memory to a server platform -2

Memory latencies in UMA and NUMA platforms

Memory latencies in UMA platforms

- All processors in a multiprocessor system share the **same memory space**.
- Memory is **connected through the MCH**.
- All processors have the **same memory access path** (e.g. via the MCH and the FSB).
- Consequently, **all processors access memory with the same latency**.



Memory latencies in NUMA platforms

- All processors in a multiprocessor system share the **same memory space**.
- The memory that a processor can access directly through its own MC, is called **local memory**.
- On the other hand, the memory that a processor accesses through an MC that belongs to another processor, it's referred to as **remote memory**.
Accesses to the remote memory involve more latency due to the need to communicate across processor boundaries.
- As a result, memory **access times vary** depending on whether the local or the remote memory is accessed.
This leads to **non-uniform memory accesses**.

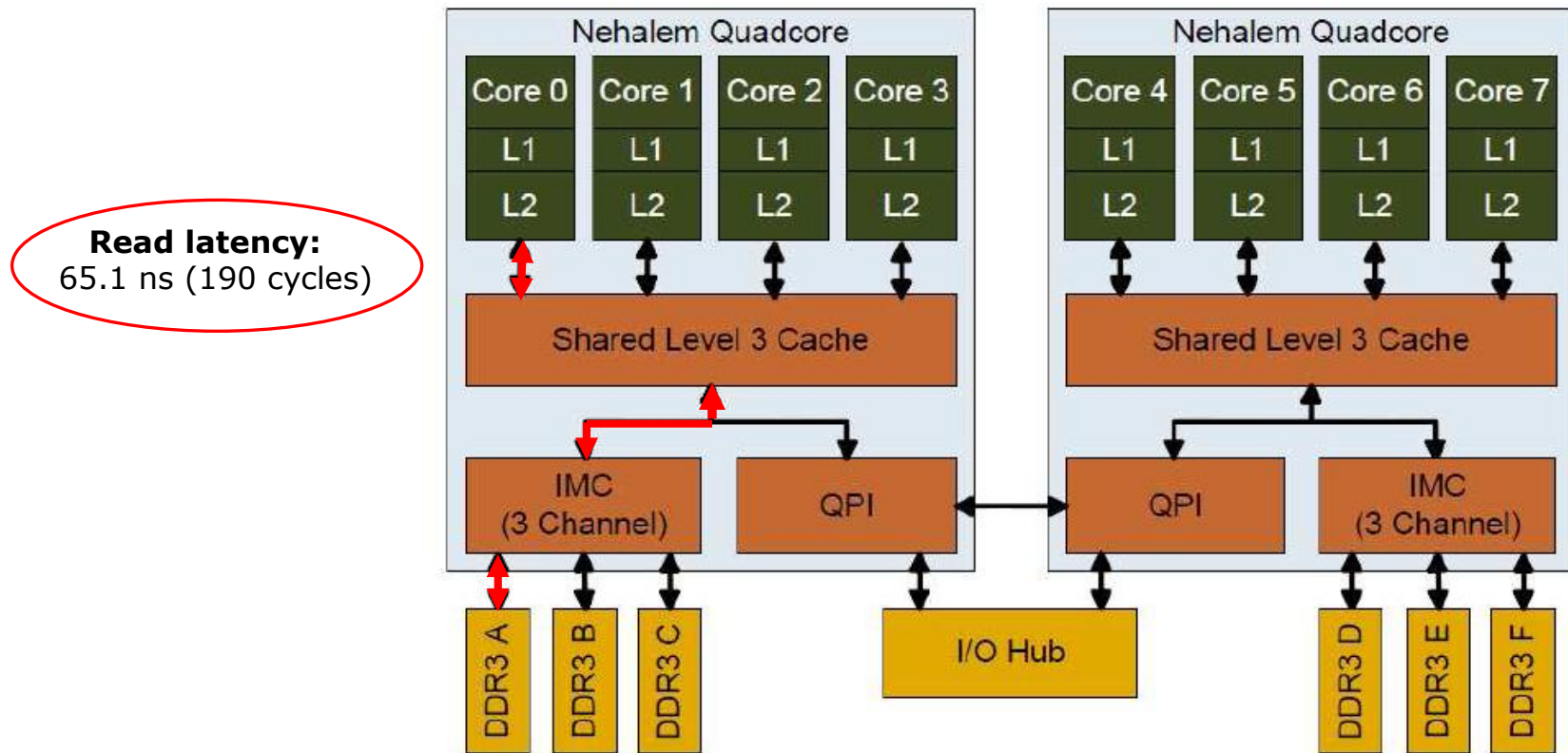
Let's take a look at an **example** of memory latencies in NUMA platforms.

1. Introduction to Intel's Core-based 2S server processors (8)

Example of measured read latencies of uncached data in NUMA platfoms [174]

Read latencies depend on whether the referenced data is kept in the own or remote memory module.

a) Read latencies when referenced data is kept in the local memory.

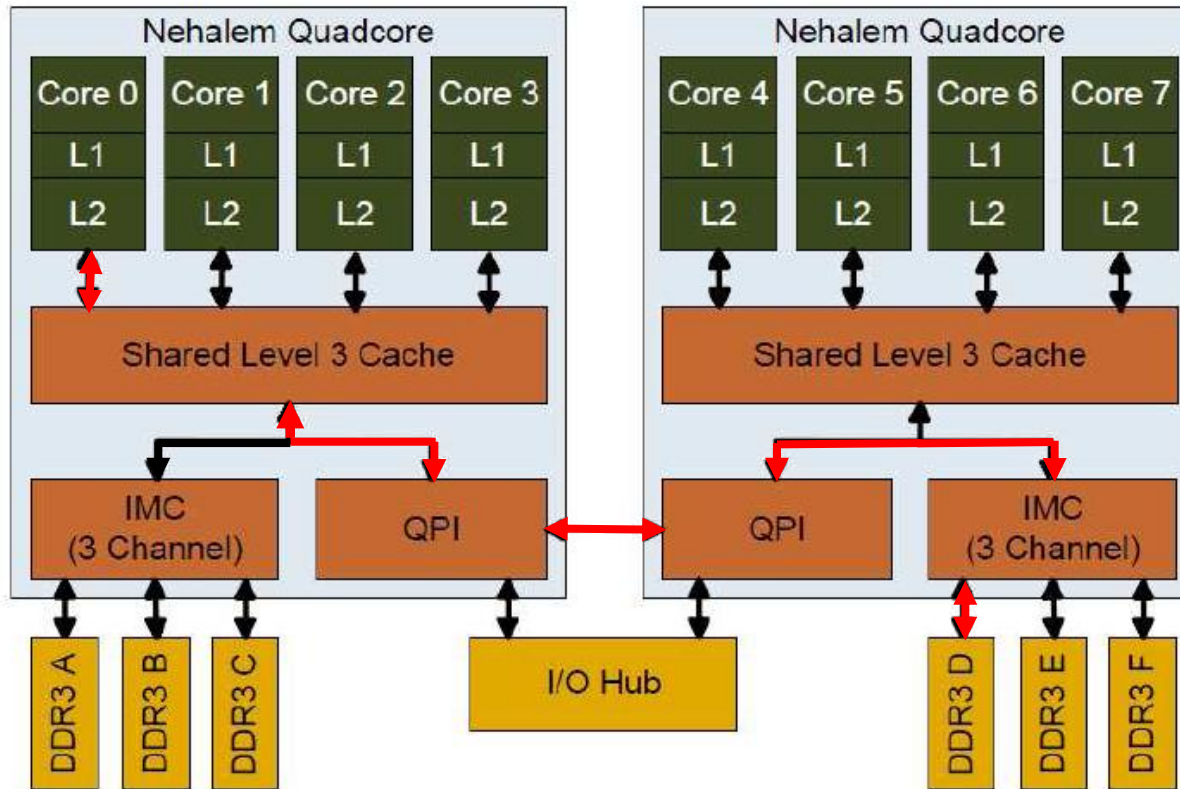


IMC: Integrated MC

1. Introduction to Intel's Core-based 2S server processors (9)

Example of measured read latencies of uncached data in NUMA platfoms -2

b) Read latencies when referenced data is kept in the remote memory
[174]



Read latency:
106 ns (~310 cycles)

Read latency is increased now by the **inter-processor access penalty** of 41 ns.

As the above example shows, **read latencies for remote data** can be **significantly longer** than those for data stored in local memory (106 ns vs. 65.1 ns).

1. Introduction to Intel's Core-based 2S server processors (10)

The 2nd phase of the architectural evolution of Intel's 2S server processors -1

The 2nd phase of the evolution began when Intel defined three performance segments in its server processor portfolio to better address the diverse workload demands of the market.

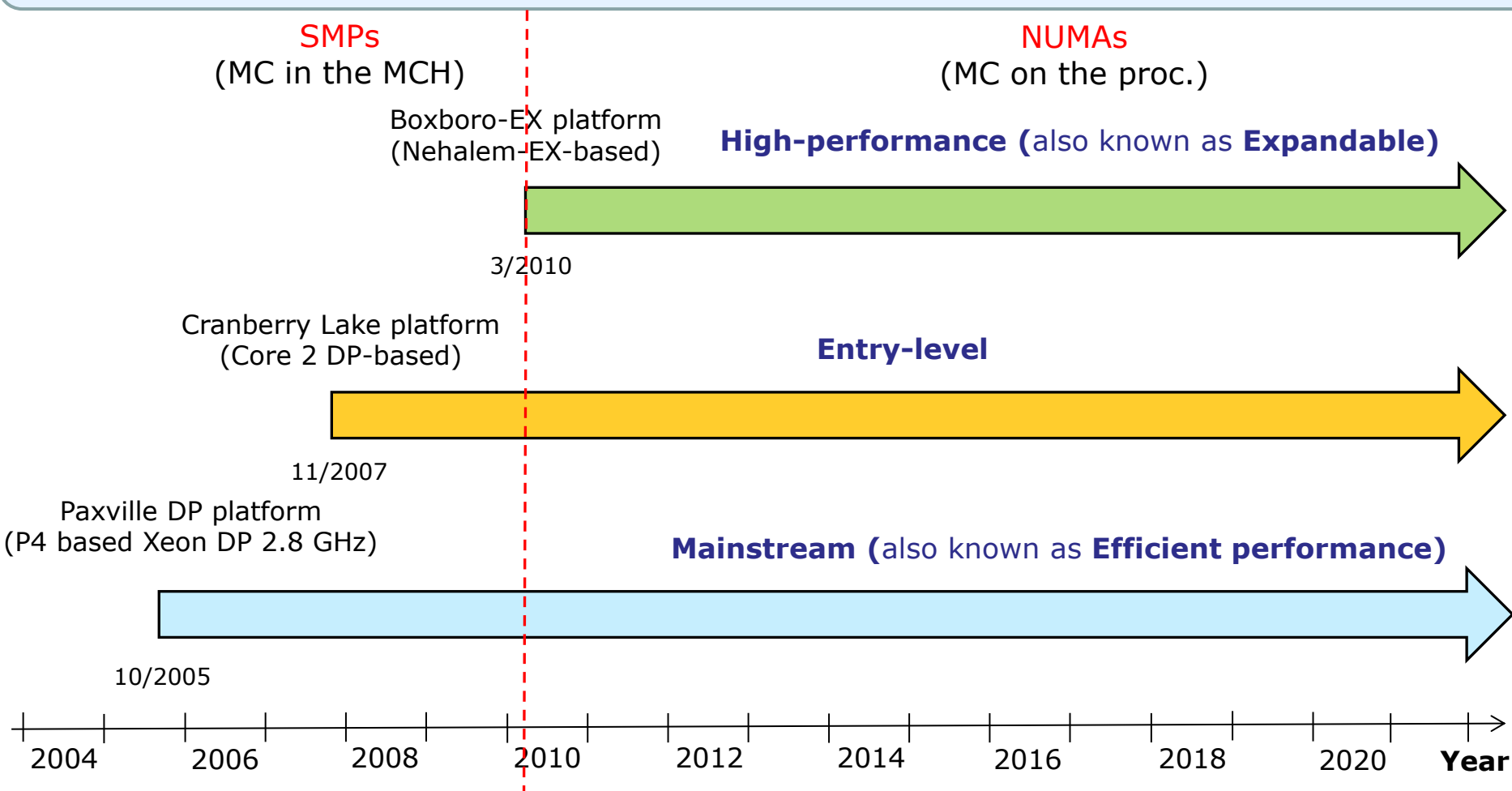


Figure: Introduction of performance segments in Intel's server processors

1. Introduction to Intel's Core-based 2S server processors (11)

The 2nd phase of the architectural evolution of Intel's 2S server processors -2

- In the **second phase of the evolution**, the company introduced **distinct processor designs tailored to each of the three performance segments**.
- Designs for different performance segments typically incorporate different processor types, employed varying memory connection technologies, and supported distinct MCHs.
- **Within each performance segment**, (e.g. for high-performance server processors) **a unified architecture** was applied **across all socket counts** (1-4).
- In the 2nd phase of the evolution, a key architectural shift occurred: **the memory controller became directly integrated into the processor**, rather than into the MCH. This transition led to **NUMA** (Non-Uniform Memory Access) **platforms**, as discussed previously.
- Server processors of this phase were based on the **Nehalem through Broadwell** generations, spanning product launches between **2010-2016**.

1. Introduction to Intel's Core-based 2S server processors (12)

The 3rd phase of the architectural evolution of Intel's 2S server processors

- In the 3rd phase of the evolution, the company continued to standardize its server platform and processor designs by unifying its designs across all socket counts and performance segments.
- The unified designs were designated as Scalable server processors.

This standardization simplified the development and implementation of the server platforms and processors.

- The 3rd phase began in 2017, with the launch of Skylake-SP-based server processors.
- These designs have become fundamental in Intel's server processor offerings and are widely used today.

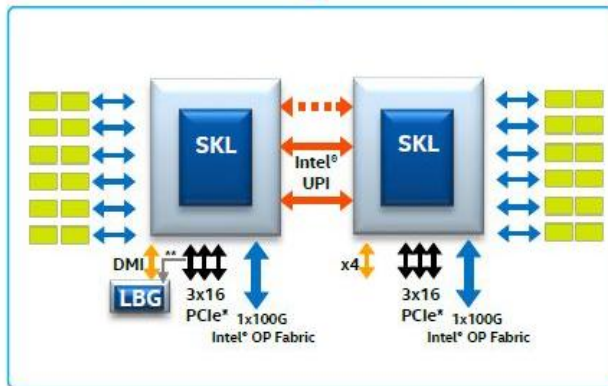
Given their ongoing relevance, we provide a brief overview of these processors subsequently.

1. Introduction to Intel's Core-based 2S server processors (13)

Example of a scalable server design: the Skylake-SP-based Purely platform [20]

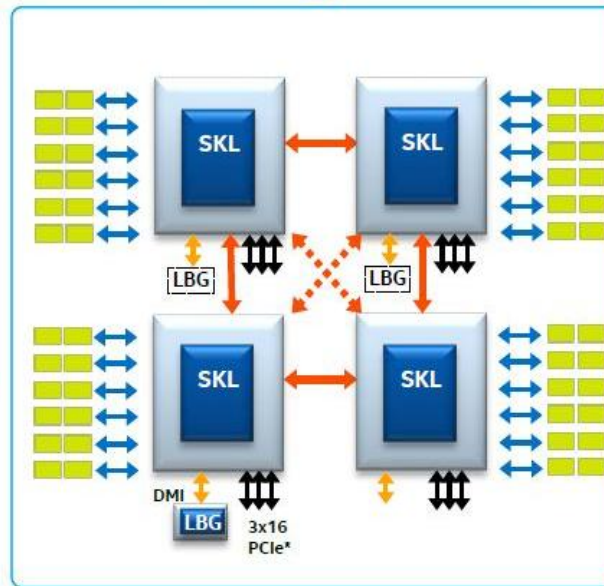
It is **unified** for all processor counts and performance segments.

2S Configurations



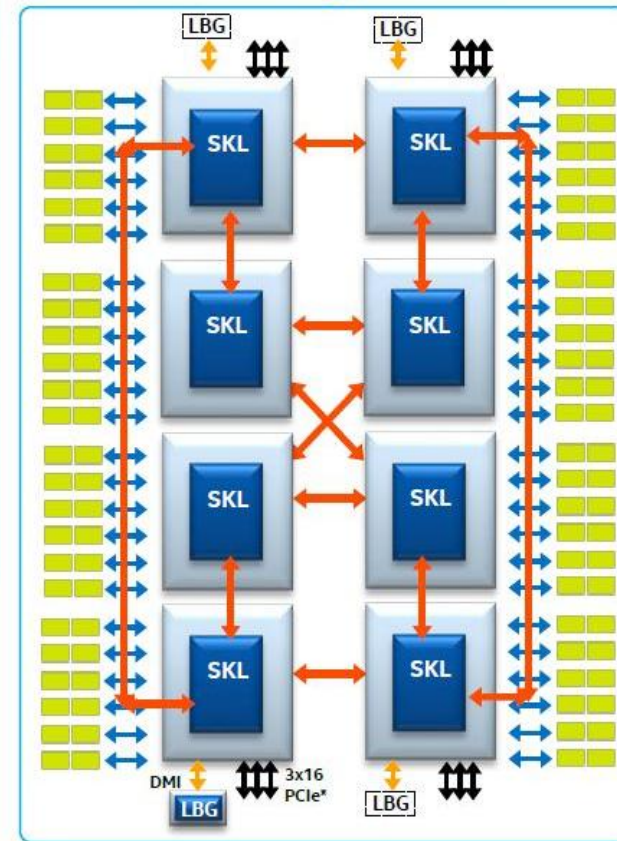
(2S-2UPI & 2S-3UPI shown)

4S Configurations



(4S-2UPI & 4S-3UPI shown)

8S Configuration



LBG: Lewisburg (PCH)

- **Unified designs** have the same **processor series**, the same kind of **memory connection** with 6 memory channels/socket, and the **same chipset**.
- The number of the **UPI links** (enhanced QPI) is scalable.

OP: OmniPath (It was Intel's communication architecture proposed to interconnect processor clusters. In 2019 Intel announced its cancellation.)

1. Introduction to Intel's Core-based 2S server processors (14)

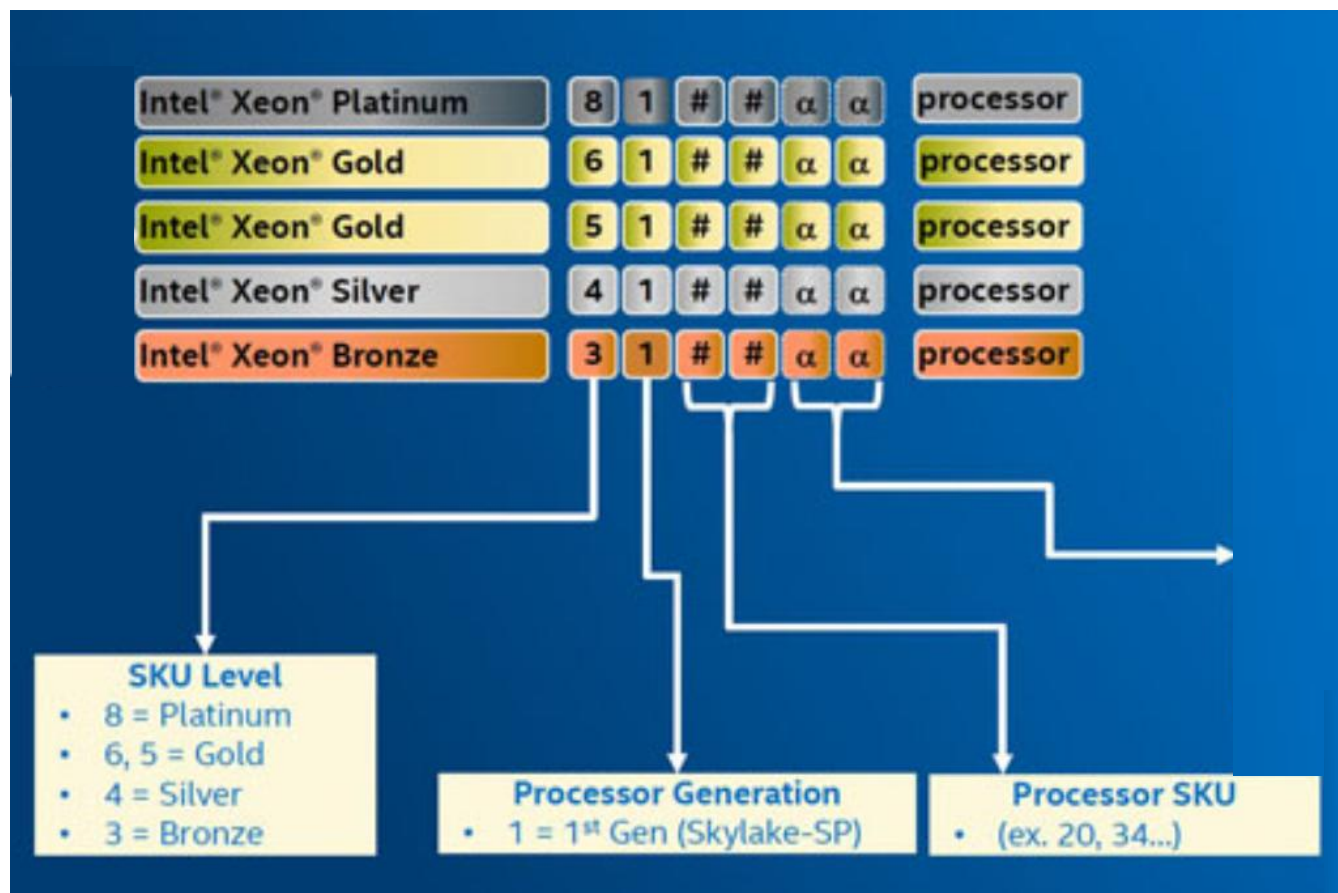
Evolving scalable server processors from the 1st up to the 5th generation

Platform	HE scalable 2S server generations	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket (up to)	TDP (up to) W	Socket
Purley	1 st gen. Xeon Platinum 8176/8180 (Skylake-SP)	14 nm	7/2017	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2400	205	LGA 3647
	2 nd gen. Xeon Platinum 8276/80 (Cascade Lake-SP)	14 nm	4/2019	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2400	205	LGA 3647
	Xeon Platinum 9282 (Cascade Lake-AP)	14 nm	4/2019	2x28C	40x PCIe 3.0 (on proc.)	12x DDR4-2933	400	BGA 5903
	Xeon Gold 6238R/6258R (Cascade Lake R-SP)	14 nm	2/2020	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2933	165/205	LGA 3647
Whitley	3 rd gen. Xeon Platinum 8380 (Eis Lake-SP)	10 nm	4/2021	40C	64xPCIe4.0 (on proc.)	8x DDR4-3200	270	LGA 4189
Eagle Stream	4 th gen. Xeon Platinum 8490H (Sapphire Rapids)	Intel 7	1/2023	60C	80xPCIe5.0 (on proc.)	8x DDR5-4800	350	LGA 4677
	5 th gen. Xeon Platinum 8592/93 (Emerald Rapids)	Intel 7	12/ 2023	64C	80xPCIe5.0 (on proc.)	8x DDR5-4800	330	LGA 4677

Intel 7 is actually a 10 nm technology.

1. Introduction to Intel's Core-based 2S server processors (15)

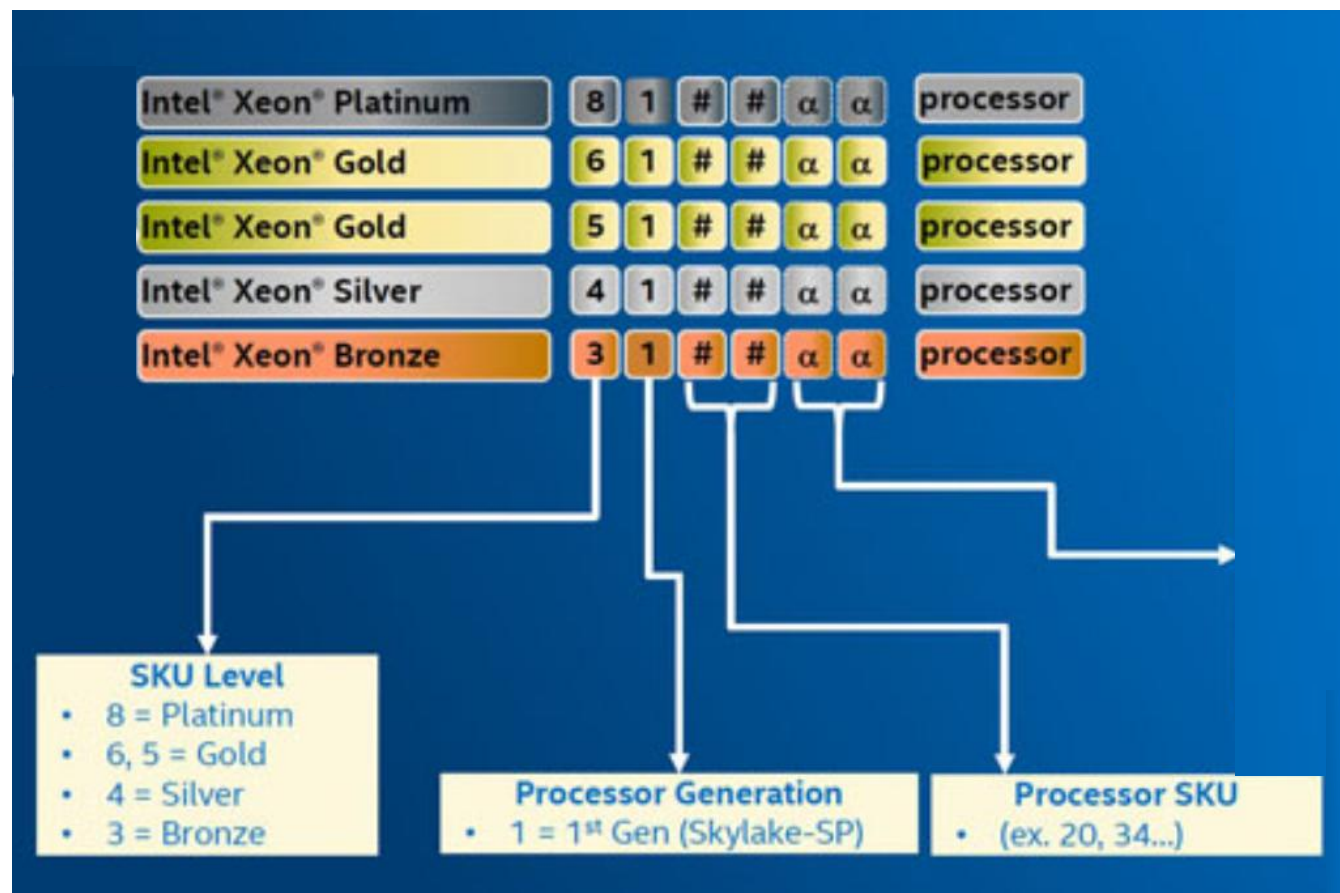
How did Intel name its scalable server processors from the 1st to the 5th gen.? -1



- The **first digit** of the model number represents the **performance class**. There are four performance classes, ranging from Xeon Platinum (highest) to Xeon Bronze (entry-level), as shown above.
- The **second digit** indicates the scalable server **generation**.

1. Introduction to Intel's Core-based 2S server processors (16)

How did Intel name its scalable server processors from the 1st to the 5th gen.? -2



The following two digits in the model designations represent the model identifier, commonly referred to as the SKU (Stock Keeping Unit).

1. Introduction to Intel's Core-based 2S server processors (17)

How did Intel name its scalable server processors from the 1st to the 5th gen.? -1

The final two characters indicate **special features** of the model [89]

(These are not discussed, mentioned for informational purposes only.)



Workload Optimized		Built-in Accelerators*					Up to Socket/ UPI	SKU Shelf
		AMX	DLB	DSA	IAA	QAT		
P	Cloud – IaaS	✓	1	1	1	1	2S/3 UPI	P
V	Cloud – SaaS	✓	1	1	1	1	2S/3 UPI, 1S	P
				1	1		2S/3 UPI	G6
M	Media Transcode	✓		1	1			G6
H	DB and Analytics	✓	4	4	4	4	8S/4 UPI	P
			2	1	1	2	4S/3 UPI	G6
N	Network/5G/Edge (High TPT / Low Latency)	✓	4	4	0	4	2S	P
			2	1	0	2	2S, 1S	G6
								G5
S	Storage & HCI	✓	4	4	0	4	2S/4 UPI	G6
			2	1	0	2	2S/3 UPI	G5
T	Long-life Use/High Tcase	✓		1			2S/2 UPI	S
U	1-Socket*	✓		1			1S	G6
Q	Liquid cooling	✓		1			2S/3 UPI	P
								G6
+	Feature +	✓	1	1	1	1		

1. Introduction to Intel's Core-based 2S server processors (18)

New advancement beginning with the 6th generation: Introducing two types of server classes: **performance-oriented** and **efficiency-oriented classes**

The **two** introduced **classes** of scalable server processors are:

- **performance-oriented models**, built up of performance-oriented **P cores**, and
- **efficiency-oriented models**, which utilize **E-cores**.

These server classes are **identified by the letters P and E**, set after the 4-digit model designations, such as 6780E or 6780P.

- Intel's dual server classes approach closely resembles **AMD's earlier concept of designing two parallel series of server processors**: the performance-oriented EPYC Genoa series, based on **Zen 4** cores and the efficiency-oriented EPYC Bergamo series, built on **Zen 4c** cores.
- Both companies now meet the diverse demands of current server workloads by offering **dual server classes**.

1. Introduction to Intel's Core-based 2S server processors (19)

6th generation scalable server processors with performance-oriented and efficiency-oriented server classes

Platform	HE scalable 2S server models	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket	TDP (up to) W	Socket
Beechnut City	6 th gen. 6780E (Xeon 6) (Sierra Forest-SP)	Intel 3	6/2024	144C (E-cores) (Crestmont) (no SMT)	88x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA 4710
	6 th gen. 6780P (Granite Rapids-SP)	Intel 3 / Intel 7	9/2024	86C (P-cores) (Redwood Cove) (SMT)	96x PCIe 5.0 (on proc.)	8x DDR5-6400	350?	LGA-4710
Avenue stream	6 th gen. Xeon 6 6980E (Sierra Forest-AP)	Intel 3 / Intel 7	2/2025	288C (E-cores) (Cresmont) (no SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529
	6 rd gen. 6980P (Granite Rapids-AP)	Intel 3 / Intel 7	9/2024	128C (P-cores) (Redwood Cove) (SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529

Remarks

Sierra Forrest: Efficiency-oriented, based on E-cores

Granite Rapids: Performance-oriented, based on P-cores

P-cores support simultaneous multithreading (SMT) while E-cores do not.

SP: Scalable Performance (8 memory channels) with the LGA-4710 socket

AP: Advanced Performance (12 memory channels) with the LGA-7529 socket

1. Introduction to Intel's Core-based 2S server processors (20)

Changes in the naming of the 6th gen. scalable processors

Beginning with the 6th generation, Intel made **two important changes** to the naming of its scalable server processors, which will be discussed next.

a) Intel **discontinued** using the “scalable processor” expression in the model names.

Instead, it adopted a **simplified naming convention**: server processors are now identified **solely** by their **generation number** (e.g., *Intel Xeon 6th gen.*) without the “Scalable” designation, e.g. 6th gen.

b) Up to the 5th generation scalable processors, there were **four performance classes**, designated by Xeon Platinum (highest) to Xeon Bronze (entry-level).

The new naming scheme **also** maintains **four performance classes**, which are simply **numbered as 9 (highest), 7, 5 to 3 (entry-level)**.

Intel identifies the **related target market segments** as follows [214]:

Series	Target market segment
6900	Most demanding cloud, AI and HPC environments
6700	Wide range of data center environments
6500	Mainstream server and edge environments
6300	Small/medium business applications

Evolving scalable server processors over the 6th generation

- We note that Intel's [next-generation](#) server processor line is named **Clearwater Forest**.
- It will use Intel's most advanced [18A technology](#) and is expected to be unveiled in H1 2026.
- [In Section 3.3](#): titled "3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor", we provide an [overview](#) of this advanced server line.

2. Exploring Intel's 2S server processors

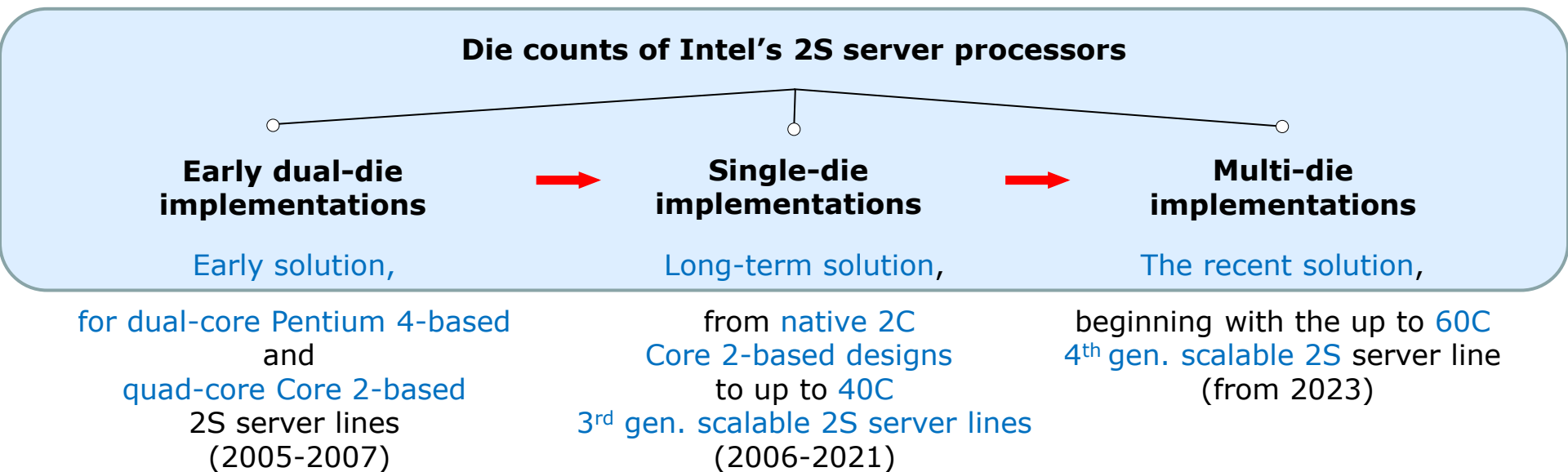
- 2.1 From how many dies are Intel's 2S server processors built up?
- 2.2 How large are Intel's 2S server dies and how many transistors do they include?
- 2.3 At what rate do core counts and memory rates rise in Intel's high-end 2S server processors?
- 2.4 Evolution of on-die interconnects
- 2.5 TDP values and socketing of Intel's high-end 2S server processors

2.1 From how many dies are Intel's 2S server processors built up?

2.1 From how many dies are Intel's 2S server processors built up? (1)

2.1 From how many dies are Intel's 2S server processors built up?

The following figure illustrates the evolution of die counts in Intel's 2S server processors:



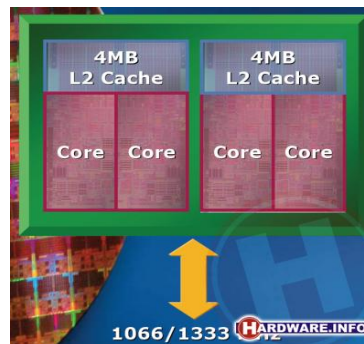
- As shown, early dual-die implementations advanced to single-die designs, which dominated server processor architectures for a long time, until 2021. In the final phase of the evolution, multi-die implementation re-emerged and gained prominence starting in 2023.
- In the following Section, we will briefly outline these key stages of the architectural evolution.

2.1 From how many dies are Intel's 2S server processors built up? (2)

a) Early dual-die server processor designs

- In the mid-2000s, at the dawn of the multicore era, Intel accelerated its entry into the server processor market by providing Pentium 4- or Core-based dual- or quad-core server processors by dual-die designs to compete with AMD's single-die dual-core designs.
- Instead of waiting for its IC technology to sufficiently advance for fabricating monolithic dual-core or quad-core implementations, Intel opted for a multi-die packaging approach by interconnecting two separate dies within the same package to avoid market loss.
- Below, we provide a few examples of such dual-die designs which Intel introduced during the early multicore era.
- Dual-core Pentium 4-based server processors implemented on two dies
Paxville DP: (2x1C), 2xDP-enhanced 64-bit Pentium 4 Prescott (10/2005)
Xeon 5000 (Dempsey): (2x1C), 65 nm shrink of the 90 nm Paxville DP 2.8 (5/2006)
- Quad-core Core 2-based server processors implemented on two dies
Xeon 5300 (Clowertown): (2x2C) 2x Core 2-based 65 nm Xeon 5100 (Woodcrest), (11/2006)
Xeon 5400 (Harpertown): (2x2C), 2x Penryn-based 45 nm 2C Harpertown (11/2007)

Example: The quad-core Xeon 5300 implemented on two dual-core Core 2 dies (Clowertown) [90], [91]

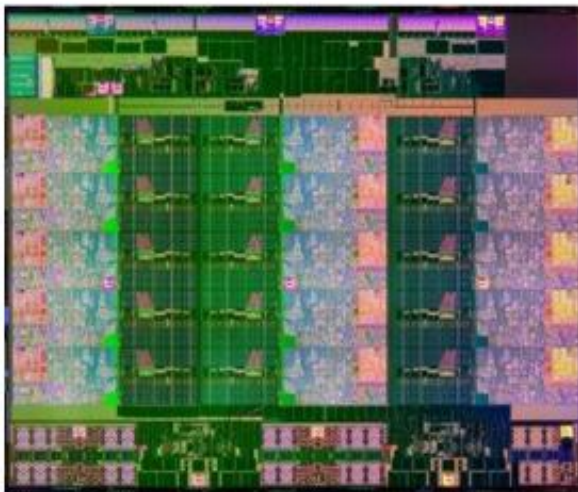


2.1 From how many dies are Intel's 2S server processors built up? (3)

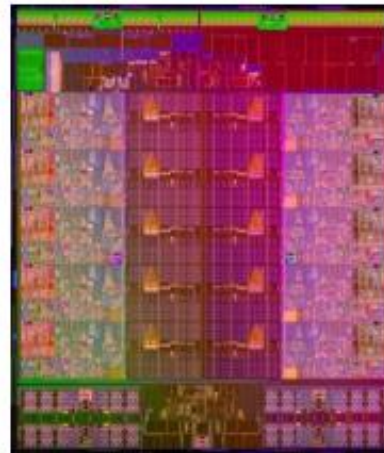
b) Single-die implementations -1

- From 2006 to 2021, from the Core 2-based Xeon 5100 2C series (Woodcrest) to the Ice Lake-SP-based 3rd gen. Xeon Scalable 2S processors with up to 40 cores Intel typically implemented its 2S server processors on a **single die**.
- However, beginning with the 15-core Ivy Bridge-EX line (2014), Intel **implemented three distinct die layouts instead of one, to reduce the fabrication costs of its low core count server processors**, as depicted below.

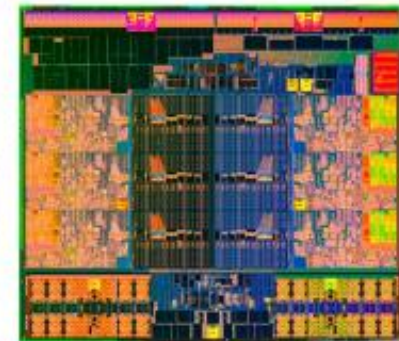
In processor models providing **fewer cores than are present on the die**, **only the required number of cores would be activated**, such as 8 cores on a 10-cores die.



HCC (XCC) layout
(High/Extreme Core Count)
for up to 15 cores



MCC layout
(Medium Core Count)
for up to 10 cores



LCC layout
(Low Core Count)
for up to 6 cores

Figure: Three monolithic die layouts of the Ivy Bridge-EX line (2014) [95]

2.1 From how many dies are Intel's 2S server processors built up? (4)

Another example of a single-die implementation

- Another example is Intel's 1st generation scalable 2S server processor, known as **Skylake-S**, introduced in 2017, which offered up to 28 cores, and was manufactured using 3 distinct die layouts, featuring up to 16, 28, and 40 cores, as depicted below.

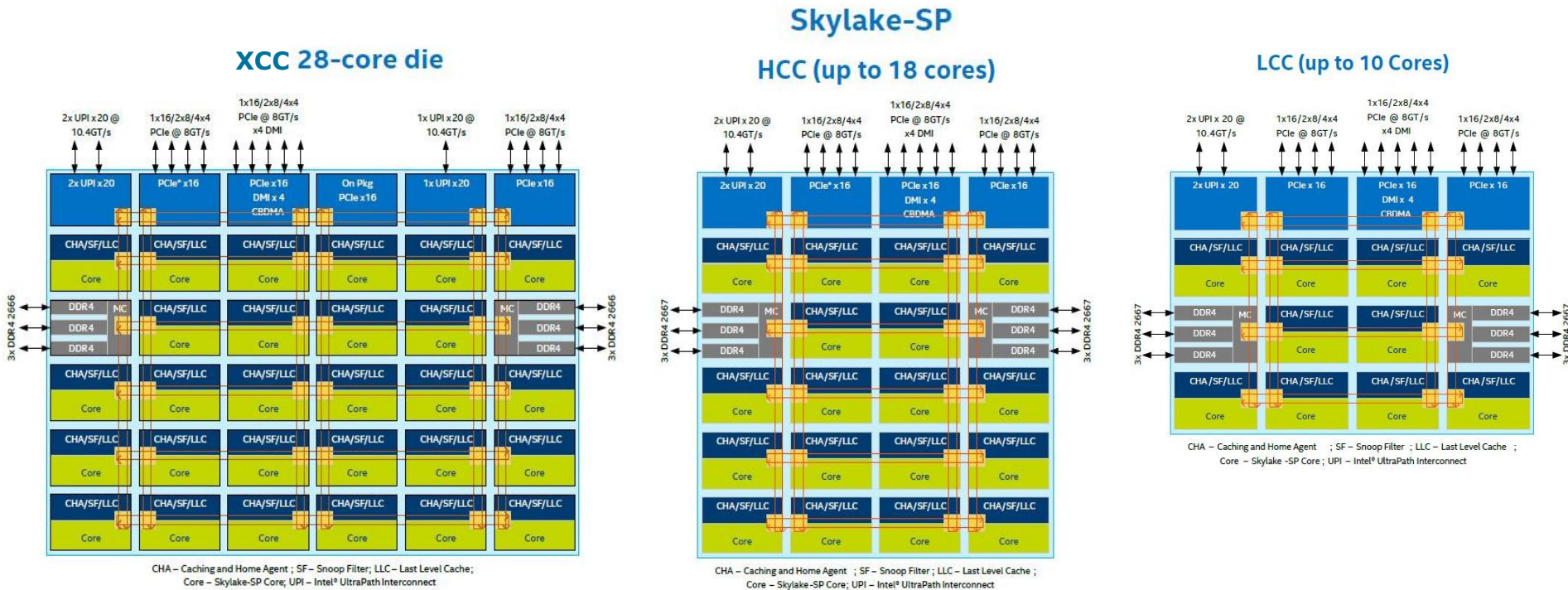


Figure: The 3 monolithic die layouts of the 3rd gen. scalable Ice Lake-SP processor [175]

- Despite the differences in core counts, all three layouts maintain a scalable implementation style by implementing the **same number of memory channels (6) and PCIe lanes.**

2.1 From how many dies are Intel's 2S server processors built up? (5)

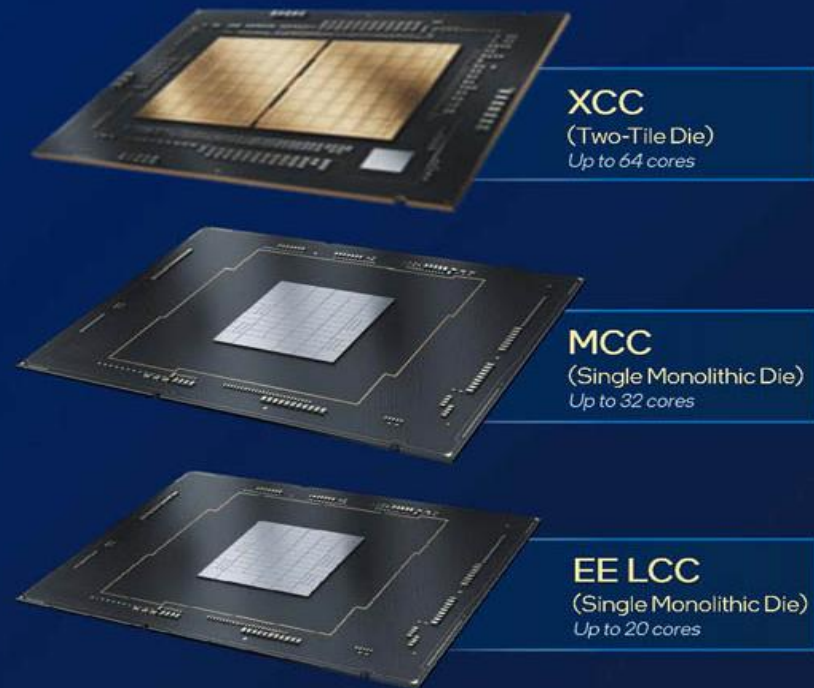
c) Intel's transition from single-die to multi-die designs -1 [195]

- Beginning with the 4th generation scalable processors, launched in 2023, Intel transitioned to **multi-die designs** in its 2S server processors, as illustrated in the Figure below [195].

4. gen. Xeon processors (01/2023) Sapphire Rapids



5. gen. Xeon processors (12/2023) Emerald Rapids (12/2023)



- It is important to note that Intel's 4th and 5th gen. multi-die scalable implementations are **homogeneous** multi-die designs, which are composed of **identical dies (tiles)**.
- In contrast, **heterogeneous** multi-die designs incorporate **non-identical tiles**.

2.1 From how many dies are Intel's 2S server processors built up? (6)

Example: The 4th gen. homogeneous up to 4-die Sapphire Rapid multi-core (2023)

- This design has **three variants**: a **single-die option** for **up to 32 cores** and two **quad-die options**, which incorporate **up to 60 cores**, as shown in the Figure below.

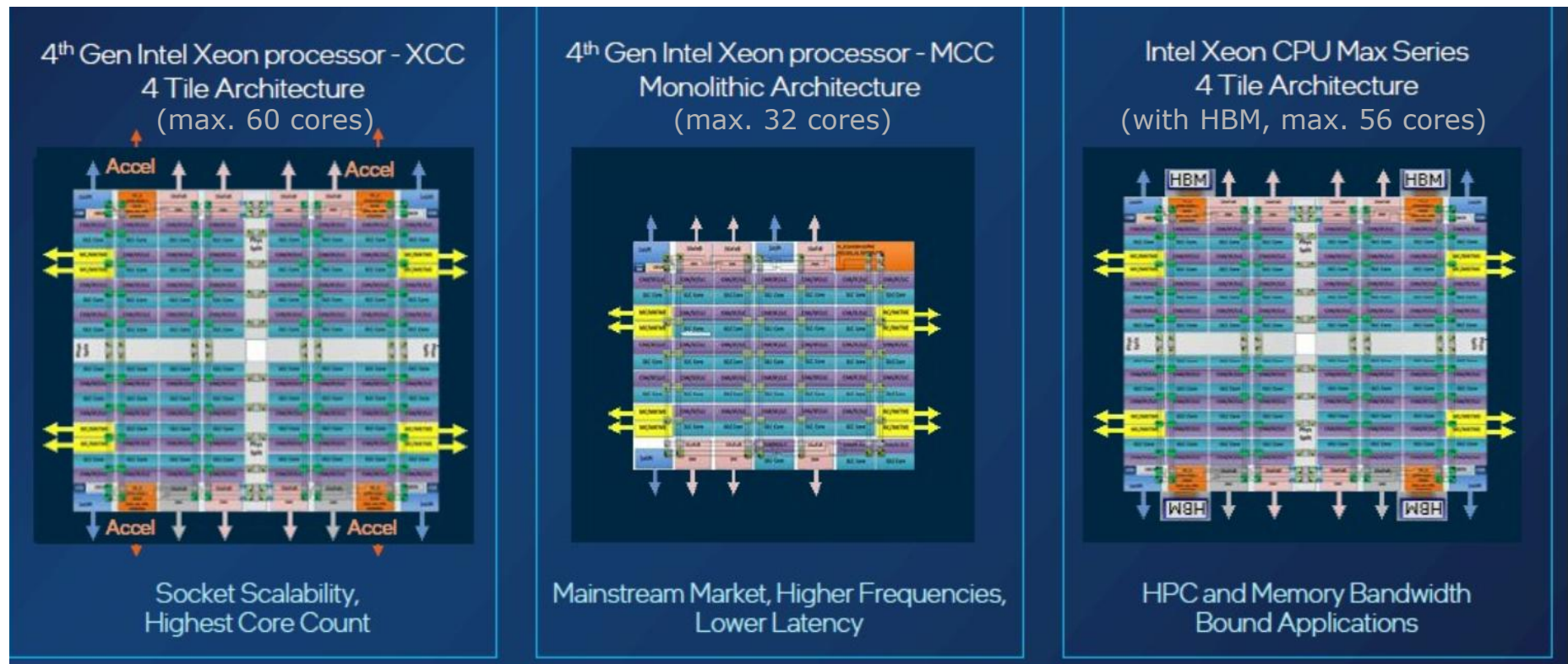


Figure: Die configurations of Sapphire Rapids processors [176]

- One of the dual-die configurations (as shown on the right) includes **on-package HBM (High-Bandwidth Memory)**, but it only has up to 56 cores.
- All die configurations provide **4x128-bit DDR5 interfaces** (equaling 8x64-bit memory channels).

2.1 From how many dies are Intel's 2S server processors built up? (7)

Benefits and downsides of multi-die designs (in general)

- Multi-die designs lead to higher manufacturing yield, which results in lower fabrication costs.
- However, these designs also have downsides, as the dies need to be interconnected and packaged together, as discussed in Section 3.2 of the Chapter "Intel's Core-based client processors".

2.1 From how many dies are Intel's 2S server processors built up? (8)

Intel's technology to interconnect dies: the 2.5D EMIB technology

- We note that Intel's multi-die scalable server processors make use of Intel's **EMIB (Embedded Multi-die Interconnect Bridge)** technology for interconnecting dies.
- This technology employs **short die-to-die silicon bridges for the interconnections**, while the **Si-bridges** are **embedded** within the organic package, as illustrated below.

For more details, we refer to Section 4 of the Chapter "Intel's Core-based client processors".

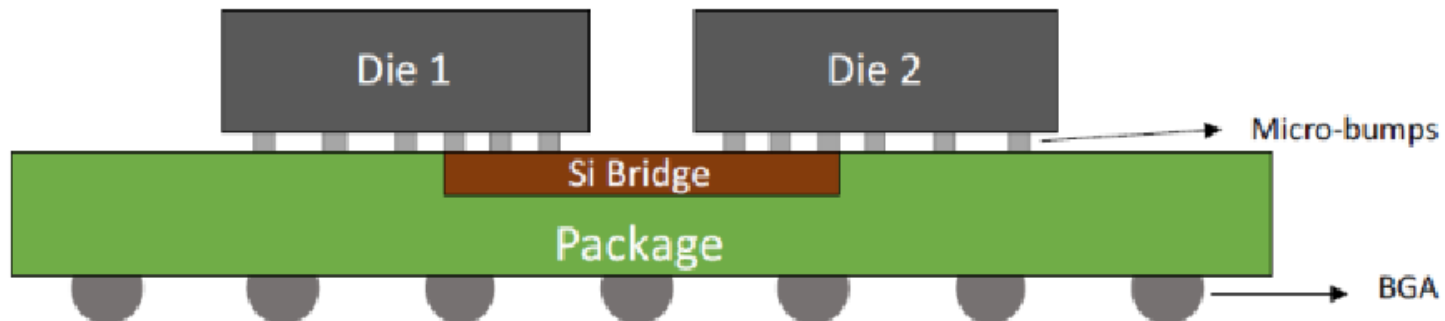


Figure: Intel's EMIB technology [38]

- We note also that the EMIB technology was introduced by Intel in **2017**, in its Stratix 10 FPGA chip.

2.1 From how many dies are Intel's 2S server processors built up? (9)

d) Intel's shift from homogeneous to heterogeneous multi-die designs -1

- Compared to homogeneous multi-die design **heterogeneous multi-die designs** offer additional benefits in **lowering manufacturing costs**.
- This advantage arises from the fact that in heterogeneous multi-die designs, the dies implement different functionalities and **dies with less demanding functionalities can be manufactured using less advanced and more cost-effective IC technologies**.
- For example, consider a scenario where the performance-critical CPU die is produced using 3 nm technology, while the graphics die and the IO die are fabricated using 10 nm technology.

This approach clearly leads to an overall reduction in manufacturing costs, as discussed in more detail in Section 3.2 of the Chapter titled "Intel's Core-based client processors".

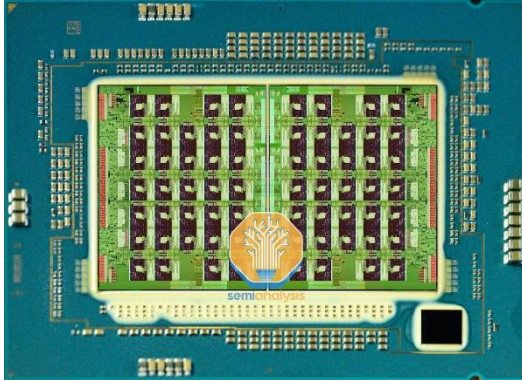
2.1 From how many dies are Intel's 2S server processors built up? (10)

Intel's shift from homogeneous to heterogeneous multi-die designs -2

Basic segmentation options of multi-die server processors

Homogeneous multi-die 2S server processor designs

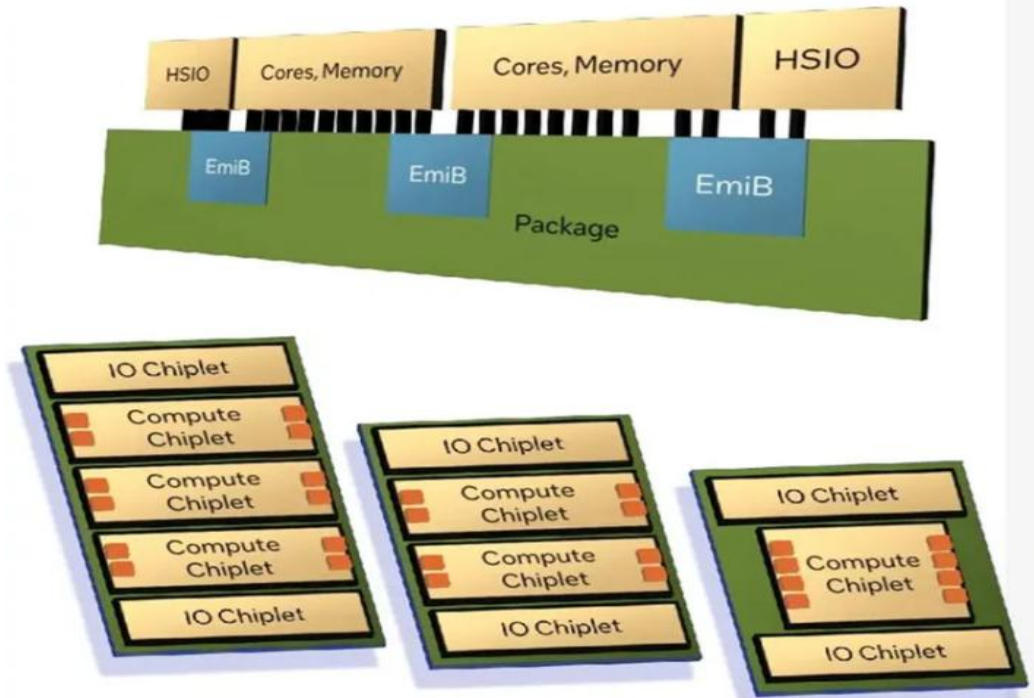
The processor is partitioned into functionally identical parts providing the same functionalities, which are then interconnected and co-packaged.



Intel's 5th gen. homogeneous multi-die Emerald Rapids 2S server processor (2023) [196]

Heterogeneous multi-die 2S server processor designs

The processor is segmented into multiple functional components, which are then interconnected and co-packaged.



Intel's 6th gen. heterogeneous multi-die Sierra Forest/Granite Rapids SP/AP 2S server processors (2024) [197]

2.1 From how many dies are Intel's 2S server processors built up? (11)

Comparing Intel's and AMD's progression towards heterogeneous multi-die 2S server designs -1

Intel

**dual-die
dual-core
designs**



**single-die
dual-core
designs**



**homogeneous
multi-die
design**



**heterogeneous
multi-die
design**

*Xeon DP 2.8 GHz
P4 Prescott
2M-based
(2005)*

*Xeon 5100
(Woodcrest)
Core 2-based
(2006)*

*4th gen.
scalable server
Sapphire Rapids
(2023)*

*6th gen.
Xeon 6
Sierra Forest-SP/AP
Granite Rapids-SP/AP
(2024)*

AMD

**single-die
dual-core
designs**



**homogeneous
multi-die
design**



**heterogeneous
multi-die
designs**

*K8-based
Opteron 200
Italy
(2005)*

*Zen-based
Naples
(2017)*

*Zen 2-based
Rome
(2019)*

2.1 From how many dies are Intel's 2S server processors built up? (12)

Comparing Intel's and AMD's progression towards heterogeneous multi-die 2S server designs -2

- In 2005, both Intel and AMD launched dual-core 2S server processors. Intel initially employed a dual-die design to reduce the time to market, which was soon replaced by a single-die implementation.
In contrast, AMD started with a more advanced single-die design from the start for their dual-core server processors in 2005.
- By 2017, with its Zen family, AMD first introduced homogeneous and, by 2019, heterogeneous multi-die designs to reduce manufacturing costs, as mentioned before.
- A few years later, in 2023 and 2024, respectively, Intel also adopted these design techniques to capitalize on their advantages.
- Ultimately, by 2024, both Intel's and AMD's client and server processors arrived at heterogeneous multi-die designs, although they employ different techniques for interconnecting and packaging multiple dies.

2.2 How large are Intel's 2S server dies and how many transistors do they include?

2.2 How large are Intel's 2S server dies and how many transistors do they include? (1)

2.2 How large are Intel's 2S server dies, and how many transistors do they include? -1

- Up to the Skylake family (2017), Intel disclosed die sizes and transistor counts for its processors.

This information is presented in the Table below.

2.2 How large are Intel’s 2S server dies and how many transistors do they include? (2)

How large are Intel’s 2S server dies, and how many transistors do they include? -2

The Table below shows published data on die sizes and transistor counts.

Platform4	Core	Techn.	Intro.	High performance Core 2-based multicore 2S server processors	Core count up to	Trans. count [10 ⁶ /10 ⁹]	Die size [mm ²]
Bensley	Pentium 4 Prescott 2M DP	65 nm	5/2006	Xeon 5000 (Dempsey)	2x1C	2x188	2x81
	Core2	65 nm	6/2006	Xeon 5100 (Woodcrest)	2C	291	143
	Core2	65 nm	11/2006	Xeon 5300 (Clovertown)	2x2C	2x291	2x143
	Penryn	45 nm	11/2007	Xeon 5400 (Harpertown)	2x2C	2x410	2x107
Boxboro-EX	Nehalem	45 nm	3/2010	Xeon X6500 (Beckton/Nehalem-EX)	8C	2.3	513
	Westmere	32 nm	4/2011	Xeon E7-28xx (Westmere-EX)	10C	2.6	512
Brickland	Ivy Bridge	22 nm	2/2014	Xeon E7-28xx v2 (Ivy Bridge-EX)	15C	4.31	541
	Haswell	22 nm	9/2014	Xeon E5-2600xx v3 (Haswell-EP)	18C	5.69	662
	Broadwell	14 nm	4/2016	Xeon E5-26xx v4 (Broadwell-EP)	22C	3.2	456
Purley	Skylake	14 nm	7/2017	Xeon Platinum 8176/80 (Skylake-SP)	28C	8.0?	~678
	Cascade Lake	14 nm	4/2019	Xeon Platinum 8276/80 (Cascade Lake-SP)	28C	n.a.	n.a.
	Cascade Lake	14 nm	4/2019	Xeon Platinum 9282 (Cascade Lake-AP)	2x28C	n.a.	n.a.
	Cascade Lake R	14 nm	2/2020	Xeon Gold 6238R/6258 (Cascade Lake R)	28C	n.a.	n.a.
Whitney	Ice Lake	10 nm	4/2020	Xeon Platinum 8380 (Ice Lake-SP)	40C	n.a.	~640
Eagle Stream	Sapphire Rapids	Intel 7	1/2023	Xeon Sapphire Rapids MCC monolithic die	32C	n.a.	~770
	Emerald Rapids	Intel 7	11/2023	Xeon Emerald Rapids (a single die)	64C	n.a.	~700

2.2 How large are Intel's 2S server dies and how many transistors do they include? (3)

How large are Intel's 2S server dies, and how many transistors do they include? -3

- However, **after the Skylake-based processors**, Intel ceased publishing die size and transistor counts, citing competitive and strategic reasons.
- As a result, current information on die sizes is limited to third-party estimates, reverse engineering, or leaks, whereas transistor counts for newer processors remain largely unavailable.

2.2 How large are Intel’s 2S server dies and how many transistors do they include? (4)

How large are Intel’s 2S server dies, and how many transistors do they include? -4

The Table below shows published data on die sizes and transistor counts.

Platform4	Core	Techn .	Intro.	High performance Core 2-based multicore 2S server processors	Core count up to	Trans. count [10 ⁶ /10 ⁹]	Die size [mm ²]
Bensley	Pentium 4 Prescott 2M DP	65 nm	5/2006	Xeon 5000 (Dempsey)	2x1C	2x188	2x81
	Core2	65 nm	6/2006	Xeon 5100 (Woodcrest)	2C	291	143
	Core2	65 nm	11/2006	Xeon 5300 (Clovertown)	2x2C	2x291	2x143
	Penryn	45 nm	11/2007	Xeon 5400 (Harpertown)	2x2C	2x410	2x107
Boxboro-EX	Nehalem	45 nm	3/2010	Xeon X6500 (Beckton/Nehalem-EX)	8C	2.3	513
	Westmere	32 nm	4/2011	Xeon E7-28xx (Westmere-EX)	10C	2.6	512
Brickland	Ivy Bridge	22 nm	2/2014	Xeon E7-28xx v2 (Ivy Bridge-EX)	15C	4.31	541
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	Cascade Lake	14 nm	4/2019	Xeon Platinum 9282 (Cascade Lake-AP)	2x28C	n.a.	n.a.
	Cascade Lake R	14 nm	2/2020	Xeon Gold 6238R/6258 (Cascade Lake R)	28C	n.a.	n.a.
Whitney	Ice Lake	10 nm	4/2020	Xeon Platinum 8380 (Ice Lake-SP)	40C	n.a.	~640
Eagle Stream	Sapphire Rapids	Intel 7	1/2023	Xeon Sapphire Rapids MCC monolithic die	32C	n.a.	~770
	Emerald Rapids	Intel 7	11/2023	Xeon Emerald Rapids (a single die)	64C	n.a.	~700

2.2 How large are Intel's 2S server dies and how many transistors do they include? (5)

Restriction on the die size

The previous Table shows that the **die size is limited to about 800 mm²**.

Why is this the case?

2.2 How large are Intel's 2S server dies and how many transistors do they include? (6)

Why is the die size restricted? -1

- The answer lies in the **IC fabrication** during which complex circuit patterns are created on a Si wafer by using **dozens of photomasks** in sequence.
- Each photomask defines a specific layer of the circuit.
- To create these patterns, the **Si surface** is first **coated with a photo-resistive material** (fény érzékeny), which reacts to light exposure.
- A **light source**, typically **UV**, is then used **to project the mask's pattern** onto the wafer.
- Depending on the type of photoresist used, either the exposed or non-exposed regions are chemically removed.

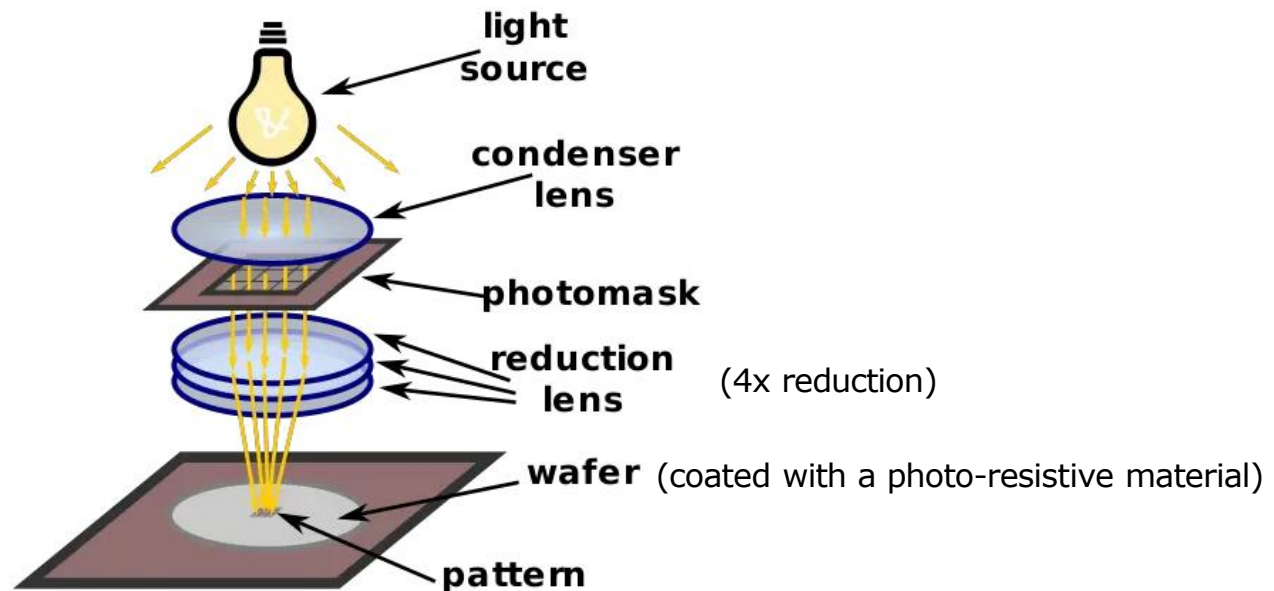


Figure: The standard process of exposing wafers using photomasks, also known as reticles [118].

2.2 How large are Intel's 2S server dies and how many transistors do they include? (7)

The lithography process [187]



Mask Making - Outline



- Part 1
 - What is a photomask?
 - Mask making steps
- Part 2
 - Writing masks with electrons
- Part 3
 - Inspection, repair, and metrology
 - Challenges for EUV masks

build uh euv Mas in the hope that euv
will soon be ready for

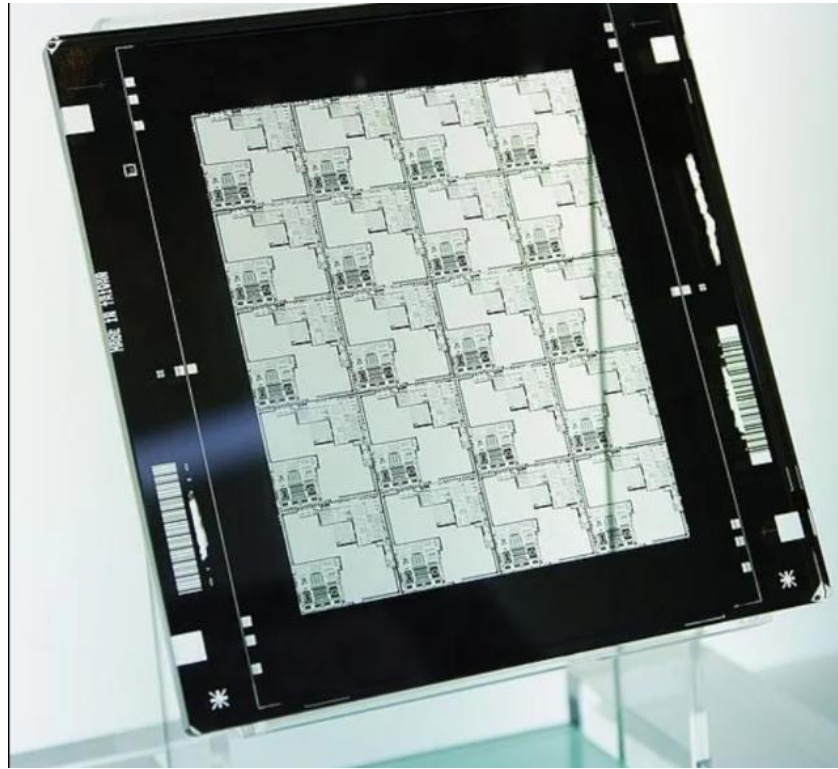
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2



2.2 How large are Intel's 2S server dies and how many transistors do they include? (8)

Example of a photomask [119]



- The **standard photomask size** is currently **104 x 132 mm**.
- However, **optical limitations**, such as aberrations and imperfections that increase towards the edges of the mask, notably **reduce the usable exposure area**.
- To achieve **high-resolution projections**, the photolithography system uses a **4× reduction lens**, which projects the mask image onto the wafer at 1/4 scale.
- As a result, the **actual exposed area on the wafer** becomes **26 × 33 mm²**.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (9)

Why is the die size restricted? -2

- The entire wafer will be exposed by **stepping and repeating** the exposure, as illustrated below.

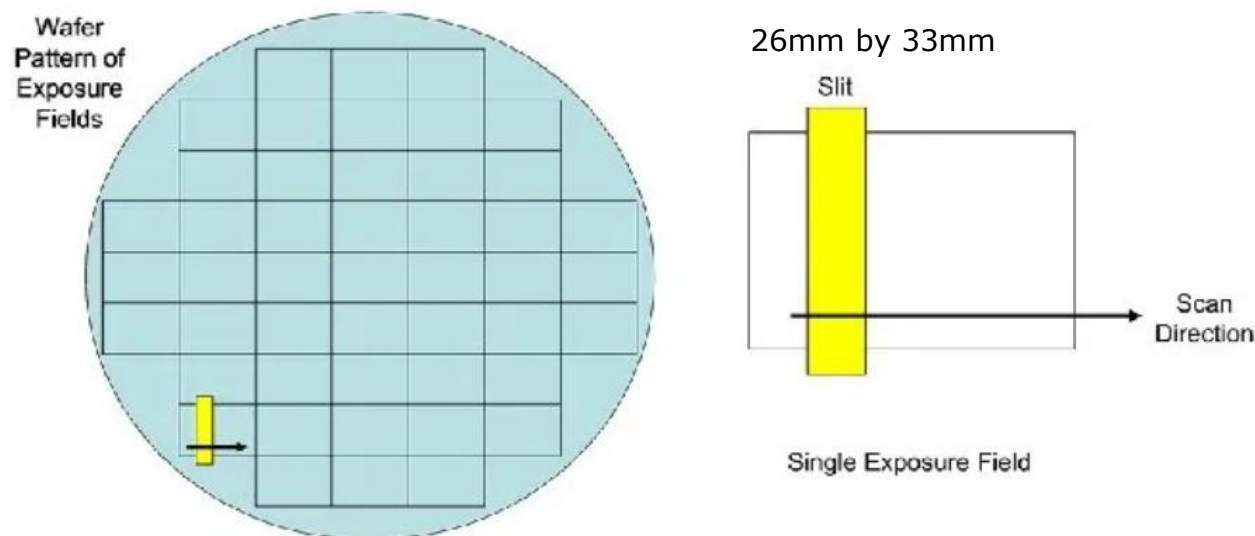


Figure: Stepping and repeating the exposure across the entire wafer [120]

- Fabs typically use **300 mm (12") wafers**.
- Since two neighboring exposures cannot be properly aligned, the **area of one exposure limits the maximal die size, to the exposed area on the wafer**, which is $26 \times 33 \text{ mm}^2 = \text{totaling } 858 \text{ mm}^2$.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (10)

Considerations regarding the die size in IC manufacturing -1

- There are **two important aspects** to consider regarding the die size.
- **One aspect is the max. die size**, which is **constrained by limitations of the photolithography**. During fabrication, each die is produced through a **series of exposures** on a silicon wafer, as previously discussed.
Neighboring exposures cannot be perfectly aligned, however, due to inherent limitations of both the stepper and the scanner system.
As a result, **the maximum die size is limited to the area of a single exposure**, which is currently **26 mm × 33 mm**, yielding a **maximum die size of 858 mm²**.
- **The other important aspect** of choosing the die size is the **economics of IC fabrication**. This aspect follows from the fact that on **larger dies, linearly more defect spots** occur. Since even a single defect can render an entire die unusable, larger dies suffer from lower yields, which directly raises the manufacturing costs per functional chip. Therefore, in semiconductor production, a **balance has to be found between die size and acceptable yield rates**, which determines the **economically optimal die size**. This balance **often calls for multi-die designs**, as smaller dies improve yields and lower costs.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (11)

Considerations regarding the die size in IC manufacturing -3

- According to the above data, the **largest die** in Intel's Core family appears to be the Sapphire Rapids 32-core MCC die, **measuring about 770 mm²**.
- In comparison, a **single tile of the up to 60-core 4th gen. Sapphire Rapids XCC** (Extreme Core Count) processor measures around **400 mm²**.

It follows that a **4-die multi-chip Sapphire Rapids** processor has a total silicon area of about **1600 mm²** (16 cm²) [122].

The aggregate die size of 1600 mm² far exceeds the maximum exposure area of current photolithography systems, which is limited to 858 mm², consequently, **this processor couldnot be fabricated as a monolithic die**; instead, it **relies on multi-die implementation**.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (12)

2.2.b How many transistors do Intel's high-end (HE) 2S server dies include? -1

Available data on **transistor counts** of Intel's 2S server processors is shown below.

Platform ⁴	Core	Techn.	Intro.	High-performance Core-based multicore 2S Xeon server processors	Core count up to	Trans. count [10 ⁶ /10 ⁹]	Die size [mm ²]
Bensley	Pentium 4 Prescott 2M DP	65 nm	5/2006	Xeon 5000 (Dempsey)	2x1C	2x188	2x81
	Core2	65 nm	6/2006	Xeon 5100 (Woodcrest)	2C	291	143
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	Broadwell	14 nm	4/2016	Xeon E5-26xx v4 (Broadwell-EP)	22C	7.2	456
Purley	Skylake	14 nm	7/2017	Xeon Platinum 8176/80 (Skylake-SP)	28C	8.0	~678
	Cascade Lake	14 nm	4/2019	Xeon Platinum 8276/80 (Cascade Lake-SP)	28C	n.a.	n.a.
	Cascade Lake	14 nm	4/2019	Xeon Platinum 9282 (Cascade Lake-AP)	2x28C	n.a.	n.a.
	Cascade Lake R	14 nm	2/2020	Xeon Gold 6238R/6258 (Cascade Lake R)	28C	n.a.	n.a.
Whitney	Ice Lake	10 nm	4/2020	Xeon Platinum 8380 (Ice Lake-SP)	40C	n.a.	~640
Eagle Stream	Sapphire Rapids	Intel 7	1/2023	Xeon Sapphire Rapids MCC monolithic die	32C	n.a.	~770
	Emerald Rapids	Intel 7	11/2023	Xeon Platinum 8592 (Emerald Rapids)	64C	n.a.	~2x700

2.2 How large are Intel's 2S server dies and how many transistors do they include? (13)

How many transistors do Intel's high-end (2S) server dies include? -2

- Nevertheless, it is possible to **estimate transistors count on Intel's recent dies by taking into account the transistor densities**, which were discussed in the Chapter "Intel – and its position in the semiconductor industry" Section 2.2.

Based on this analysis, the **Intel 7 technology** achieves a transistor density of approximately **180 million transistors/mm²**.

- Applying this figure to Intel's **largest 32-core Sapphire Rapids MCC die**, measuring 770 mm², yields an estimated transistor count of **nearly 140 billion transistors**.

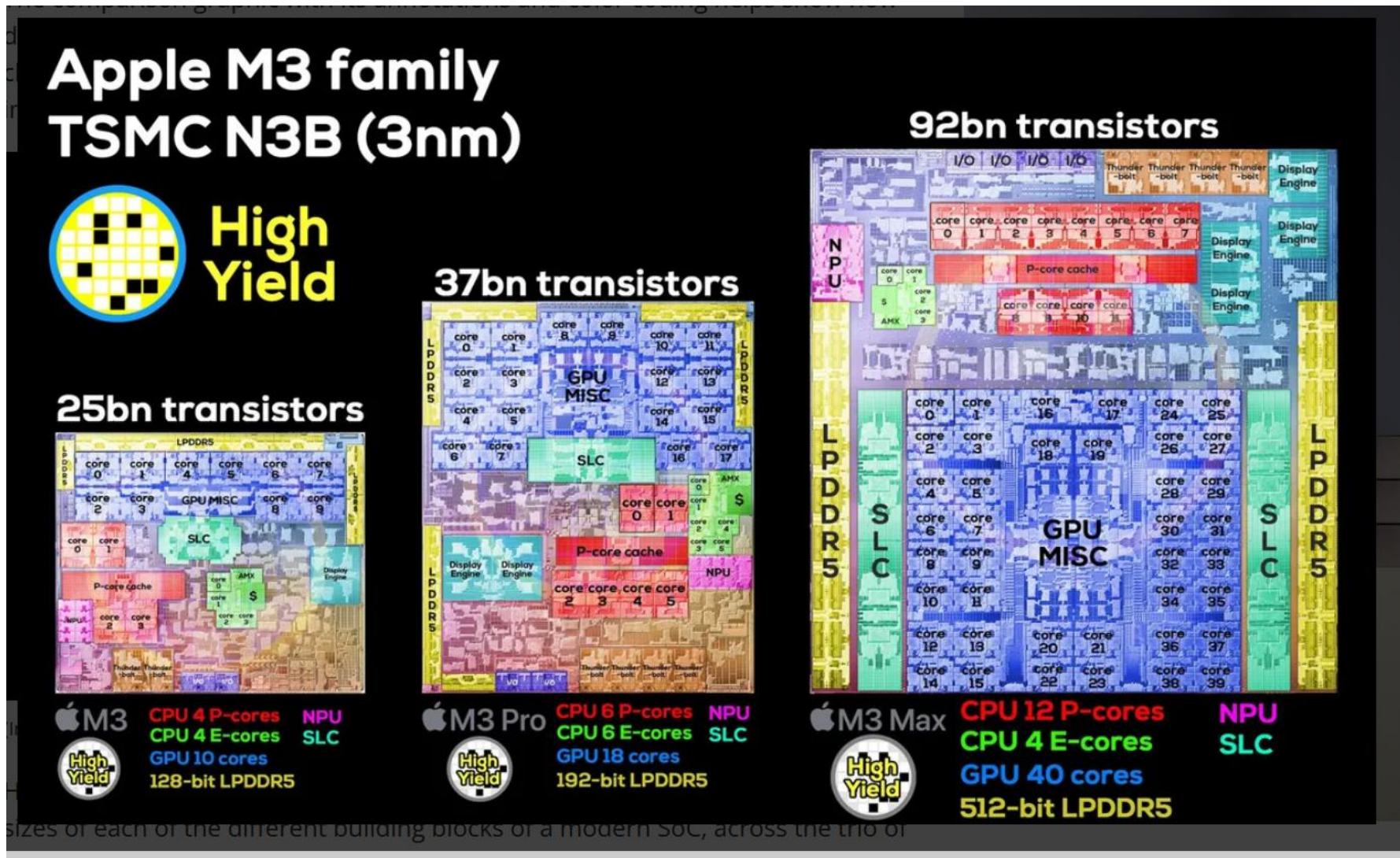
However, this figure should be treated **with caution**.

The die includes a substantial portion of "dark silicon" that is not utilized.

Dark silicon: unoccupied Si area.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (14)

For comparison: Apple M3 family dies (Source: tomshardware)



SLC: System Level Cache

AMX: Apple Matrix Coprocessor

250 – 300 mm²

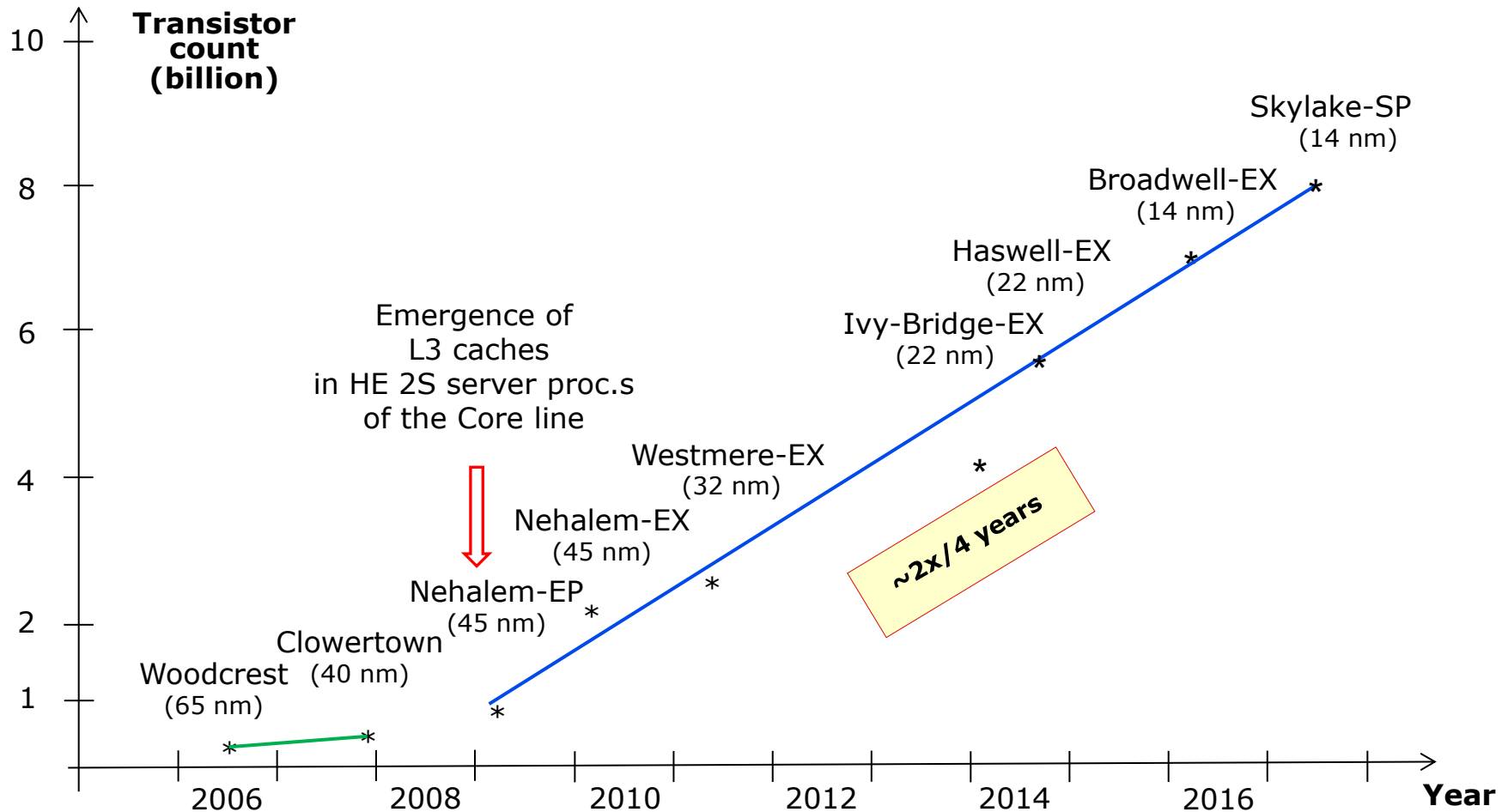
2.2 How large are Intel's 2S server dies and how many transistors do they include? (15)

At what rate do transistor counts rise in Intel's 2S server processors? -1

Based on the data presented in the Table above, we can visualize the **rate at which transistor counts have increased over successive processor generations** in the following Figure.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (16)

Rising transistor counts in Core-based high-end (HE) 2S server processors -1



Above graph shows that after L3 caches were introduced, **transistor counts doubled approximately every four years**, deviating from **Moore's law**, which predicts a doubling every two years.

What is the reason for this deviation? We will discuss it next.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (17)

Why do transistor counts in Intel's high-end server processors increase at a slower rate than predicted by Moore's law? -1

- Initially, Intel increased both the transistor counts and die sizes of its server processors, as shown in the corresponding Table.

Larger dies **reduced the power density** (W/mm^2), i.e. they **improved cooling**.

- However, once the **die size reaches its limit** (around 800 mm^2), about the time when Intel introduced L3 caches for their server processors; **die sizes could not be enlarged any more**, for reasons discussed previously.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (18)

Why do transistor counts in Intel's high-end server processors increase at a slower rate than predicted by Moore's law? -2

- Subsequently, transistor counts can be doubled further by scaling the transistor geometry by 0.7; however, then certain constraints arise.

To understand the reasons for it, let's assume that each scaled-down transistor dissipates only half as much heat as its larger, unscaled counterpart.

Then twice as many scaled-down transistors would produce the same amount of heat as the unscaled transistors.

To put it another way, in this case, during scaling, the power density (W/mm^2) remains unchanged.

- However, in practice, power dissipation does not scale as favorably as previously assumed.

Let's presume, e.g., that the scaled-down transistors dissipate 0.7 times as much heat as its larger unscaled counterparts.

With twice as many transistors, this would increase the power density (W/mm^2) to 1.4 times its previous value.

- In this scenario, during scaling, the power dissipation becomes the limiting factor for the transistor count, and fewer than twice as many scaled-down transistors can be integrated on the same silicon die to prevent overheating, even though more scaled-down transistors could be integrated onto the die.

2.2 How large are Intel's 2S server dies and how many transistors do they include? (19)

Why do transistor counts in Intel's high-end server processors increase at a slower rate than predicted by Moore's law? -3

- When **power density becomes the primary constraint on increasing transistor counts**, as is the case in server-grade processors, after introducing L3 caches, **doubling transistor counts is only feasible if the power consumption of the scaled transistors is halved compared to the unshrunk ones.**
Otherwise, the power density (W/mm^2) would exceed thermal limits.
- Seemingly, in developing the current IC technology, it typically takes about **two scaling steps** (halving transistor geometry), **which requires 4 years.**
- Consequently, **under power density constraints, a doubling of transistor counts requires 4 years rather than 2 years predicted by Moore's law.**

2.3 At what rate do core counts and memory speeds rise in Intel's Core-based 2S server processors?

- 2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors?
- 2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors?

2.3.1 At what rate do core counts rise
in Intel's high-end 2S server processors?

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (1)

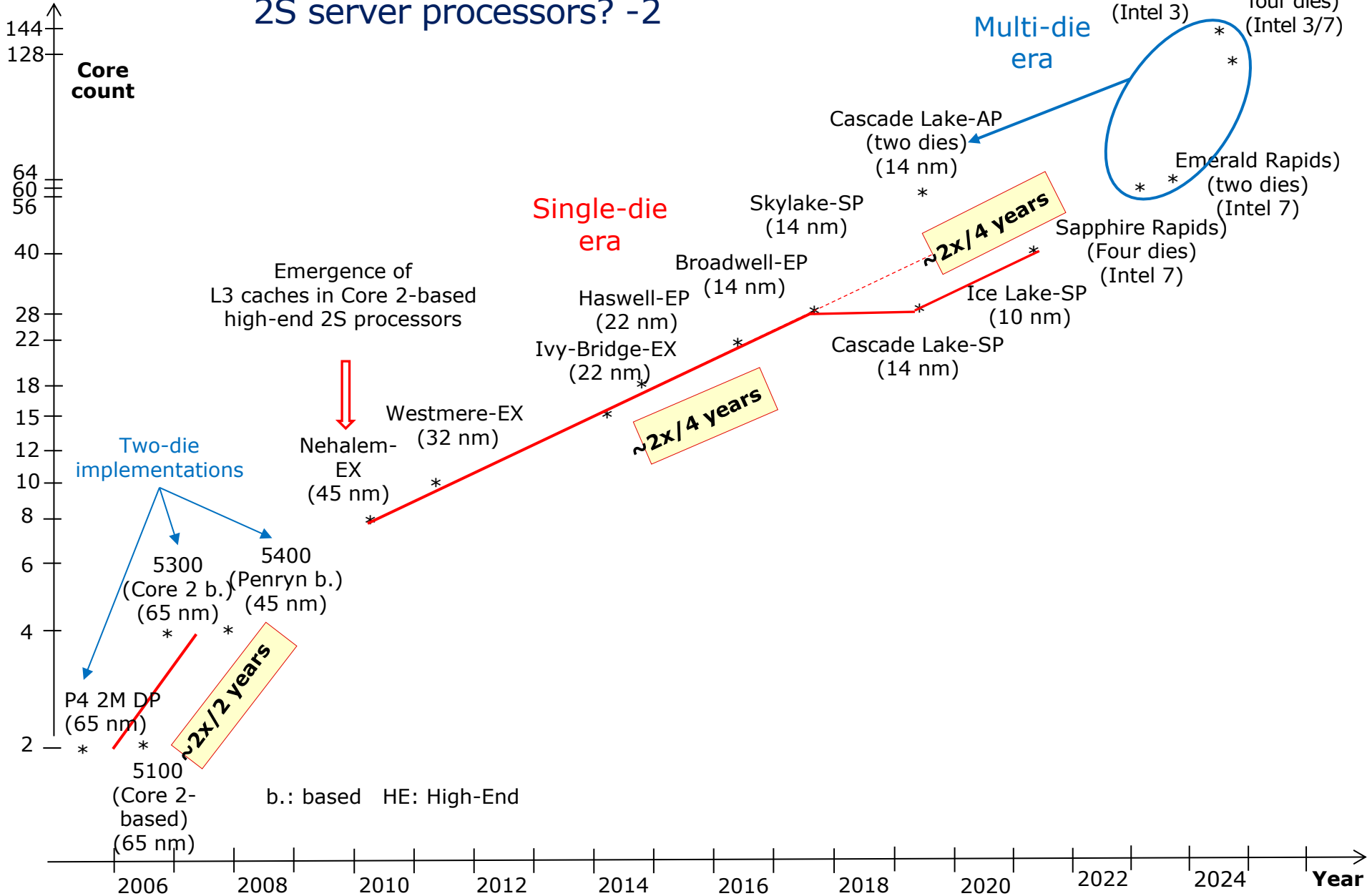
2.3.1 At what rate do core counts rise in Intel's Core-based 2S server processors? -1

- Next, let's focus on the **number of CPU cores** provided in high-end 2S server processors and discuss the **rate at which core counts are increasing**.
- The core counts discussed are included in the Tables summarizing the key features of 2S server processors.

Using this data, the following graph illustrates the increase in core counts of Intel's HE 2S server processors.

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (2)

At what rate do core counts rise in Intel's high-end 2S server processors? -2



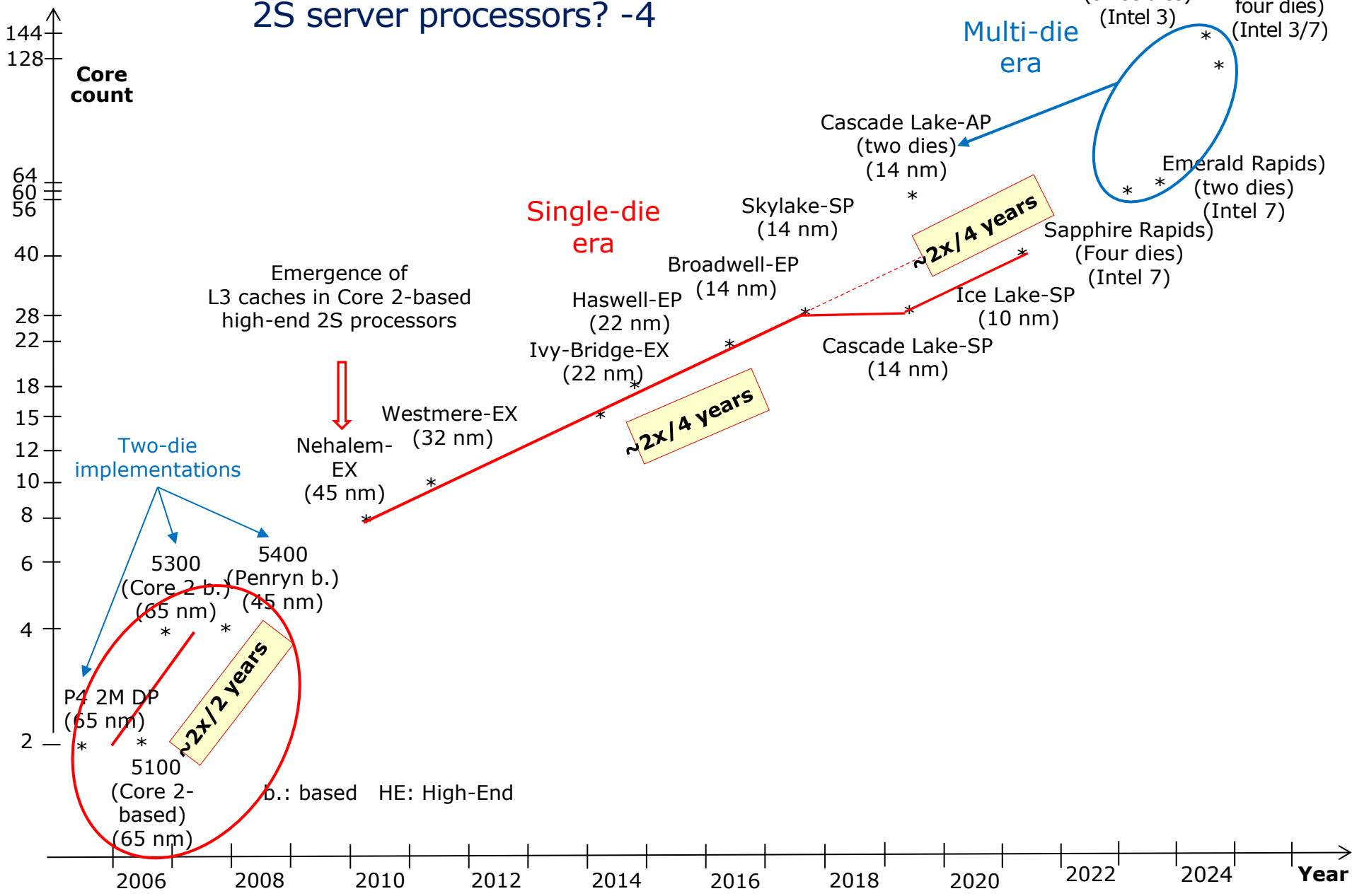
2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (3)

At what rate do core counts rise in Intel's high-end 2S server processors? -3

- The graph is composed of **five segments**, each with different characteristics.

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (4)

At what rate do core counts rise in Intel's high-end 2S server processors? -4



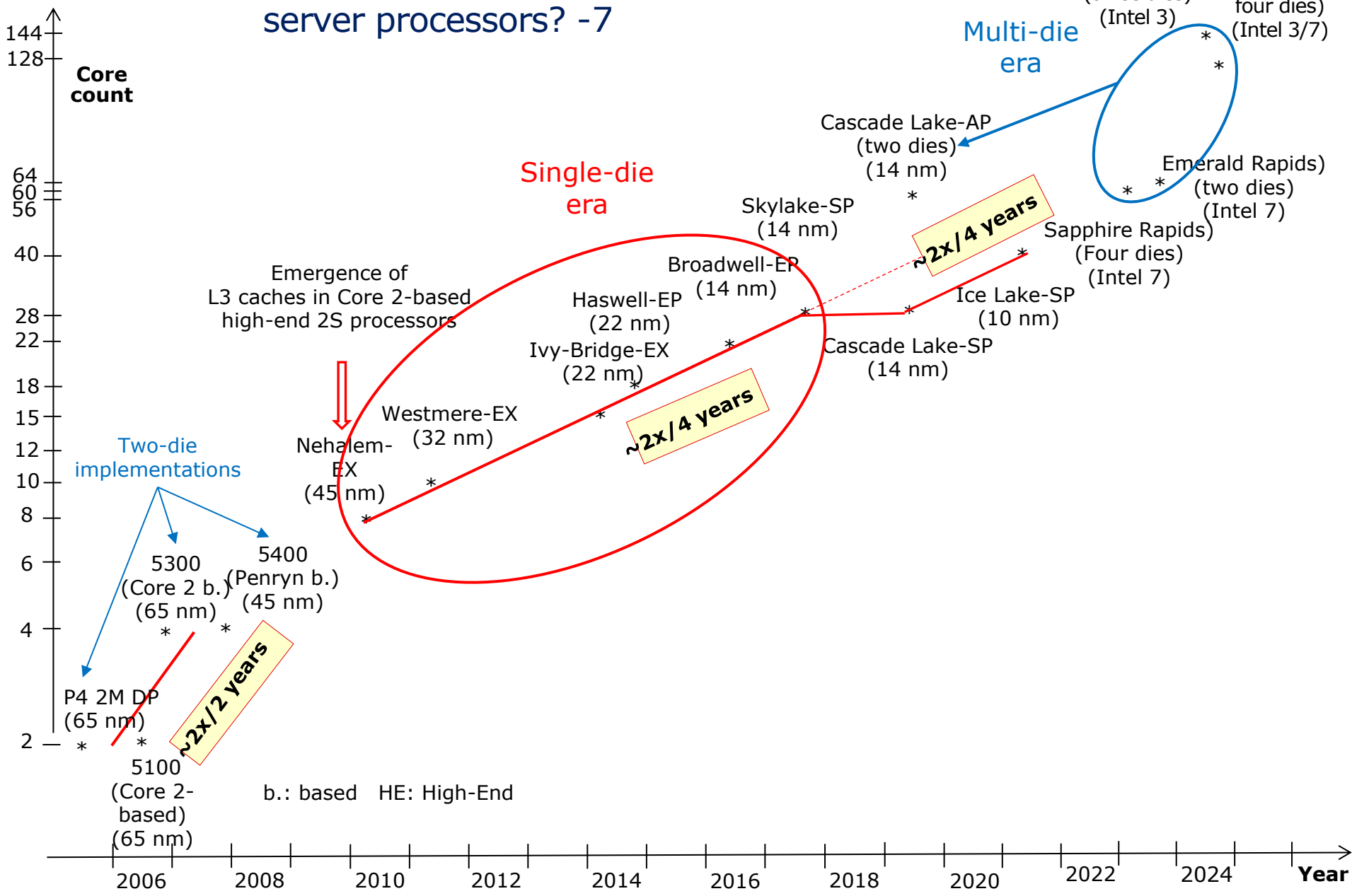
2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (5)

At what rate do core counts rise in Intel's high-end 2S server processors? -3

- Initially, from 2006, core counts doubled approximately in two years (as seen in the graph), in accordance with Moore's revised law, which predicts doubling transistor counts on a die every two years.

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (6)

At what rate do core counts rise in Intel's high-end 2S server processors? -7



2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (7)

At what rate do core counts rise in Intel's high-end 2S server processors? -5

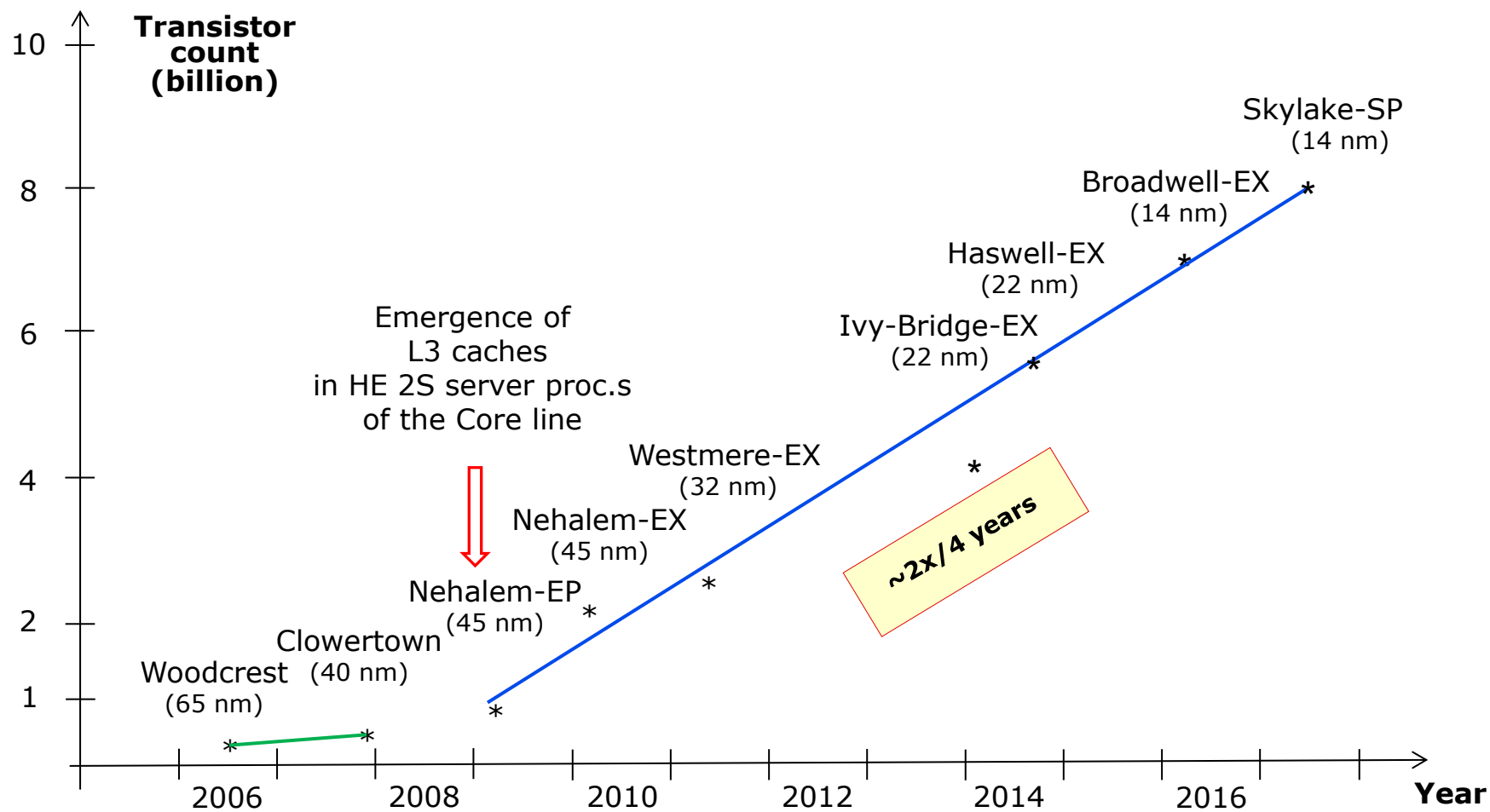
- Following the introduction of L3 caches in 2010, the required transistors count increased rapidly, causing the server die sizes to approach their maximum limit.

As a consequence, maintaining power density became the primary constraint on the rate of increasing transistor counts, as discussed previously.

- Therefore, rather than doubling every two years as predicted by Moore's Law, transistor counts could now only be expected to double approximately every four years, as previously discussed and illustrated in the following Figure.

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (8)

At what rate did transistor counts rise in Intel's high-end (HE) 2S server processors between 2006 and 2016



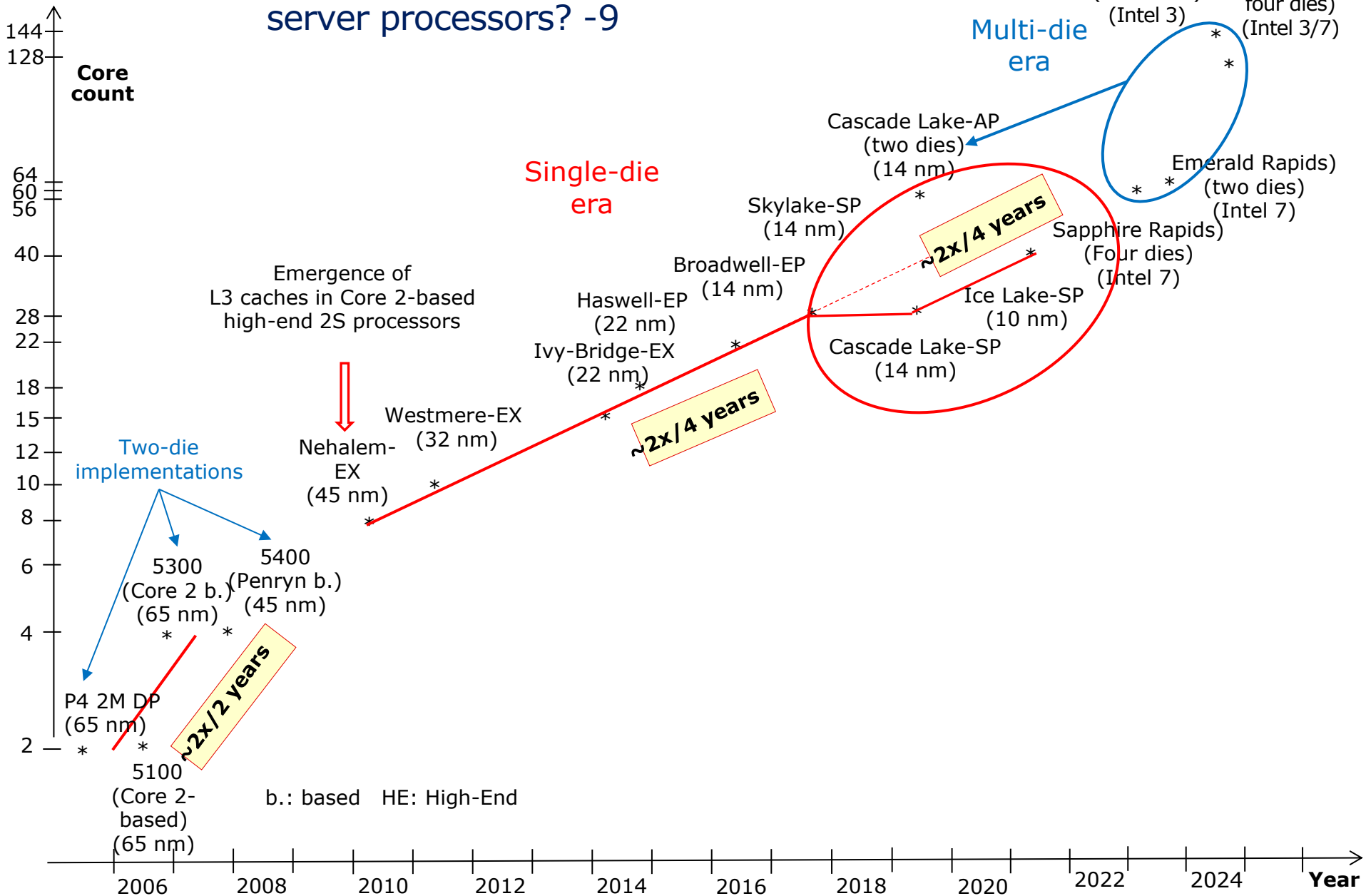
2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (9)

At what rate do core counts rise in Intel's high-end 2S server processors? -6

- Slowing down the rate at which transistor counts could be doubled has caused a **corresponding slowdown in rising core counts** as well, as indicated in the related Figure.

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (10)

At what rate do core counts rise in Intel's high-end 2S server processors? -9



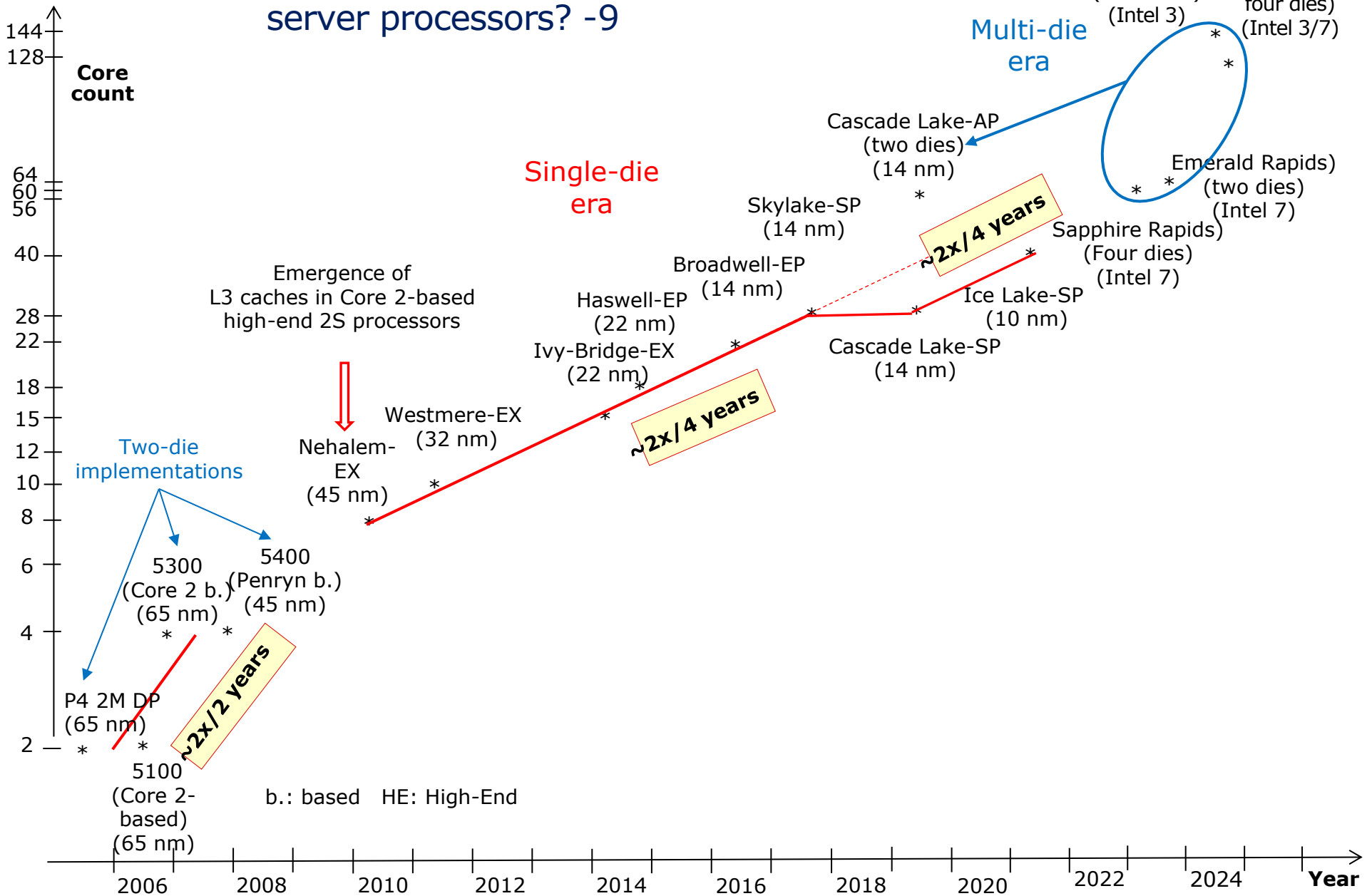
2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (11)

At what rate do core counts rise in Intel's high-end 2S server processors? -8

- In the mid-2010s, Intel faced unexpected challenges in mastering the 10 nm IC technology, resulting in a temporary halt in increasing core counts for a few years.
- Once Intel mastered this technology, core counts began to rise again at a rate similar to previously observed.

2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (12)

At what rate do core counts rise in Intel's high-end 2S server processors? -9



2.3.1 At what rate do core counts rise in Intel's high-end 2S server processors? (13)

At what rate do core counts rise in Intel's high-end 2S server processors? -8

- Finally, with the [advent of the multi-die era](#), primarily [after 2022](#), the earlier patterns governing the increase of core counts became outdated, and **no new regularity** could yet be identified.

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors?

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (1)

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? -

- This Section begins by examining [the rate at which memory speeds increase](#) in Intel's high-end 2S server processors.
- Following that, we will [compare the growth rates of core counts and memory speeds](#) within these processors.
- We will show that there is a disparity between their growth rates and discuss [how to manage this disparity](#) to avoid per-core memory bandwidth bottlenecks.

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (2)

At what rate do memory speeds rise in Intel's high-end 2S server processors? -2

The graph below provides the answer to this question.

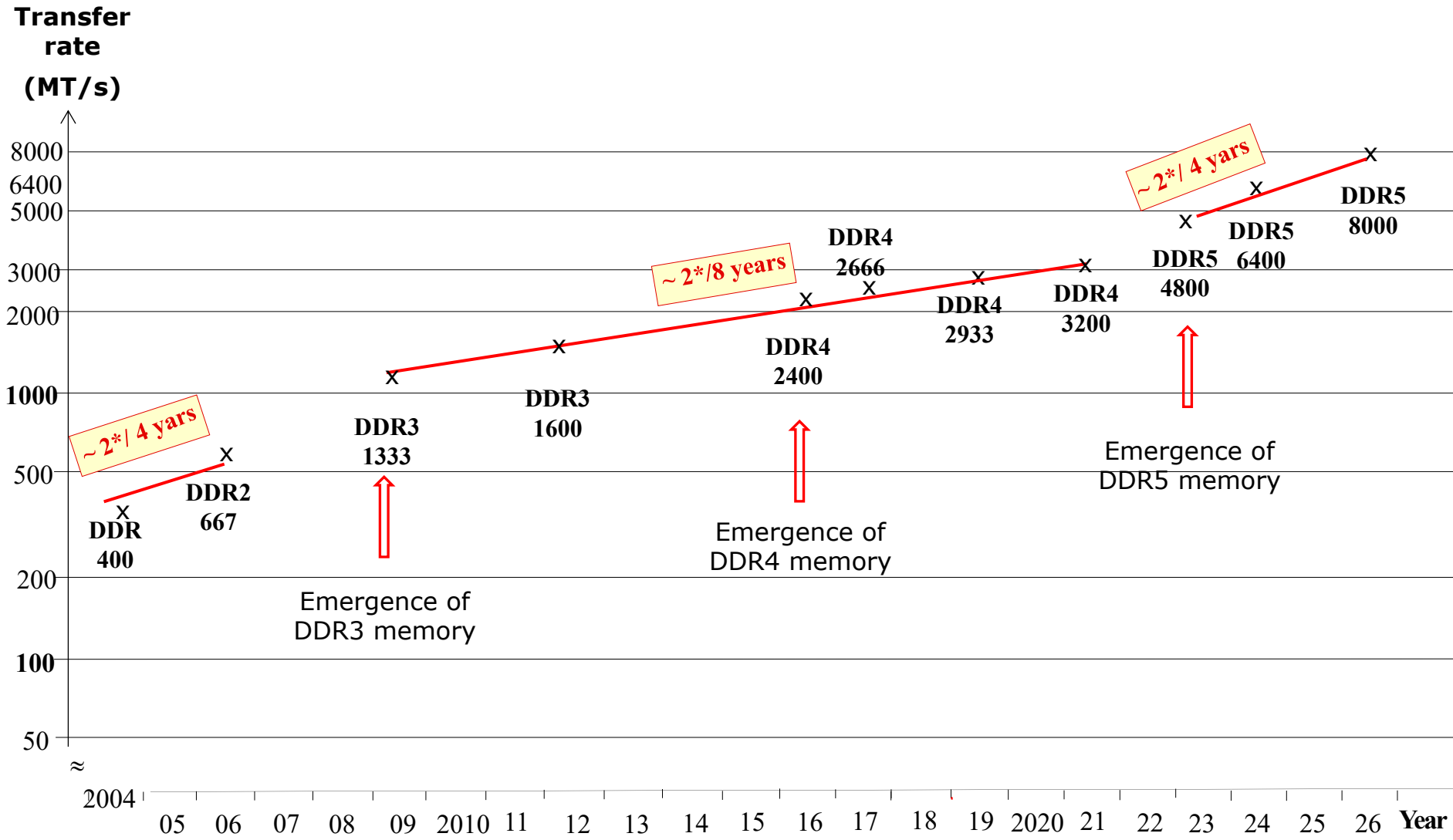


Figure: Memory speeds in Intel's high-end 2S server processors

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (3)

At what rate do memory speeds rise in Intel's HE 2S server processors? -3

- As illustrated in the Figure above, memory speeds have increased at two distinct rates over time:
- In the early memory generations – specifically with DDR and DDR2 memories – transfer rates approximately doubled every four years.
- However, following the introduction of DDR3 memories, around 2009, the pace of advancement slowed considerably, and the time needed for doubling the memory transfer rates lengthened to about eight years.
- This deceleration is primarily attributed to the increased architectural complexity of DDR3 and its successors, including tighter timing margins, more sophisticated signal integrity requirements, and greater power management challenges.
- With the arrival of DDR5 memories, memory speed began to rise at a higher rate as before, with the new rate of doubling in approximately 4 years again.

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (4)

Let's compare the rates at which core counts and memory speeds rise

A comparison of both growth rates - prior to the arrival of DDR5 memories - reveals that in this time **memory transfer speeds have increased at about half the rate of CPU core counts.**

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (5)

What happens to per-core memory bandwidth when CPU core counts increase faster than memory speeds?

- The **per-core memory bandwidth** (BW_M/n_C) is:

$$BW_M/n_C = n_M * w * T_M / n_C$$

where

n_M : number of memory channels

w : width of the memory channels (usually 8 byte)

T_M : transfer rate of the memory

n_C : core count

- When memory speeds (T_M) grow at a slower rate than CPU core counts (n_C), the ratio of T_M/n_C becomes smaller, i.e. the **per-core memory bandwidth decreases**.
- However, a reduction in per-core memory bandwidth (BW_M/n_C) will lead to a **memory bandwidth bottleneck, unless appropriate countermeasures are taken**.

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (6)

How can per-core memory bandwidth be preserved when memory transfer rates increase at half the rate of the core counts across processor generations?

In the expression describing the per-core memory bandwidth

$$BW_M/n_C = n_M * w * T_M / n_C$$

the per-core memory bandwidth BW_M/n_C) can be preserved when the expression

$$n_M * T_M / n_C$$

remains constant, i.e., when sufficient additional memory channels will be implemented to compensate for increasing core counts.

Here, we assumed that the memory channel width (w) does not vary across generations.

2.3.2 At what rate do memory speeds rise in Intel’s high-end 2S server processors? (7)

How many memory channels (n_M) are required per socket to preserve the per-core memory bandwidth of a reference platform in this case? [97]

- As stated in [97], if memory speeds rise at half the rate of core counts, a high-end 2S server with n_C cores will require approximately $\sqrt{2} * \sqrt{n_C}$ memory channels per socket, to maintain per-core memory bandwidth equivalent to that of a selected reference platform.

- The Table on the right presents the number of memory channels required per socket (n_M) according to the provided expression.
- For instance, a 32-core processor necessitates 8 memory channels in order to maintain the per-core memory bandwidth as compared to the assumed reference platform.

n_C	n_M
2	2.0
4	2.8
6	3.5
8	4.0
10	4.5
12	4.9
14	5.3
16	5.6
20	6.3
24	6.9
32	8.0
48	9.8
64	11.3

Table: The required number of memory channels (n_M) per socket needed to preserve the per-core memory bandwidth of a high-end 2S server with n_C cores to a selected platform, which is taken as a reference. (It is assumed that CPU cores and memory speeds are increasing at the rates previously discussed).

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (8)

One of the key challenges in designing server processors -1

- As core counts in successive generations of server processors rise at a rate twice that of memory speeds, **it becomes necessary to add more memory channels per socket.**

This compensates for the higher growth rate of core counts compared to that of memory speeds.

- **Nonetheless, it is a quite challenging task to increase the number of memory channels** due to electrical and physical constraints, which are not discussed in this Chapter.
- Therefore, **ensuring an adequate number of memory channels for preserving the memory bandwidth per socket, is one of the key challenges in server processor and platform design.**
- The subsequent Table highlights Intel's achievements in **preserving per-core memory bandwidth** (relative to a chosen reference platform) in its high-end 2S server platforms.

2.3.2 At what rate do memory speeds rise in Intel’s high-end 2S server processors? (9)

Intel’s achievement in maintaining per-core bandwidth in its HE 2S server platforms

Platform 605	48re	Techn.	Date of Intro.	High-end multicore 2S server lines	Core count (up to)	Per socket mem. channels (up to)	Per core BW rel. to ref. platform
Ref. platform	Pentium 4 Prescott DP	90 nm	6/2004	90 nm Nocona	1C	1x DDR2/400	1.0
	Pentium 4 Presc.	90 nm	10/2005	Xeon DP 2.8 (Paxville DP)	2x1C	1x DDR2/400	0.5
Bensley	Pentium 4 Presc.	65 nm	5/2006	Xeon 5000 (Dempsey)	2x1C	2x DDR2/667	1.7
	Core2	65 nm	6/2006	Xeon 5100 (Woodcrest)	2C	2x DDR2/667	1.7
	Core2	65 nm	11/2006	Xeon 5300 (Clowertown)	2x2C	2x DDR2/667	0.8
	Penryn	45 nm	11/2007	Xeon 5400 (Harpertown)	2x2C	2x DDR2/667	0.8
Tylersburg	Nehalem	45 nm	3/2009	Xeon 55xx (Gainestown)	4C	3x DDR3/1333	2.5
Boxboro-EX	Nehalem	45 nm	3/2010	Xeon X6500 Nehalem-EX)	8C	8x DDR3/1066	2.7
	Westmere	32 nm	4/2011	Xeon E7-28xx (Westmere-EX)	10C	8x DDR3/1333	2.7
Brickland	Ivy Bridge	22 nm	2/2014	Xeon E7-28xx v2 (Ivy Bridge-EX)	15C	8x DDR3/1600	2.1
Grantley	Haswell	22 nm	9/2014	Xeon E5-2600xx v3 (Haswell-EP)	18C	4x DDR4/2133	1.2
	Broadwell	14 nm	4/2016	Xeon E5-26xx v4 (Broadwell-EP)	22C	4x DDR4/2400	1.1
Purley	Skylake	14 nm	7/2017	Xeon Platinum (8176/80 Skylake-SP)	28C	6x DDR4/2666	1,4
	Cascade Lake	14 nm	4/2019	Xeon Platinum 8276/80 (Cascade Lake-SP)	28C	6x DDR4/2933	1.6
Whitney	Ice Lake	10 nm	4/2021	Xeon Platinum 8380 (Ice Lake-SP)	40C	8x DDR4-3200	1.6
Eagle Stream	Sapphire Rapid	Intel 7	1/2023	Xeon Platinum 8490H (Sapphire Rapid)	60C	8x DDR5-4800	1.6

2.3.2 At what rate do memory speeds rise in Intel's high-end 2S server processors? (10)

How far could Intel preserve per-core bandwidth in its HE 2S server platforms?

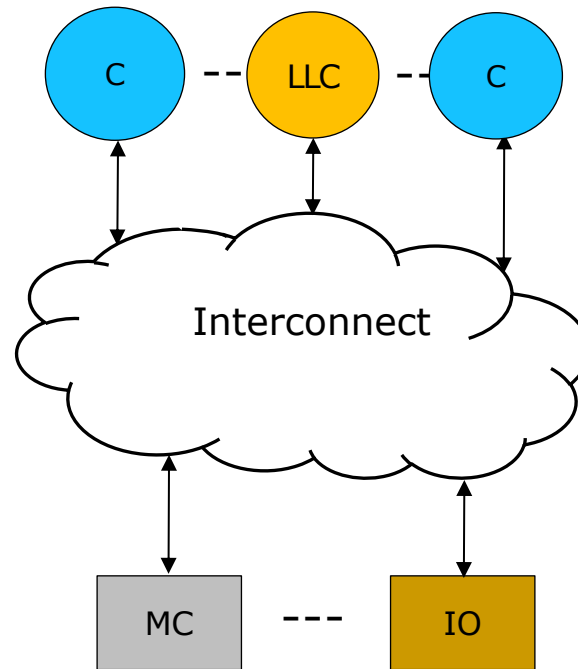
- The Table above showcases Intel's success in maintaining per-core memory bandwidth relative to a selected reference platform (identified by dark blue color).
- As the data indicates, Intel broadly achieved its goal, albeit with certain deviations, for instance, the Boxborough-EX platform delivered substantially higher per-core memory bandwidth than what was necessary according to the established criteria.

2.4 Evolution of on-die interconnects

2.4 Evolution of on-die interconnects (1)

2.4 Evolution of on-die interconnects -1

On-die interconnects link together functional units implemented on the die, such as cores, the Last Level Cache (LLC), memory- or IO-controllers, as illustrated in the Figure below.



C: Core

LLC: Last Level Cache (monolithic implemented)

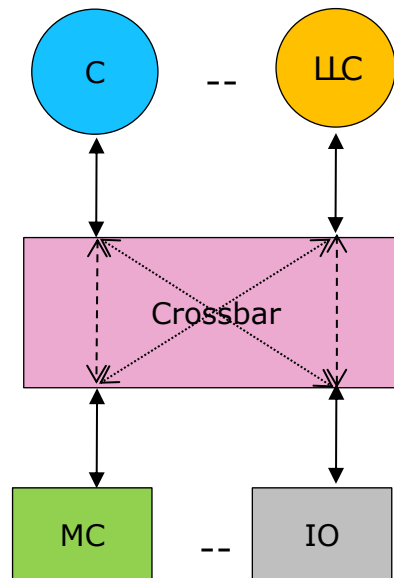
MC: Memory Controller

Figure: Task of the on-die interconnect in a server processor

2.4 Evolution of on-die interconnects (2)

Evolution of on-die interconnects -2

- During the evolution of multicore server processors also on-die interconnects evolved.
- First multicore servers, with 2 to 4 cores, typically made use of **crossbar switches** (as shown below) to link together functional units.



Examples

*Nehalem and Westmere-based
server processors (2009-2011)*

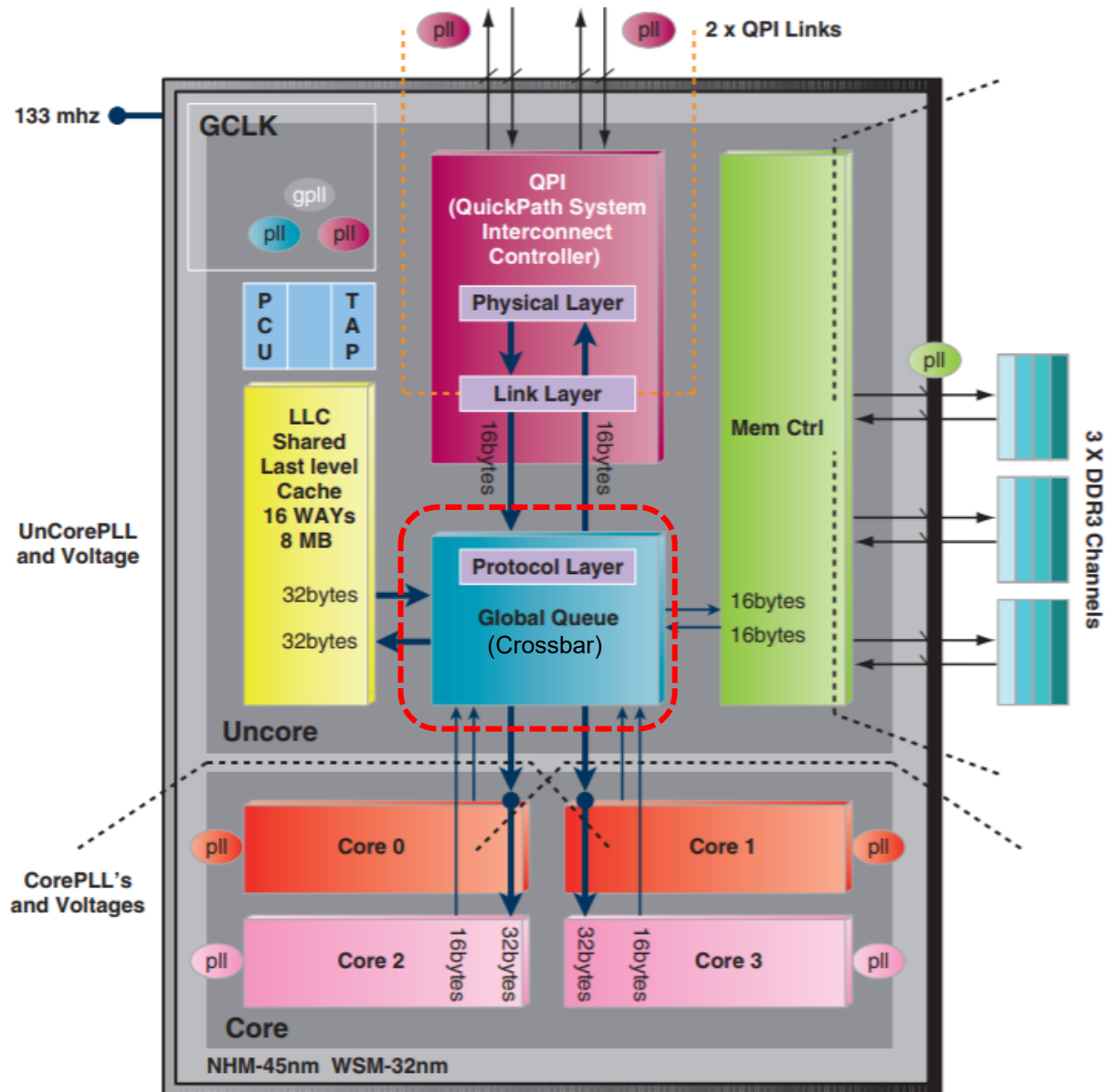
Figure: Principle of a crossbar-based on-die interconnect

- These server processors already introduced **L3 caches**, also known as **LLC** (Last Level Cache), however, these caches were still built as (unsliced) **monolithic** units.

2.4 Evolution of on-die interconnects (3)

Example: Crossbar interconnect in Intel's Nehalem-based DP-server processors (2008) [55]

In the Nehalem-DP server the functional units (cores, LLC, MC) as well as the 2. processor) are interconnected by a **crossbar switch**.



2.4 Evolution of on-die interconnects (4)

Intel's transition from crossbar-based to ring bus-based on-die interconnects in their Sandy Bridge-based server processors -1

- The **cost** of implementing crossbar switches increases **quadratically with the number of ports**.

To address this issue, Intel transitioned to **cache-coherent ring bus interconnects** starting with its **8-core Nehalem-EX** server processor line in 2010.

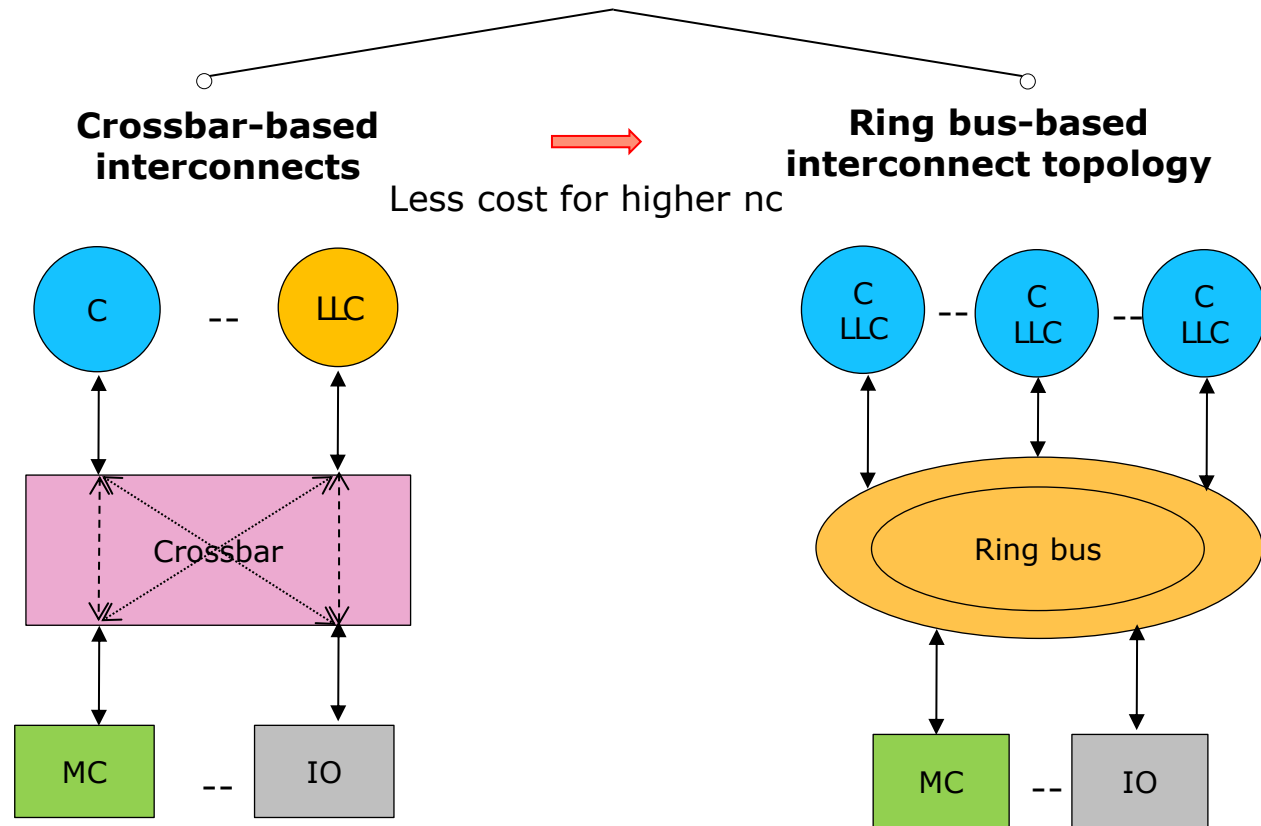
At the same time, Intel introduced **sliced Last-Level Caches (LLC)** to allow **concurrent access** to multiple slices, thereby improving performance.

- The **CPU cores** were connected to the ring bus **through their associated L3 cache slices**, as illustrated in the example on the following slide.

2.4 Evolution of on-die interconnects (5)

Intel's transition from crossbar-based to ring bus-based on-die interconnects in their Nehalem-EX server processors -2

On-die interconnects of Intel's multicore server processors



Examples

Nehalem-EP and Westmere-EP server processors (2010-2011)

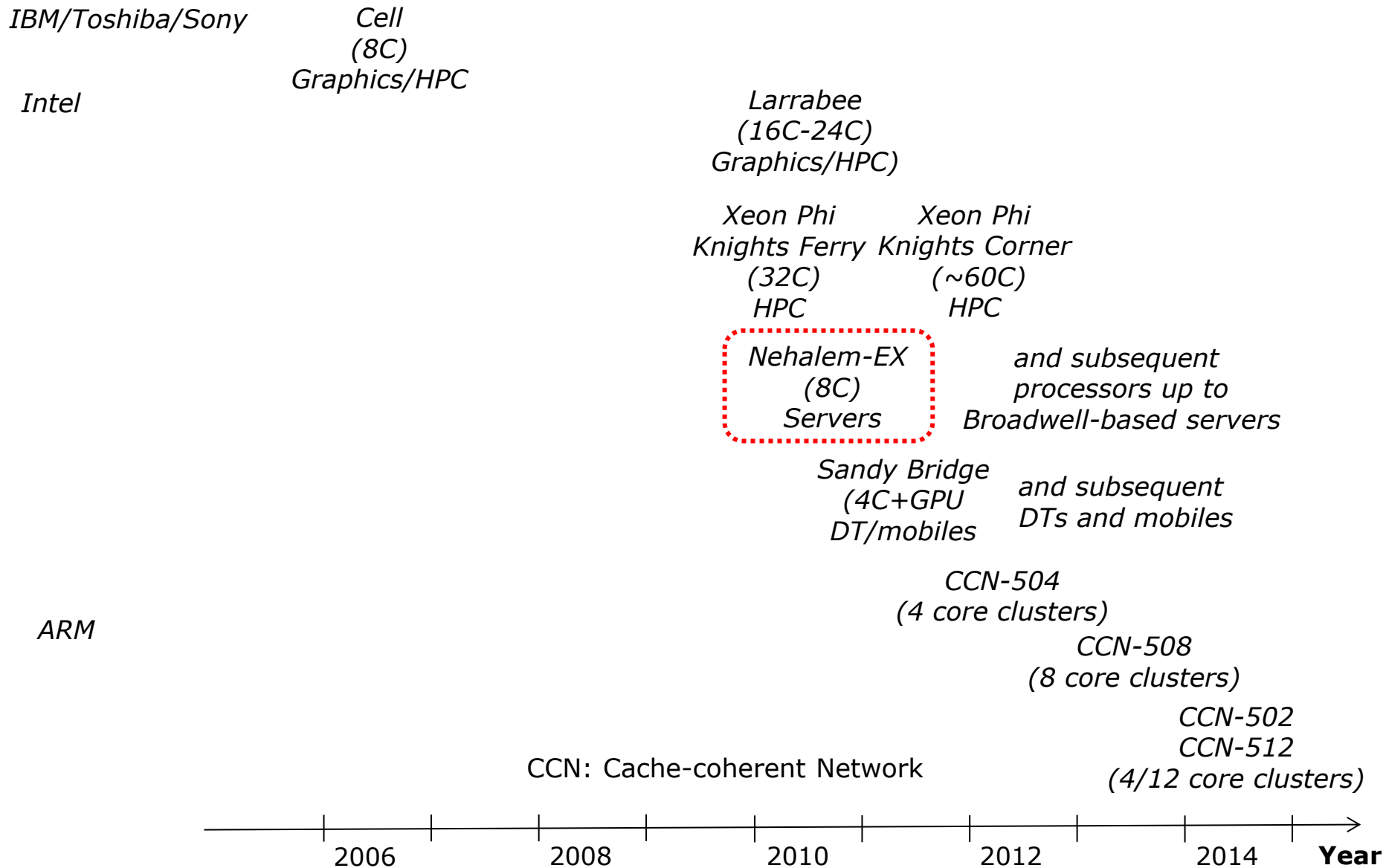
Nehalem-EX-based and subsequent server processors (since 2010) until Broadwell-based server processors (2016)

Monolithic implemented LLC (L3) is assumed.

Sliced LLC (L3) is assumed.

2.4 Evolution of on-die interconnects (6)

Emergence and use of on-die ring buses in general -1



2.4 Evolution of on-die interconnects (7)

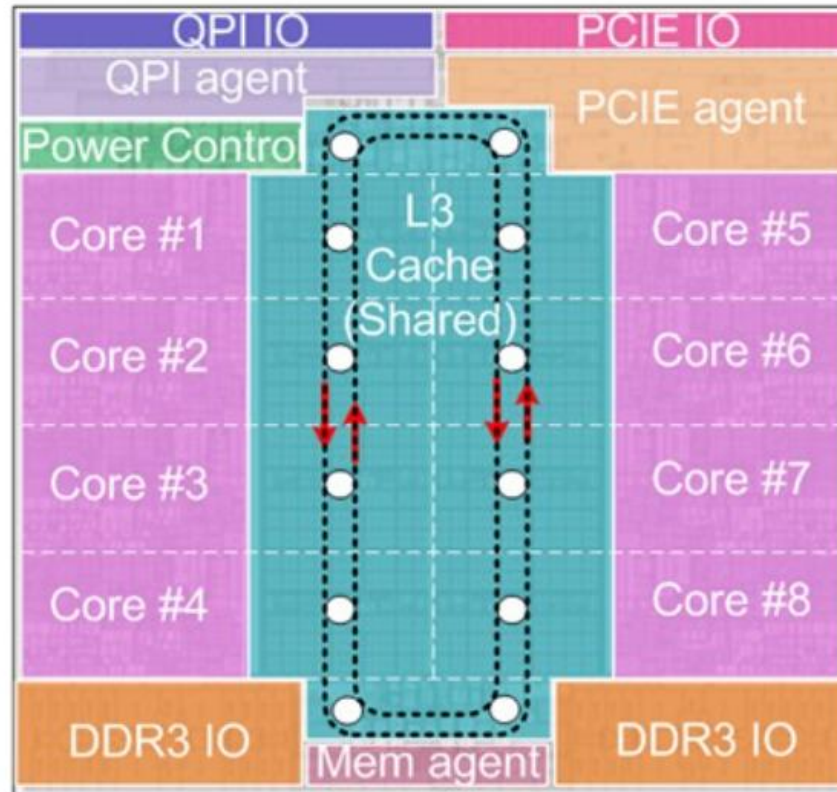
Emergence and use of on-die ring buses in general -2

The **first ring bus** was introduced in the **Cell processor**, a joint development of IBM, Toshiba and Sony **in 2006**.

It later gained **widespread use in Intel** processors and **ARM's CCNs** (Cache-Coherent Networks) during **the early 2010's**, as illustrated in the next Figure.

2.4 Evolution of on-die interconnects (8)

Example: Ring bus-based on-die interconnect in Intel's Sandy Bridge-EP (Xeon E5-26xx) 8-core server processor (2012) [38] -2



The interconnect ring incorporates **two 32-bit data paths operating in opposite directions**, enabling the system to dynamically select the shortest route for data transfers between ring bus ports.

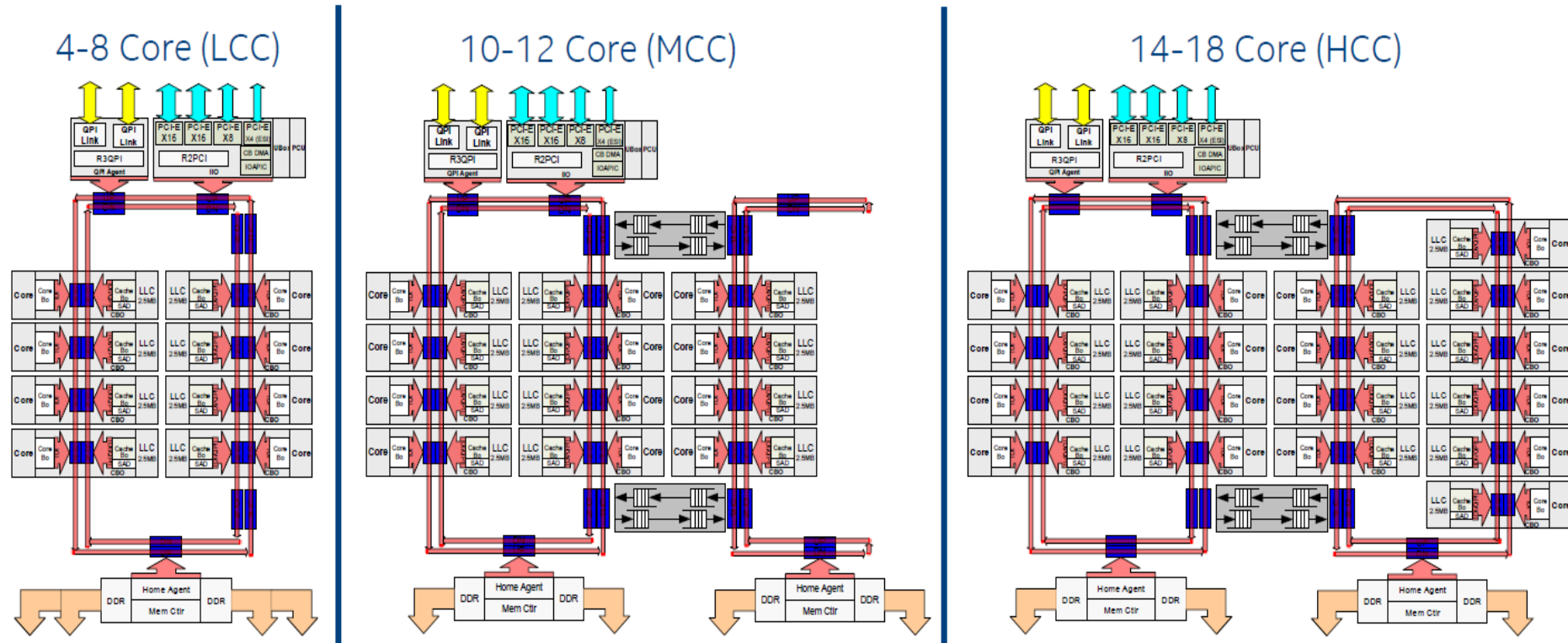
2.4 Evolution of on-die interconnects (9)

Introducing dual ring buses in Intel's 2S server processors for higher core counts ($n_c \geq 10$)

- For server processors with ≥ 10 cores Intel introduced even dual ring buses (first seen in the Haswell-EP line in 2016) to reduce core-to-core latencies during data transfers.
- It is important to note that the Haswell-EP server processor has 3 different die configurations, which vary depending on the core count and implement different ring bus architectures, as shown subsequently.

2.4 Evolution of on-die interconnects (10)

Example for introducing dual ring buses in Intel's 2S server processors for the up to 18 core Haswell-EP processor in 2014) [39]



Note that in dual-ring bus configurations, both buses are interconnected by buffers, as illustrated in the Figure above.

Downside of ring buses and introducing 2D mesh interconnects

- **Ring buses** were initially introduced for **8 cores**.
- As the number of cores increases, longer ring buses are necessary, leading to **higher core-to-core latencies**.
- However, with **core counts** over 15, even dual ring configurations became insufficient due to **unacceptable long latencies**.

This has driven the evolution toward more scalable interconnect solutions, such as the **2D-mesh interconnect**, which will be discussed next.

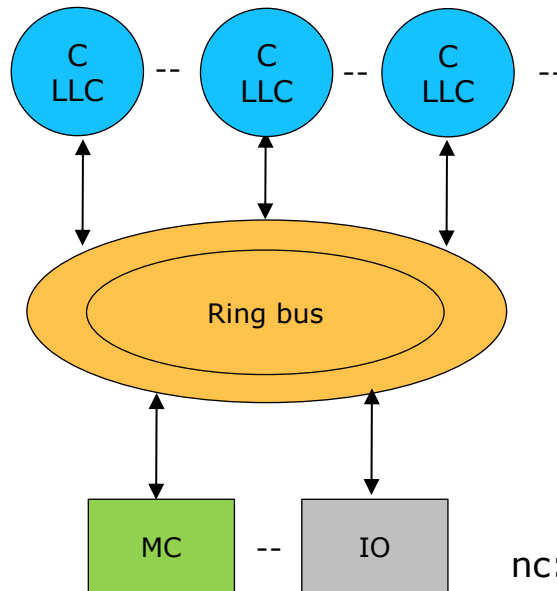
2.4 Evolution of on-die interconnects (12)

Switching from ring bus- to 2D mesh-based on-die interconnects in Intel's 2S server processors

Evolution of on-die interconnects in Intel's 2S server processors

Less cost
for higher n_c

Ring bus-based interconnect (Principle)



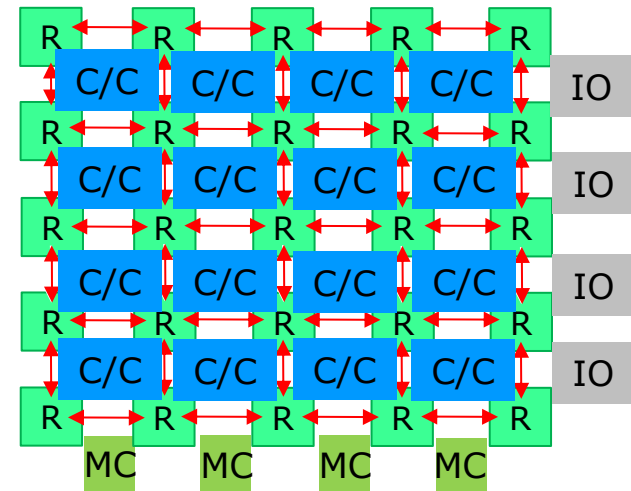
Examples

Intel Nehalem-EX-based and subsequent
server processors (since 2010) until
Broadwell-based server processors (2016)

ARM
CCN-5xx interconnect for servers (2012)

Shorter latencies for
higher n_c

2D mesh-based interconnect (Principle)



R: Router
C/C: Core+L3 cache slice

Skylake-SP based server
processors (2017)
Knights Landing for AI, (2016)

CMN-6xx Interconnect for servers (2017)

2.4 Evolution of on-die interconnects (13)

Switching from dual ring buses to a 2D mesh interconnect in Intel's up to 28-core Skylake-SP server processor line in 2017 [40]

2D mesh interconnects consist of **dual links** arranged in a **2D grid structure**, interconnected through **routers** that forward **data packets** from senders to receivers, as shown below.

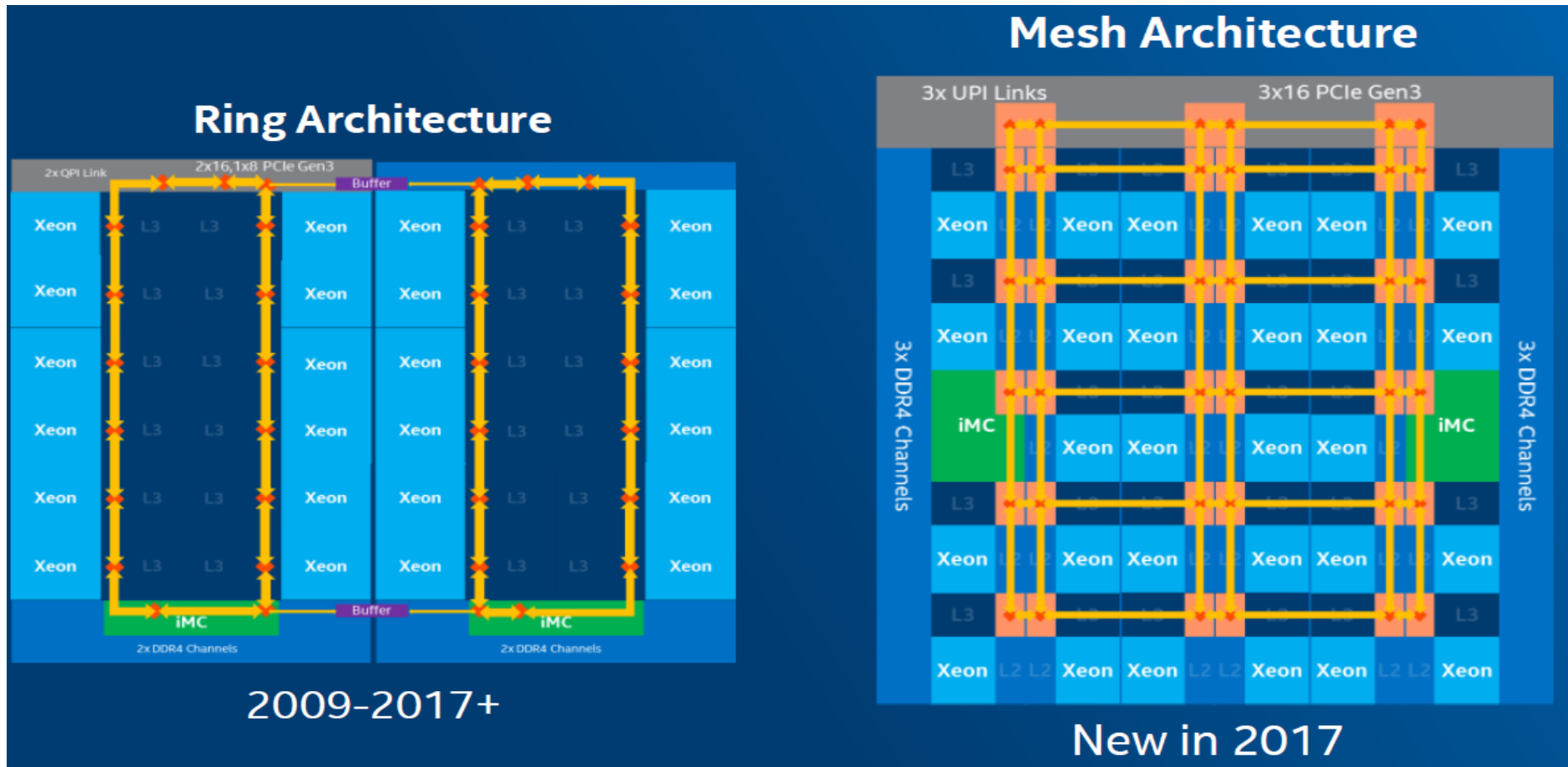
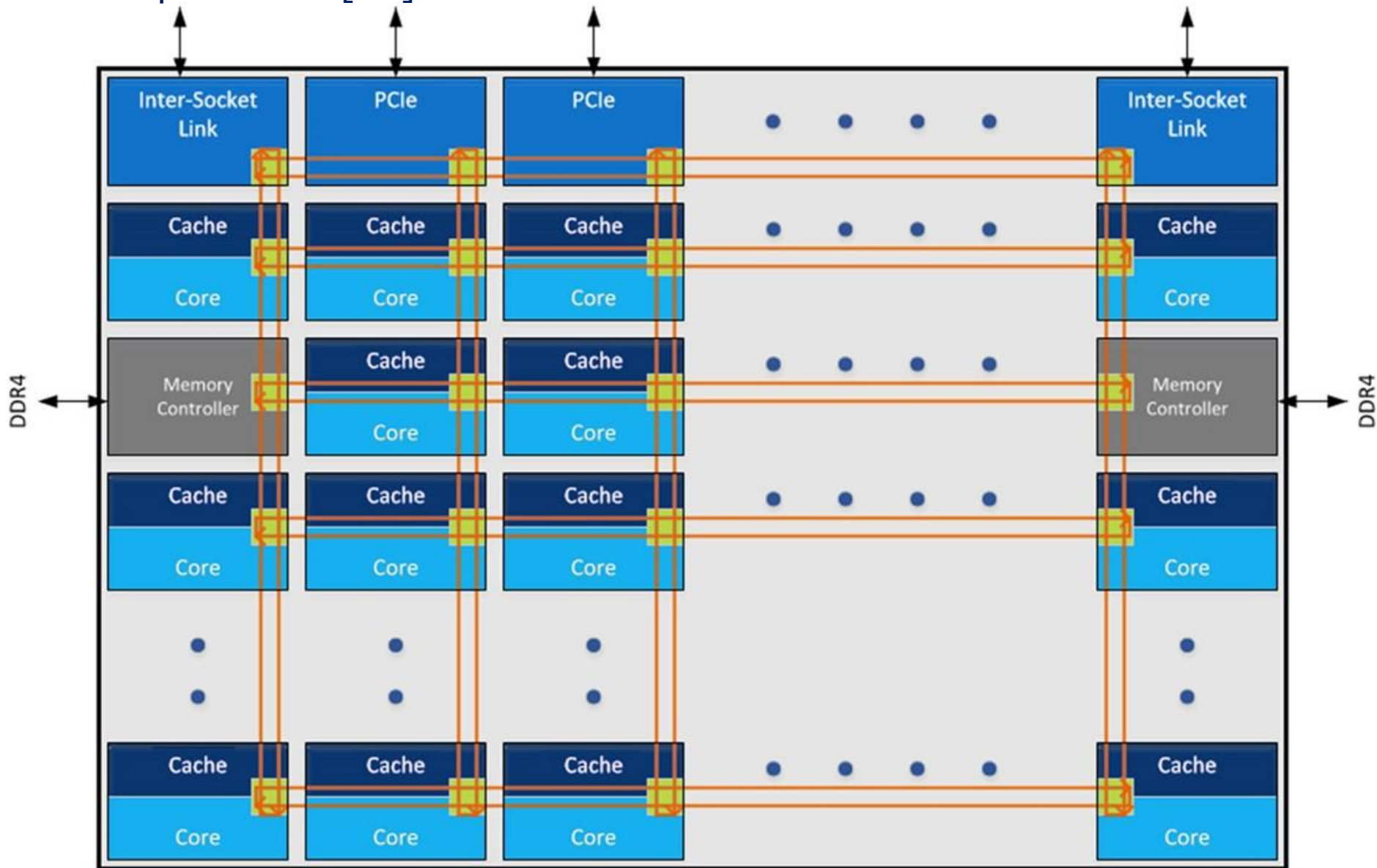


Figure: Contrasting dual ring and 2D mesh interconnects [40]

2.4 Evolution of on-die interconnects (14)

A more detailed look at Intel's (5x6) 2D mesh interconnect in its Skylake-SP server processor [41]



2.4 Evolution of on-die interconnects (15)

Switching from ring buses to 2D mesh interconnects in Intel's processors

Switching to mesh (2D grid) interconnects in Intel's processors

Ring architecture

Sony/Toshiba/IBM Cell (2006): 8 cores



Larrabee (2009). 24-32 cores (cancelled)

Xeon Phi

Knights Ferry (2010): 32 cores

Knights Corner (2012): 57-61 cores

Nehalem-EX (Beckton) (2010) up to 8 cores

All **Sandy Bridge** lines (2011) up to 8 cores

All **Ivy Bridge** lines (from 2012 on) up to 15 cores
to

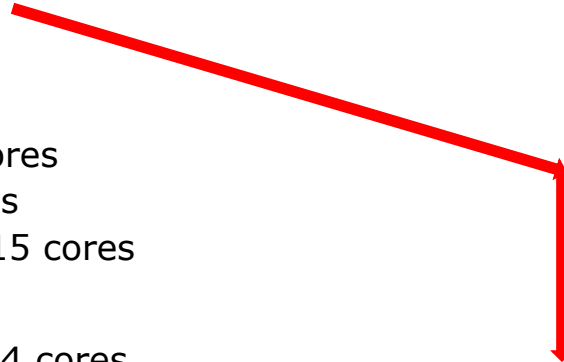
All **Broadwell** lines (from 2014 on) up to 24 cores

Skylake (2015) up to 4 cores

Mesh (2D grid) architecture

Tile processor (2007): 80 cores

SCC (2010): 48 cores



Xeon Phi

Knights Landing (2016): up to 72 cores



Skylake-X (HEDT) (2017) up to 18 cores

Skylake-SP (2017) up to 28 cores

Benefits of the mesh interconnection architecture

- Higher performance
- Lower latencies

2.4 Evolution of on-die interconnects (17)

Evolution of interconnects in Intel's Core-based 2S server and client processors, as well as in AMD's Zen processors -1

Intel

crossbar switch → **ring bus** → **dual ring buses** → **2D mesh**

2D server processors

5100
Woodcrest
2 cores
(2006)

Nehalem-EX
8 cores
(2010)

Haswell-EP
18 cores
(over 10 cores)
(2016)

Skylake-SP
28 cores
(2017)

Client processors

Core 2
2 cores
(2006)

2nd gen.
Sandy Bridge
8 cores
(2011)

AMD

Zen basic die
4 cores
(2017)

Zen 3 CCDs
8 cores
(2020)

Zen 4 CCDs
16 cores
(2023)

2.5 TDP values and socketing of Intel's high-end 2S server processors

2.5 TDP values and socketing of Intel's high-end 2S server processors (1)

2.5 TDP values and socketing of Intel's high-end 2S server processors

TDP values of Intel's high-end 2S server processors -1

First, let's recall the notion of **TDP (Thermal Design Power)**.

- **TDP (Thermal Design Power)** is the **power budget** allocated to a processor during its design.
- The **TDP** value assigned to a processor **limits its performance**.
- It is **specified in datasheets** and is usually given in **W**.
- It can be interpreted as the **maximum power consumption of the processor when running realistic, power-intensive applications**.

It's important to **distinguish** these applications from **power viruses**, which are specifically designed to **draw maximum power**.

- The **TDP value** serves as the **reference** for designing the **cooling system of the platform**.
- The **cooling system**, also known as the **thermal solution**, ensures adequate cooling that prevents the junction temperatures (T_j) of the transistors on the die from exceeding a predetermined limit (T_{jmax}), for instance, 90°C.

2.5 TDP values and socketing of Intel's high-end 2S server processors (2)

TDP values of Intel's high-end 2S server processors -2

The following Tables provide an overview of the **power consumption** (in terms of the **TDP** value) of Intel's high-end 2S server processors.

2.5 TDP values and socketing of Intel's high-end 2S server processors (3)

TDP values of Intel's early HE 2S server processors introduced prior to the scalable processors

Platform	HE multicore 2S server models	Techn .	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket (up to)	TDP (up to) W	Socket
Paxville-DP	Xeon DP 2.8 GHz (P4 Prescott 2M DP)	90 nm	10/2005	2x1C	24x PCIe 1.1 (on MCH)	2x DDR2-400 (via MCH)	135	PGA 604
Bensley	Xeon 5000 (Dempsey) (P4 Prescott 2M DP)	65 nm	5/2006	2x1C	16x PCIe 1.0 (on MCH)	4x DDR2-533 (FB-DIMM) (via MCH)	130	LGA 771
	(Woodcrest) (Core 2)	65 nm	6/2006	2C	16x PCIe 1.0 (on MCH)	4x DDR2-667 (FB-DIMM) (via MCH)	65	LGA 771
	Xeon 5300 (Clovertown) (Core 2)	65 nm	11/2006	2x2C	16x PCIe 1.0 (on MCH)	4x DDR2-667 (FB-DIMM) (via MCH)	80	LGA 771
	Xeon 5400 (Harpertown) (Penryn)	45 nm	11/2007	2x2C	24x PCIe 1.0 (on MCH)	4x DDR2-667 (FB-DIMM) (via MCH)	120	LGA 771
Boxboro-EX	Xeon E65xx Nehalem-EX) (Beckton)	45 nm	3/2010	8C	36x PCIe 2.0 (on MCH)	8x DDR3-1066	130	LGA 1567
	Xeon E7-28xx (Westmere-EX)	32 nm	4/2011	10C		8x DDR3-1333	130	
Brickland	E7-28xx v2 (Ivy Bridge-EX)	22 nm	2/2014	15C	32x PCIe 3.0 (on proc.)	8x DDR3-1600	155	LGA 2011-1

2.5 TDP values and socketing of Intel's high-end 2S server processors (4)

TDP values of Intel's HE scalable 2S server processors -1

Platform	HE scalable 2S server models	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket (up to)	TDP (up to) W	Socket
Purley	1 st gen. Xeon Gold 6138R (Skylake-SP)	14 nm	7/2017	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2666	125	LGA 3647
	2 nd gen. Xeon Platinum 9282 (Cascade Lake-AP)	14 nm	4/2019	2x28C	40x PCIe 3.0 (on proc.)	12x DDR4-2933	400	BGA 5903
	Xeon Gold 6238R/6258R (Cascade Lake-SP)	14 nm	2/2020	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2933	165/205	LGA 3647
Whitley	3 rd gen. Xeon Platinum 8380 (Eis Lake-SP)	10 nm	4/2021	40C	64xPCIe4.0 (on proc.)	8x DDR4-3200	270	LGA 4189
Eagle Stream	4 th gen. Xeon Platinum 9480H (Sapphire Rapids)	Intel 7	1/2023	56C	80xPCIe5.0 (on proc.)	8x DDR5-4800	350	LGA 4677
	5 th gen. Xeon Platinum 8592/93 (Emerald Rapids)	Intel 7	12/ 2023	64C	80xPCIe5.0 (on proc.)	8x DDR5-5600	330	LGA 4677

2.5 TDP values and socketing of Intel's high-end 2S server processors (5)

TDP values of Intel's HE scalable 2S server processors -2

Platform	HE scalable 2S server models	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket	TDP (up to) W	Socket
Beechnut City	6 th gen. 6780E (Xeon 6) (Sierra Forest-SP)	Intel 3	6/2024	144C (E-cores) (Crestmont) (no SMT)	88x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA 4710
	6 th gen. 6780P (Granite Rapids-SP)	Intel 3 / Intel 7	9/2024	86C (P-cores) (Redwood Cove) (SMT)	96x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA-4710
Avenue City	6 th gen. Xeon 6 6980E (Sierra Forest-AP)	Intel 3 / Intel 7	2/2025	288C (E-cores) (Cresmont) (no SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529
	6 th gen. Xeon 6 6980P (Granite Rapids-AP)	Intel 3 / Intel 7	9/2024	128C (P-cores) (Redwood Cove) (SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529

Remarks

SP: Scalable Performance (8 memory channels)

AP: Advanced Performance (12 memory channels)

Sierra Forrest: E-cores

Granite Rapids: P-cores

P-cores support simultaneous multithreading (SMT) while E-cores do not.

2.5 TDP values and socketing of Intel's high-end 2S server processors (6)

How much power do Intel's HE scalable 2S server processors consume (in terms of the TDP)?

- The previous Tables indicate that **high-end 2S server processors** typically exhibit **considerably higher TDP values** than typical desktop models.

Early HE 2S models started with TDP values around **130 W**, while **recent** models have reached **up to 500 W**, reflecting the growing performance demand in server environments.

2.5 TDP values and socketing of Intel's high-end 2S server processors (7)

How are Intel's HE scalable 2S server processors socketed? -1

The Tables below provide the answer.

2.5 TDP values and socketing of Intel's high-end 2S server processors (8)

How are Intel's HE scalable 2S server processors socketed? -2

Platform	HE multicore 2S server models	Techn .	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket (up to)	TDP (up to) W	Socket
Paxville-DP	Xeon DP 2.8 GHz (P4 Prescott 2M DP)	90 nm	10/2005	2x1C	24x PCIe 1.1 (on MCH)	2x DDR2-400 (via MCH)	135	PGA 604
Bensley	Xeon 5000 (Dempsey) (P4 Prescott 2M DP)	65 nm	5/2006	2x1C	16x PCIe 1.0 (on MCH)	4x DDR2-533 (FB-DIMM) (via MCH)	130	LGA 771
	(Woodcrest) (Core 2)	65 nm	6/2006	2C	16x PCIe 1.0 (on MCH)	4x DDR2-667 (FB-DIMM) (via MCH)	65	LGA 771
	Xeon 5300 (Clovertown) (Core 2)	65 nm	11/2006	2x2C	16x PCIe 1.0 (on MCH)	4x DDR2-667 (FB-DIMM) (via MCH)	80	LGA 771
	Xeon 5400 (Harpertown) (Penryn)	45 nm	11/2007	2x2C	24x PCIe 1.0 (on MCH)	4x DDR2-667 (FB-DIMM) (via MCH)	120	LGA 771
Boxboro-EX	Xeon E65xx Nehalem-EX) (Beckton)	45 nm	3/2010	8C	36x PCIe 2.0 (on MCH)	8x DDR3-1066	130	LGA 1567
	Xeon E7-28xx (Westmere-EX)	32 nm	4/2011	10C		8x DDR3-1333	130	
Brickland	E7-28xx v2 (Ivy Bridge-EX)	22 nm	2/2014	15C	32x PCIe 3.0 (on proc.)	8x DDR3-1600	155	LGA 2011-1

PGA: Pin Grid Array
LGA: Land Grid Array

2.5 TDP values and socketing of Intel's high-end 2S server processors (9)

How are Intel's HE scalable 2S server processors socketed? -3

Platform	HE scalable 2S server models	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket (up to)	TDP (up to) W	Socket
Purley	1 st gen. Xeon Gold 6138R (Skylake-SP)	14 nm	7/2017	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2666	125	LGA 3647
	2 nd gen. Xeon Platinum 9282 (Cascade Lake-AP)	14 nm	4/2019	2x28C	40x PCIe 3.0 (on proc.)	12x DDR4-2933	400	BGA 5903
	Xeon Gold 6238R/6258R (Cascade Lake-SP)	14 nm	2/2020	28C	48x PCIe 3.0 (on proc.)	6x DDR4-2933	165/205	LGA 3647
Whitley	3 rd gen. Xeon Platinum 8380 (Eis Lake-SP)	10 nm	4/2021	40C	64xPCIe4.0 (on proc.)	8x DDR4-3200	270	LGA 4189
Eagle Stream	4 th gen. Xeon Platinum 9480H (Sapphire Rapids)	Intel 7	1/2023	56C	80xPCIe5.0 (on proc.)	8x DDR5-4800	350	LGA 4677
	5 th gen. Xeon Platinum 8592/93 (Emerald Rapids)	Intel 7	12/ 2023	64C	80xPCIe5.0 (on proc.)	8x DDR5-5600	330	LGA 4677

PGA: Pin Grid Array
LGA: Land Grid Array

2.5 TDP values and socketing of Intel's high-end 2S server processors (10)

How are Intel's HE scalable 2S server processors socketed? -4

Platform	HE scalable 2S server models	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket	TDP (up to) W	Socket
Beechnut City	6 th gen. 6780E (Xeon 6) (Sierra Forest-SP)	Intel 3	6/2024	144C (E-cores) (Crestmont) (no SMT)	88x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA 4710
	6 th gen. 6780P (Granite Rapids-SP)	Intel 3 / Intel 7	9/2024	86C (P-cores) (Redwood Cove) (SMT)	96x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA-4710
Avenue City	6 th gen. Xeon 6 6980E (Sierra Forest-AP)	Intel 3 / Intel 7	2/2025	288C (E-cores) (Cresmont) (no SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529
	6 th gen. Xeon 6 6980P (Granite Rapids-AP)	Intel 3 / Intel 7	9/2024	128C (P-cores) (Redwood Cove) (SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529

Remarks

SP: Scalable Performance (8 memory channels)

AP: Advanced Performance (12 memory channels)

Sierra Forrest: E-cores

Granite Rapids: P-cores

P-cores support simultaneous multithreading (SMT) while E-cores do not.

How are Intel's HE scalable 2S server processors socketed? -5

- Intel's 64-bit multicore 2S servers generally use **LGA sockets** with **two notable exceptions**.
- The first exception is Intel's first dual-core x86-based server processor, known as **Xeon DP 2.8 GHz**, which employed **PGA** sockets.

The second exception is the **scalable Xeon Platinum 9282**, which is a **dual-die** implementation that provides 2x28 cores and has a **BGA** socket.

, It was explicitly used only by OEMs.

LGA: Land Grid Array

PGA: Pin Grid Array

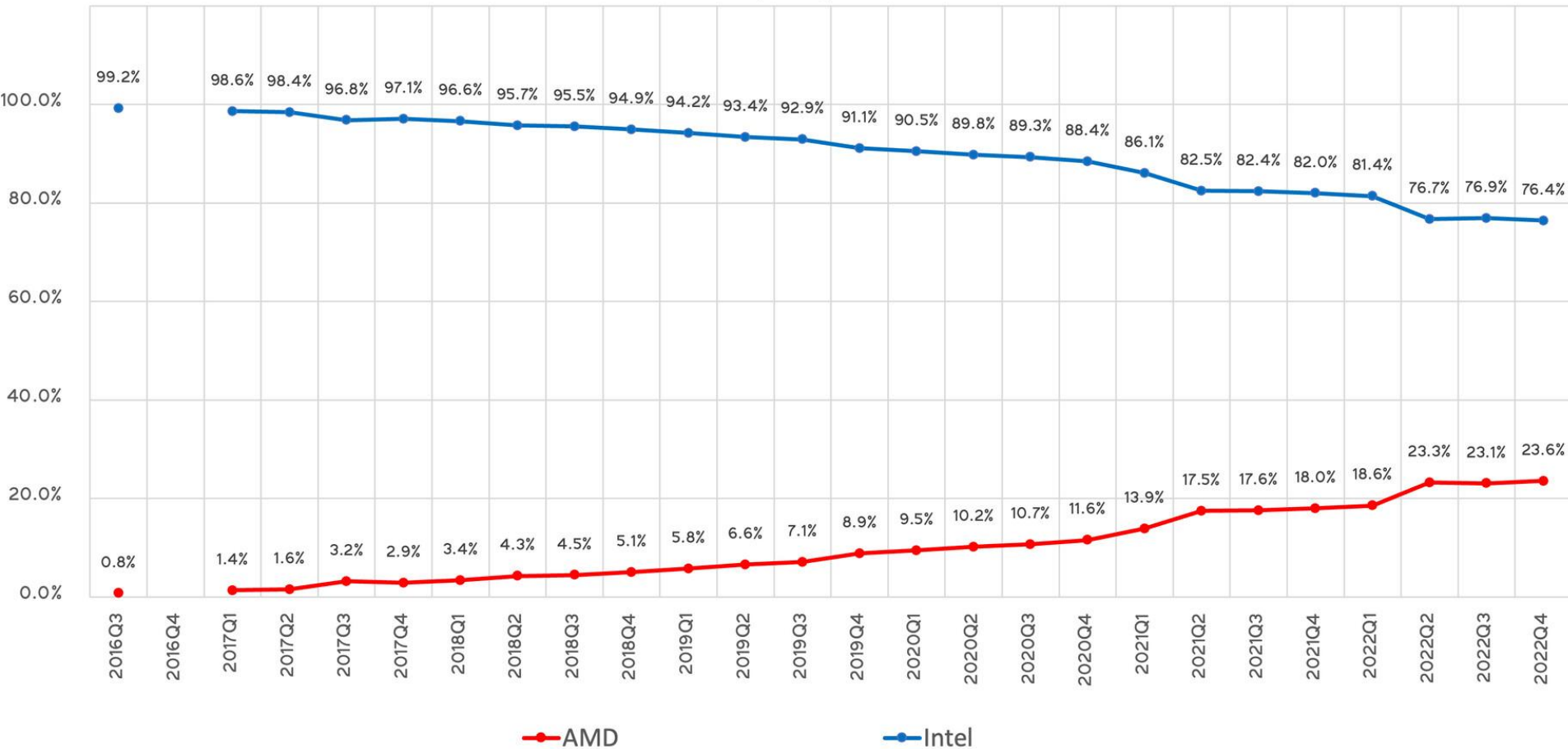
BGA: Ball Grid Array

How many contacts have Intel's 2S server sockets?

- The number of socket contacts is primarily determined by the number of memory channels and PCI lanes, which are directly connected to the processor.
- Early 2S lines were designed to support 3 memory channels and offered 48 PCIe lanes through the MCH, necessitating 771 socket contacts.
- In contrast, the most recent 2S server processors can accommodate up to 12 memory channels and 96 PCIe lanes, driving the contact count to as high as 9729.

2.5 TDP values and socketing of Intel's high-end 2S server processors (13)

AMD's and Intel's revenue share of the global server processor market -1 [178]



The Figure above illustrates **AMD's growing revenue share** in the global server processor market, climbing from nearly 0 % in 2017 to 24.2 % by Q4/2024 [178].

The Nvidia-Intel cooperation

- Announced on Sept. 18, 2025.
- NVIDIA will invest \$5 billion in Intel through a stock purchase agreement.
- As part of the partnership, Intel will design and manufacture custom CPUs, targeting both data center and client markets, which provide integrated support for NVIDIA's NVLink interconnect.

The NVLink enables tight coupling between Intel's x86 CPUs and NVIDIA's computing platforms, including GPUs.

2.5 TDP values and socketing of Intel's high-end 2S server processors (15)

Intel's stock price May – October 2025



3. Case examples for 2S server boards

- 3.1 Case example 1: 1st gen. Skylake-SP-based 2S server board (2017)
- 3.2 Case example 2: 6st gen. Granite Rapids-based 2S server board (2025)
- 3.3 Outlook: The 6+ gen. Clearwater Forest 2S server processor (to be released in 2026)

3.1 Case example 1: 1st gen. Skylake-SP-based server board (2017)

3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (1)

3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board ((2017)

- Intel's **1st gen. scalable 2S server processors** are traditional symmetrical multicores with up to **20 cores** implemented on a **single die**.
- Below are the **core features** of the **1st gen. scalable 2S Gold class server processors**, known as **Skylake-SP** processors.

Model	Family Xeon	Price	Launched	Cores	Threads	Frequency	Max Turbo	TDP	L2\$ MiB	L3\$ MiB	Mem. DDR4-	Max Mem.
<u>3104</u>	Bronze	\$ 213	07/2017	6	6	1.7 GHz		85 W	6	8.25	2133	768 GiB
<u>3106</u>	Bronze	\$ 306	07/2017	8	8	1.7 GHz		85 W	8	11	2133	768 GiB
<u>4108</u>	Silver	\$ 417	07/2017	8	16	1.8 GHz	3 GHz	85 W	8	11	2400	768 GiB
<u>4109T</u>	Silver	\$ 501	07/2017	8	16	2 GHz	3 GHz	70 W	8	11	2400	768 GiB
<u>4110</u>	Silver	\$ 501	07/2017	8	16	2.1 GHz	3 GHz	85 W	8	11	2400	768 GiB
<u>4112</u>	Silver	\$ 473	07/2017	4	8	2.6 GHz	3 GHz	85 W	4	8.25	2400	768 GiB
<u>4114</u>	Silver	\$ 694	07/2017	10	20	2.2 GHz	3 GHz	85 W	10	13.75	2400	768 GiB
<u>4114T</u>	Silver		07/2017	10	20	2.2 GHz	3 GHz	85 W	10	13.75	2400	768 GiB
<u>4116</u>	Silver	\$ 1,002	07/2017	12	24	2.1 GHz	3 GHz	85 W	12	16.5	2400	768 GiB
<u>4116T</u>	Silver		07/2017	12	24	2.1 GHz	3 GHz	85 W	12	16.5	2400	768 GiB
<u>6138R</u>	Gold	\$ 4,937	05/2018	20	40	2.0 GHz	3.7 GHz	125 W	20	27.5	2666	768 GiB

Table: Key features of the 1st gen. scalable Skylake-SP 2S server processors

3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (2)

Switching from dual ring buses to a 2D mesh interconnect in Intel's up to 28-core Skylake-SP server processor line in 2017 [40] -1

2D mesh interconnects consist of **dual links** arranged in a **2D grid structure**, interconnected through **routers** that forward **data packets** from senders to receivers, as shown below.

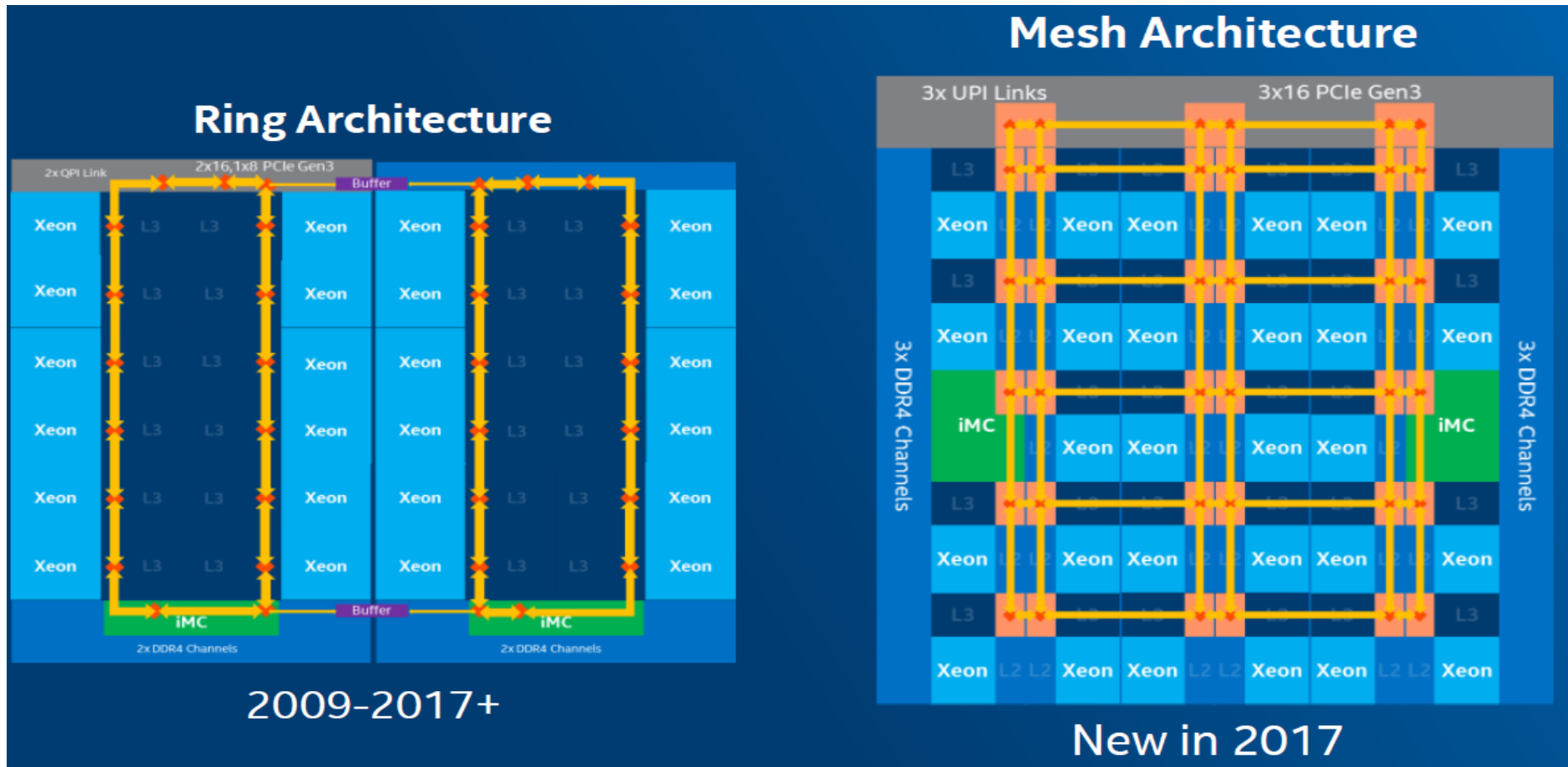


Figure: Contrasting dual ring and 2D mesh interconnects [40]

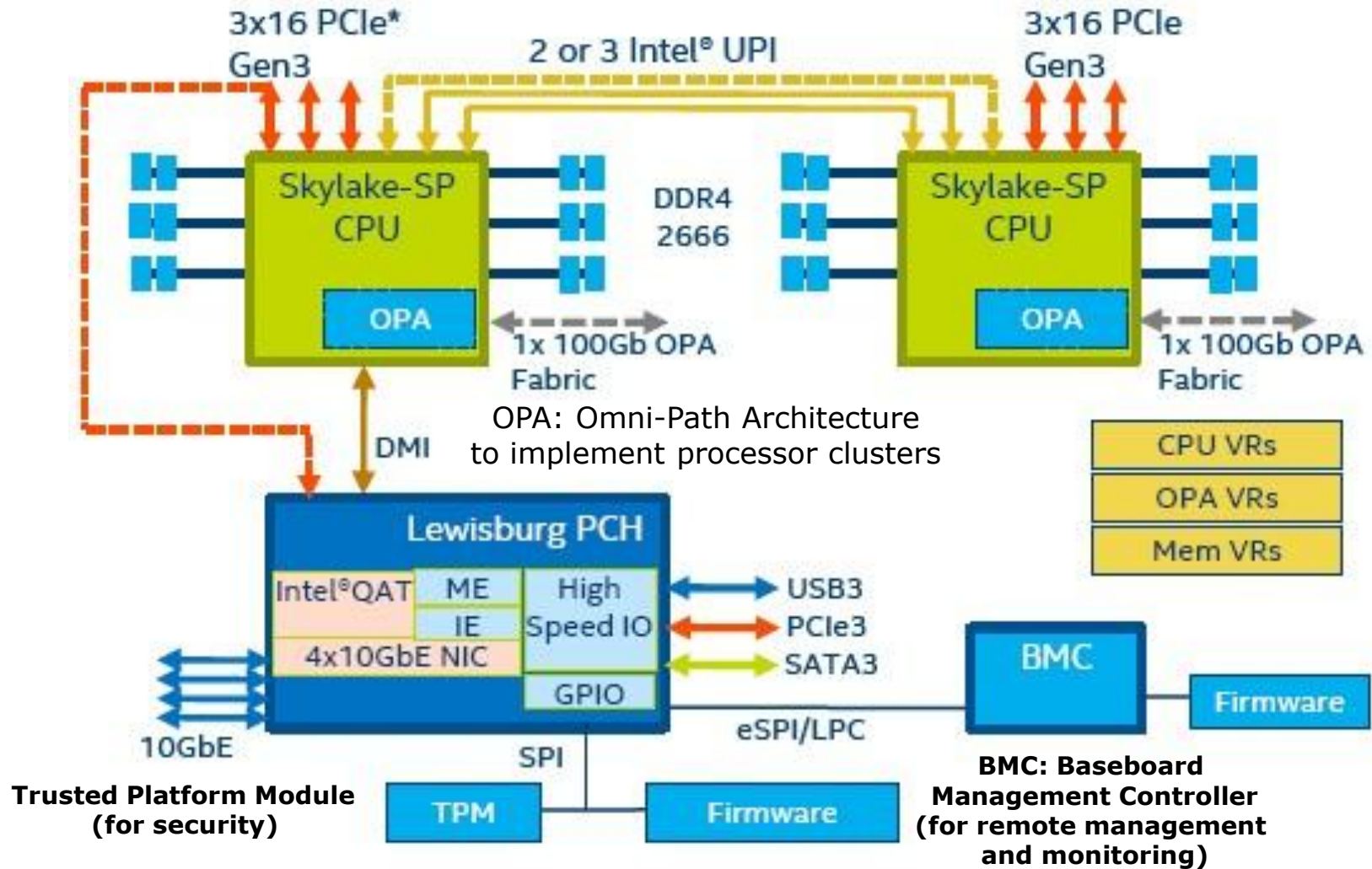
3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (3)

Switching from dual ring buses to a 2D mesh interconnect in Intel's up to 28-core Skylake-SP server processor line in 2017 [40] -2

- In a **ring interconnect**, core-to-core delays **increase linearly**.
With high core count server implementations, such as those **over 20 cores**, these delays can **significantly reduce performance**.
- Intel addresses this issue by introducing **2D mesh interconnects**, which effectively **reduce core-to-core delays**.
- Subsequently, we will present the **block diagram** of Skylake-SP processors and outline a **motherboard** designed for 1st and 2nd gen. scalable servers.

3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (4)

Block diagram of a 1st gen. Skylake-SP based 2S server platform (2017) [179]



BMC: Baseboard Management Controller	PCH: Intel® Platform Controller Hub	IE: Innovation Engine
Intel® OPA: Intel® Omni-Path Architecture	Intel QAT: Intel® QuickAssist Technology	ME: Manageability Engine
NIC: Network Interface Controller	VMD: Volume Management Device	NTB: Non-Transparent Bridge

3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (5)

Block diagram of
Supermicro's X11DPi-N
motherboard for
1st/2nd gen. scalable
2S servers with
Skylake-SP/
Cascade Lake-SP
processors [180]

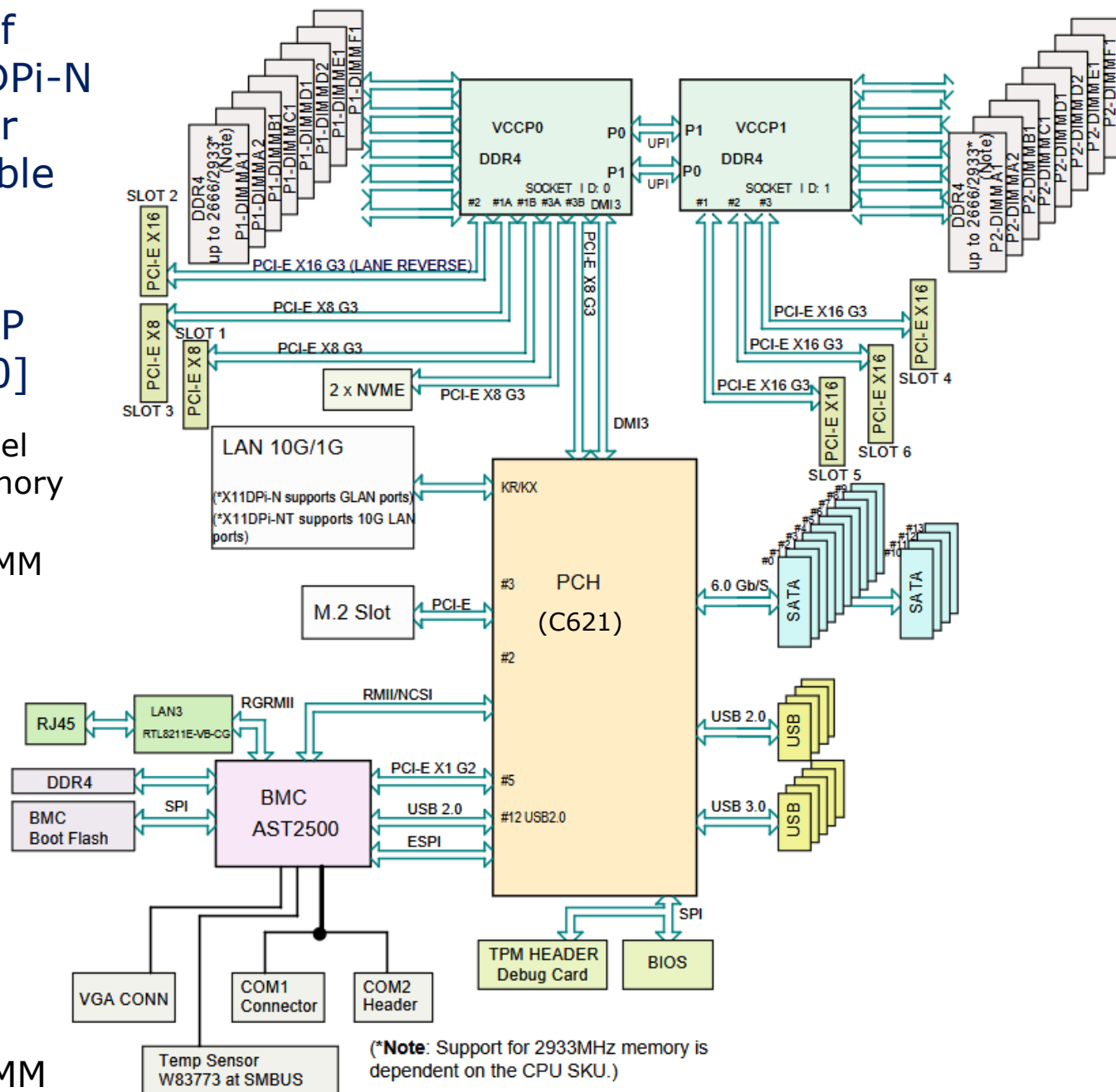
It supports 2x6 channel
RDIMM DDR4-2933 memory
with up to 4 TB size.

RDIMM: Registered DIMM
(buffered)

PCIE G3: PCIe 3.0

BMC: Baseboard
Management Controller
(for remote management
and monitoring)

RDIMM: Registered DIMM



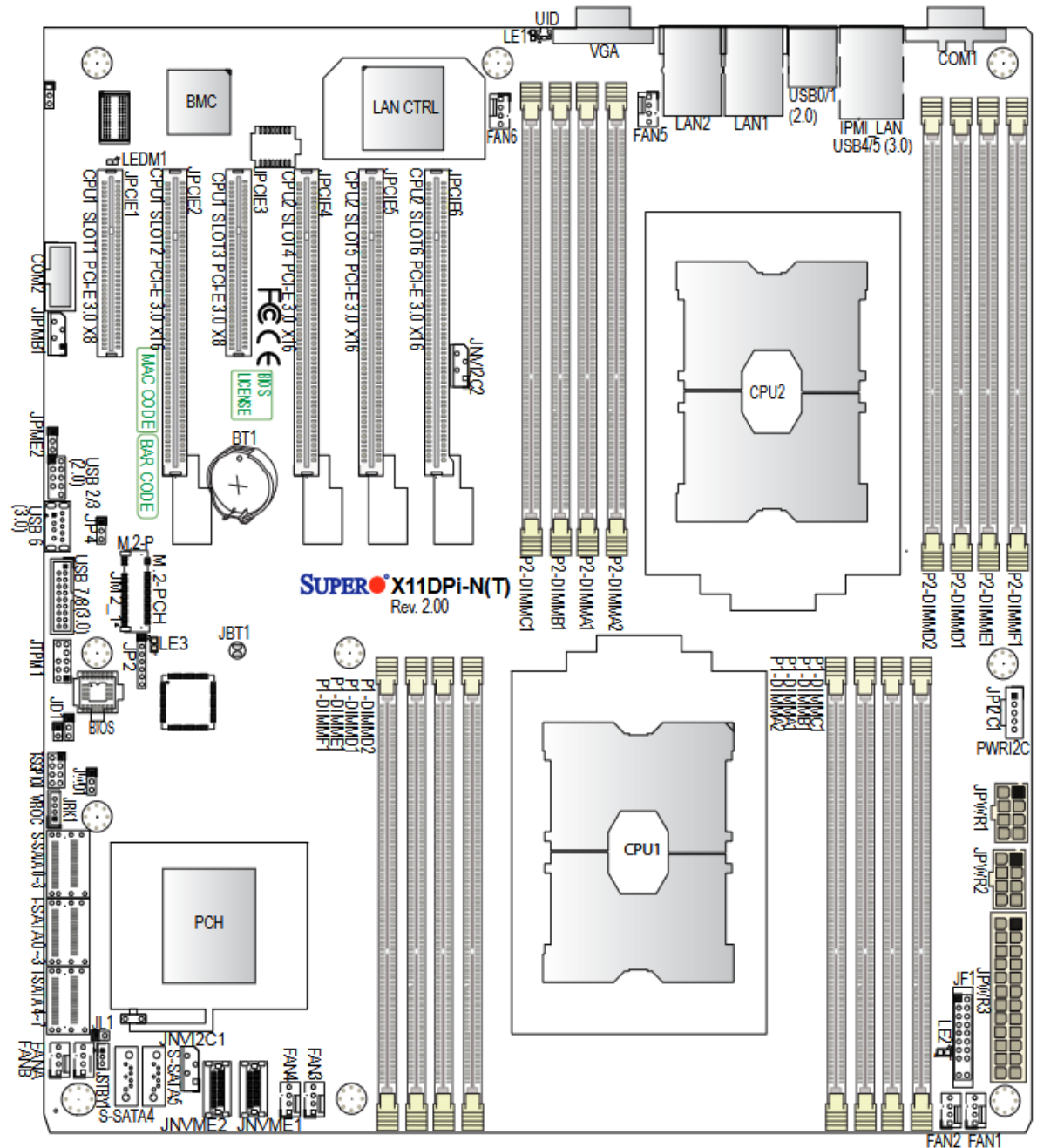
3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (6)

Remark on the number of DIMMs per processor

- There are 6 memory channels with 8 DIMM slots per processor.
- Different configurations can be used to populate the DIMM slots, which are not detailed here.
- In certain configurations, dual DIMMs can be connected to the same channel, this is the reason why there are more DIMM slots than memory channels per processor.

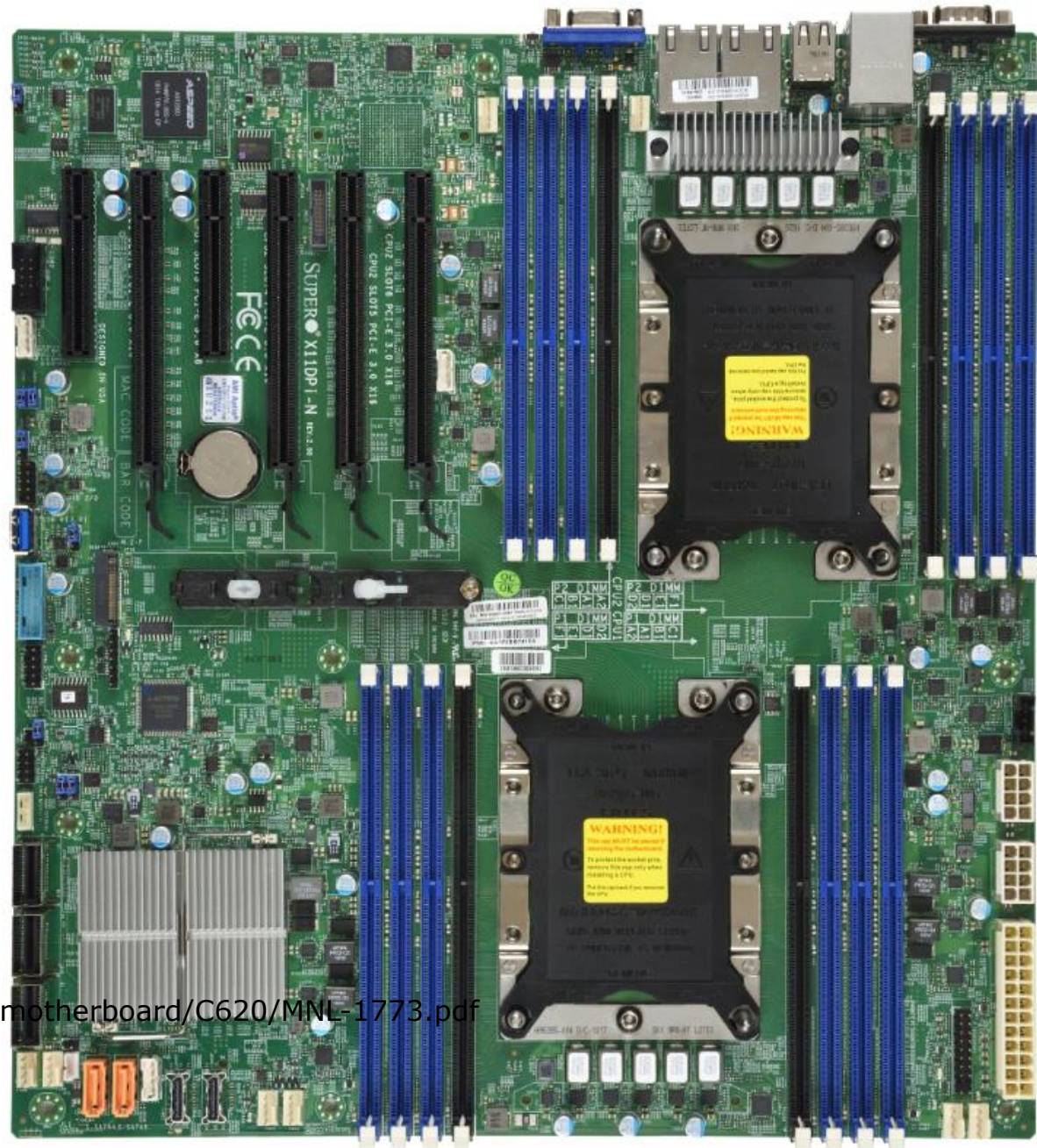
3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (7)

Layout of the
Supermicro's X11DPi-N
motherboard
for 1st/2nd gen. scalable
2S servers with
Skylake-SP or
Cascade Lake-SP
processors [180]



3.1 Case example 1: 1st gen. Skylake-SP based scalable 2S server board (8)

Picture of
Supermicro's X11DPi-N
motherboard
for 1st/2nd gen. scalable
2S servers with
Skylake-SP or
Cascade Lake-SP
processors [180]



<https://www.supermicro.com/manuals/motherboard/C620/MNL-1773.pdf>

7.2 Case example 2: 6st gen. Granite Rapids-based 2S server board (2025)

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (1)

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (2025)

- Intel's 6th gen. **Granite Rapids-SP/AP and Sierra Forest-SP/AP scalable 2S server processors** are Intel's **first heterogeneous multi-die designs**, as indicated below.

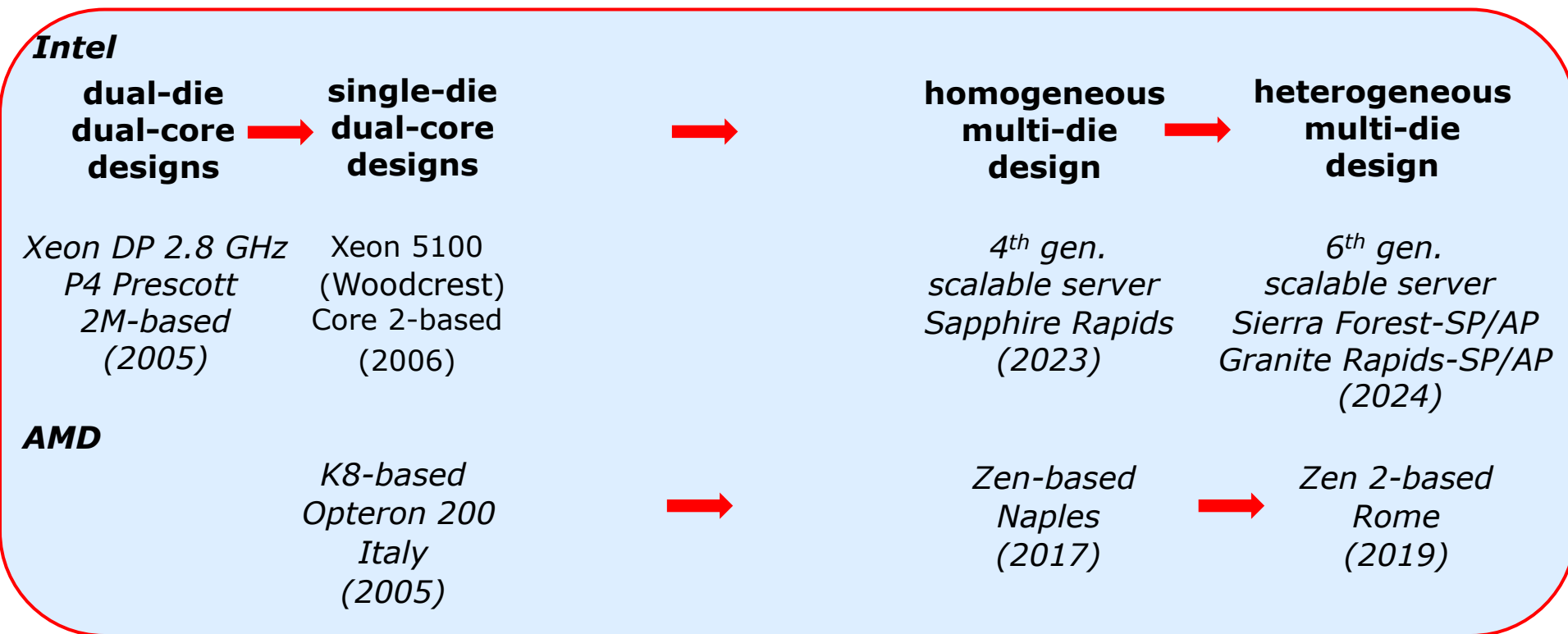


Figure: Evolution of Intel's and AMD's server processor designs concerning their die layout

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (2)

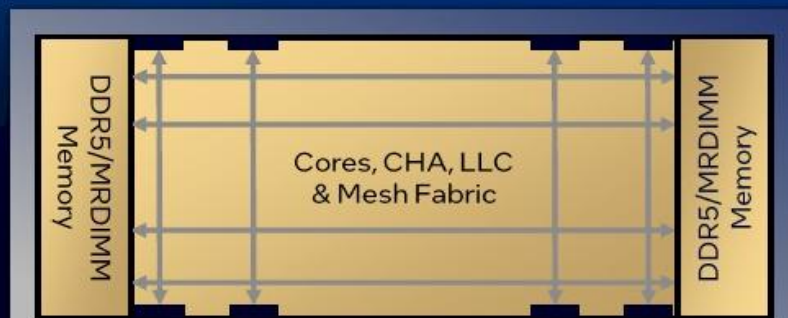
Intel's 6th gen. scalable Granite Lake-SP/AP and Sierra Forest-SP/AP processor lines [2024] -1

- These processors include **two types of dies** (tiles in Intel's terminology): **Compute dies**, and **IO dies**, as shown below.

Compute die

Intel 3 process

Monolithic mesh, cores, cache, memory controllers and EMIB technology



I/O die

Intel 7 process

Universal I/O stacks with UPI, PCIe, CXL and Accelerator Engines

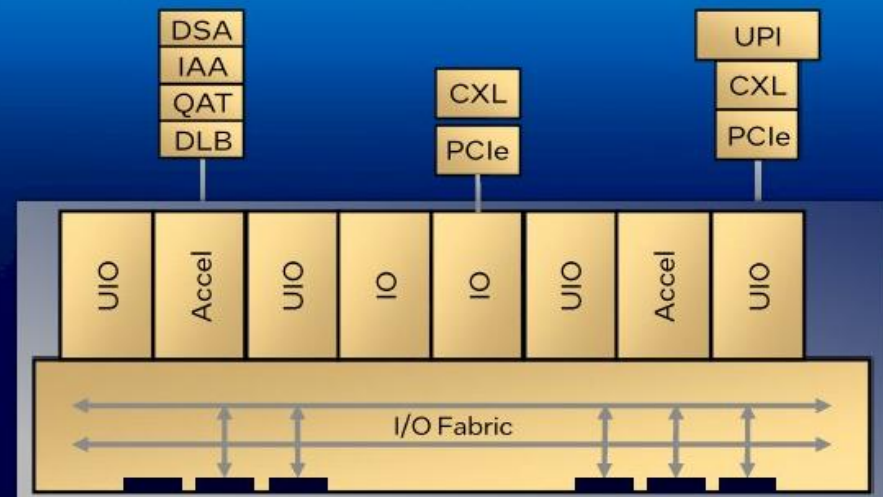


Figure: Die types of the scalable Granite Lake-SP/AP and Sierra Forest-SP/AP processors [182]

CXL: (Compute Express Link) is a high-speed, low-latency interconnect standard built on top of PCIe for memory expansion modules and accelerators, from 2019.

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (3)

Intel's 6th gen. scalable Granite Lake-SP/AP and Sierra Forest-SP/AP processor lines [2024] -2

These processors include 1 to 3 Compute dies and 2 IO dies.

The compute dies can be in 4 different configurations called UCC, XCC, HCC, and LCC, as illustrated below.

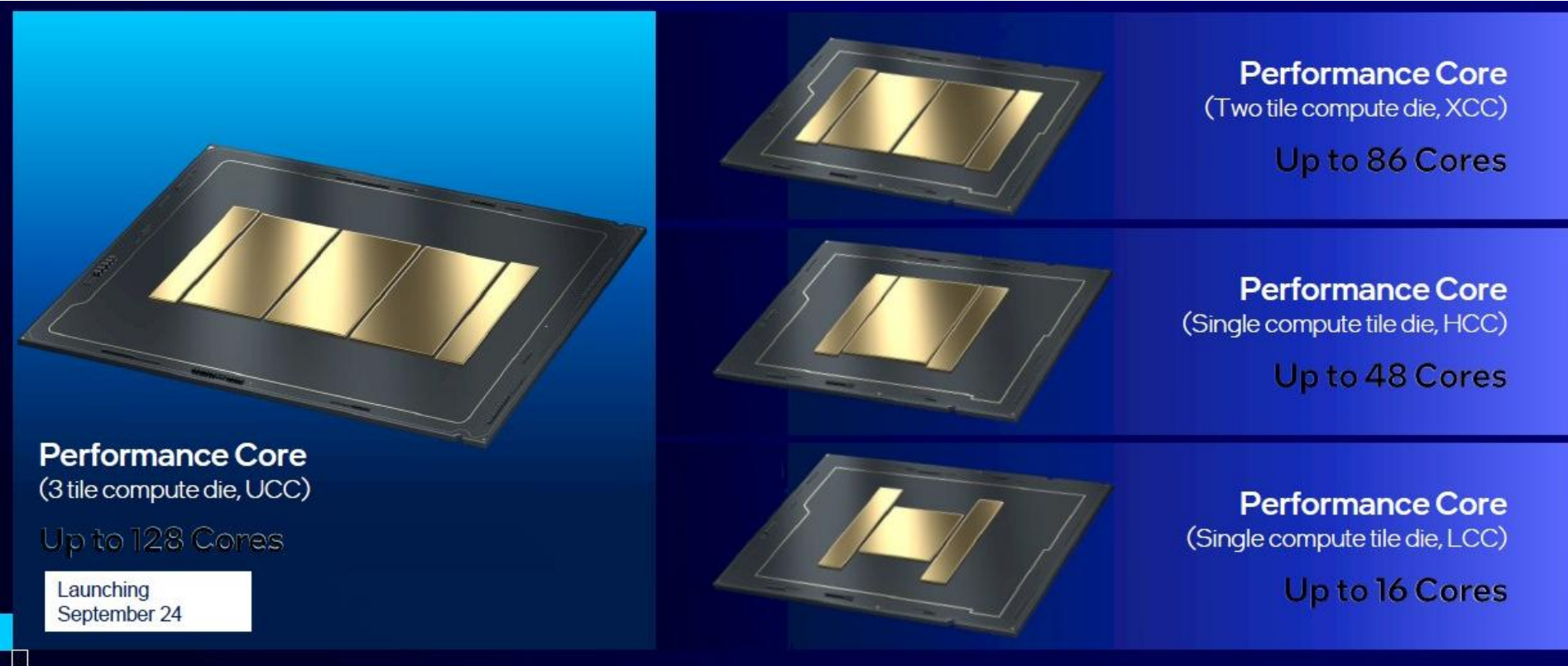


Figure: Die configurations of Sierra Forest and Granite Rapids processors [182]

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (4)

Intel's 6th gen. scalable Granite Lake-SP/AP and Sierra Forest-SP/AP processor lines [2024] -3

- The **Compute dies** are **598 mm²** large, are fabricated on the **Intel3** technology, while the **IO dies** measure **241 mm²**, and are produced with the **Intel7** process.
- The dies are **EMIB** connected.

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (5)

Contrasting key features of Intel's 6th gen. scalable Granite Lake-SP/AP and Sierra Forest-SP/AP processor lines

Platform	HE scalable 2S server models	Techn.	Date of Intro.	Core count (up to)	PCIe	No./speed of mem. channels per socket	TDP (up to) W	Socket
Beachnut City	6 th gen. 6780E (Xeon 6) (Sierra Forest-SP)	Intel 3/ Intel 7	6/2024	144C (E-cores) (Crestmont) (no SMT)	88x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA 4710
	6 th gen. 6780P (Granite Rapids-SP)	Intel 3 / Intel 7	9/2024	86C (P-cores) (Redwood Cove) (SMT)	96x PCIe 5.0 (on proc.)	8x DDR5-6400	330	LGA-4710
Avenue City	6 th gen. Xeon 6 6980E (Sierra Forest-AP)	Intel 3 / Intel 7	2/2025	288C (E-cores) (Cresmont) (no SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529
	6 th gen. Xeon 6 6980P (Granite Rapids-AP)	Intel 3 / Intel 7	9/2024	128C (P-cores) (Redwood Cove) (SMT)	96xPCIe5.0 (on proc.)	12x DDR5-6400	500	LGA 7529

Remarks

Sierra Forest includes E Cores – Granite Rapids includes P cores

SP: Scalable Performance (8 memory channels/LGA 4710) – Xeon 6700E/P series

AP: Advanced Performance (12 memory channels/LGA7529) – Xeon 6900E/P series

P-cores support simultaneous multithreading (SMT) while E-cores do not.

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (6)

Main Features of Intel's 6th gen. Granite Rapids-AP scalable processor line [181]

Intel Xeon 6 P-core SKU Map

Roadmap SKUs with customization options

PERFORMANCE SKUs

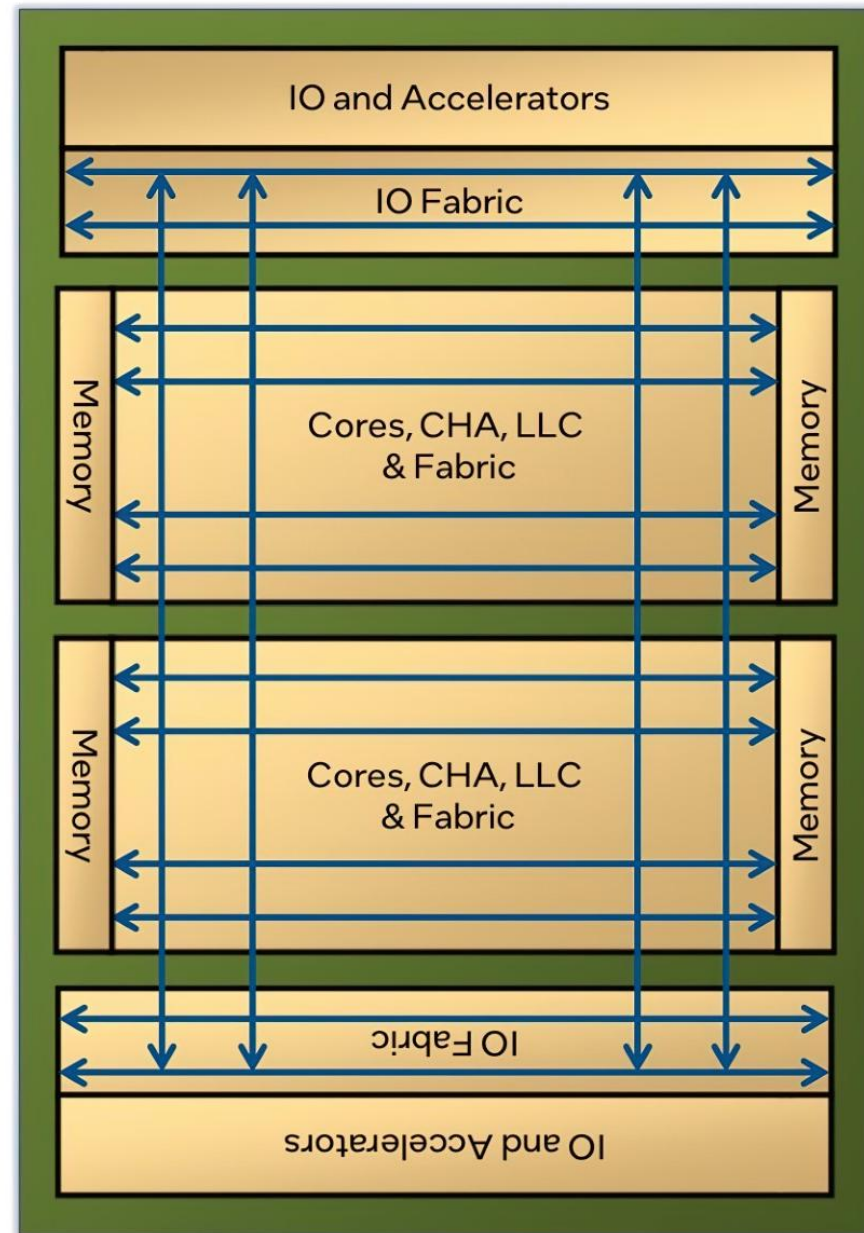
SKU	CORES	BASE (GHz)	ALL CORE TURBO (GHz)	Max TURBO (GHz)	L3 CACHE (MB)	TDP (Watts)	Max. Scala.	Memory Channels	DDR5 Memory Speed	MRDIMM Speed	Default Accel. Devices	Intel TDX Keys (Per CPU)	UPI Links Enab.	PCIe Lanes
6980P	128	2.0	3.2	3.9	504	500	2S	12	6400	8800	4/4/4/4	1024	6	96
6979P	120	2.1	3.2	3.9	504	500								
6972P	96	2.4	3.5	3.9	480	500								
6952P	96	2.1	3.2	3.9	480	400								
6960P	72	2.7	3.8	3.9	432	500								

- The 6980P/6979P include 5 dies (3 Compute dies with 44 cores and 2 IO dies, while the other models have 2 Compute dies withx44 cores each and and 2 IO dies.
- Available not-used cores are disabled.

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (7)

A logically homogeneous 2D mesh interconnects the tiles of the 6th gen. Granite Rapids-AP processors (2024) [183]

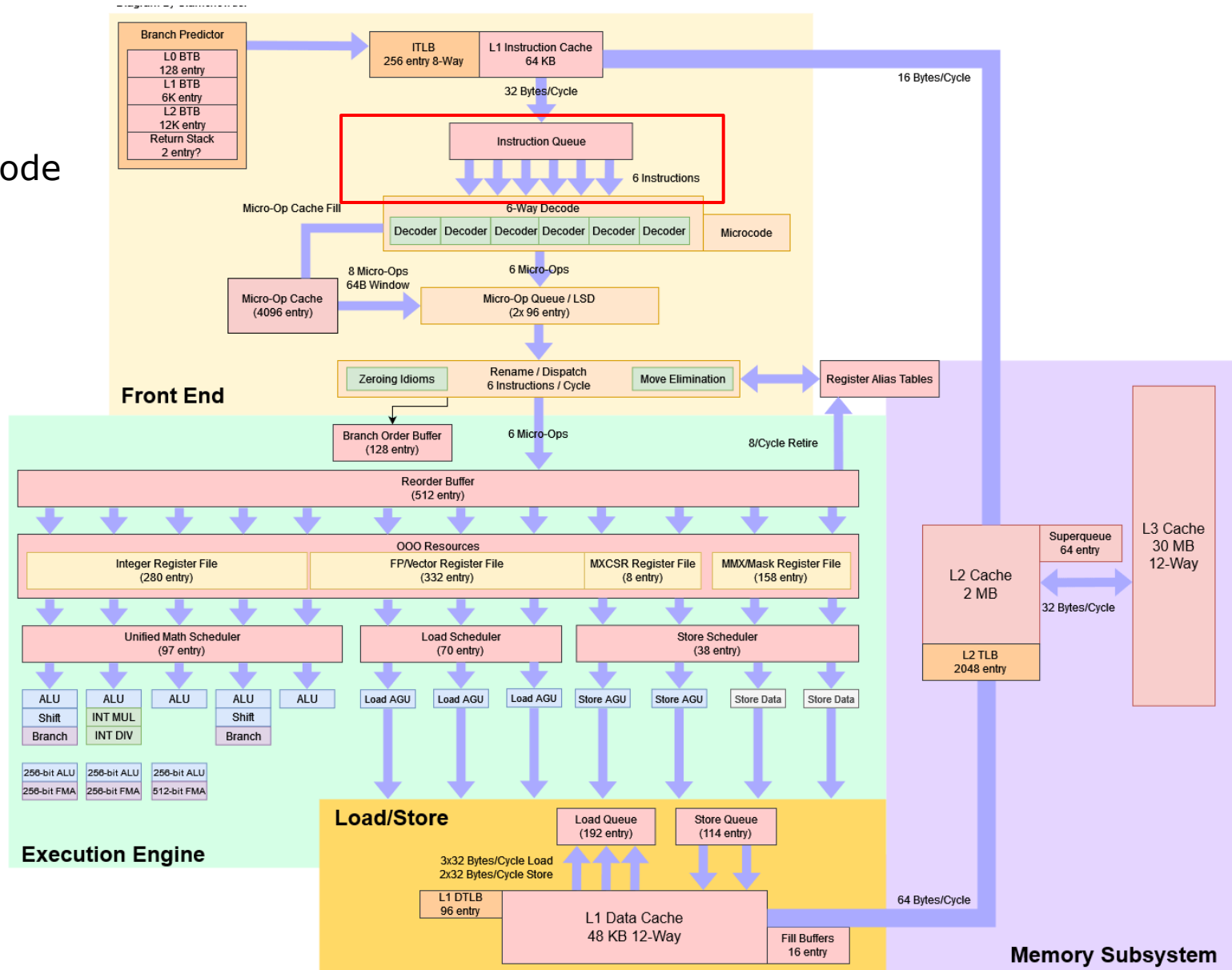
- Each mesh stop is associated with a core, a combined Caching and Home Agent (CHA) and a Last Level Cache (LLC) slice (L3 cache slice).
- The Caching Agent reduces the latency when accessing frequently used data in the cache, by keeping track of memory addresses stored in the caches of the connected processors. (Not detailed here).
- The Home Agent ensures memory coherency across multiple cores and processors.
- The Interconnect, called Fabric, consists of switches that forward data packets from sources to destinations.



3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (8)

Microarchitecture of the Redwood Cove core [215]

6-wide decode

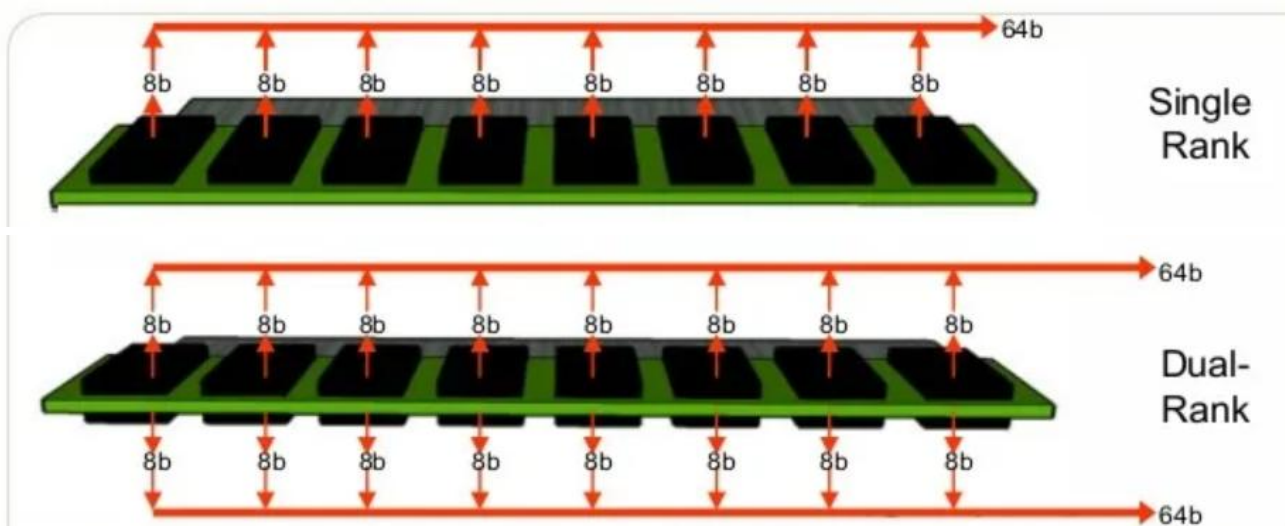


3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (9)

MRDIMMs (Multiplexed Rank DIMMs) -1

- Intel's 6th gen. scalable server processors support **two memory types**, which are **DDR5 DIMMs** and **MRDIMMs** (Intel's first implementation).
- To understand **MRDIMMs**, however, it is crucial first to discuss the concept of **memory ranks**.
- Memory ranks** are 64-bit data blocks composed of a **set of memory chips located on a memory module**, which are **accessed simultaneously** by the MC, as illustrated below.

DRAM Ranks



Source: Difference Between Dual Rank and Single Rank RAM – [Memory4less Official Blog](#)

- If the memory **provides ECC (Error Checking and Correction)** support, an **additional 8 bits** are required in addition to the 64 bits.
- An important aspect of **multi-rank memory** is that **the memory controller can only access one rank at a time**, regardless of the number of memory ranks on the module.

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (10)

MRDIMMs (Multiplexed Rank DIMMs) -2

- MRDIMMs aim to increase achievable memory bandwidth by enabling two ranks to operate in parallel while the Memory Controller on the CPU die operates at double clock speed, as illustrated below.

Problem

- New workloads benefiting from more BW/thread
- Adding DIMMs to deliver BW is expensive
- Prefer to increase memory BW @ iso capacity/core

MCR Details

- Fully platform compatible with DDR5 RDIMMs
- Mux merges two ranks at 1x speed into channel running at 2x speed
- Each Rank operates at half the channel speed
- JEDEC standardization in progress today
- Achieving 30-40% more BW than RDIMM

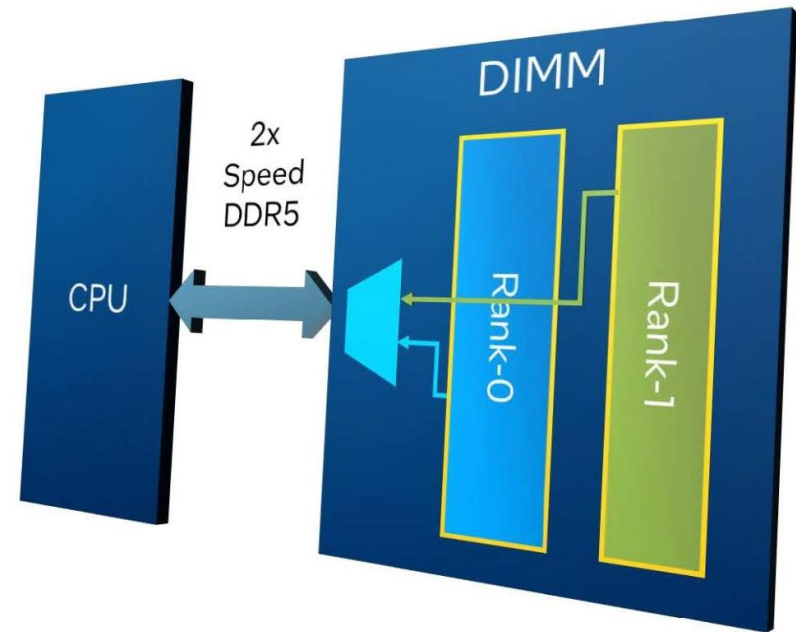
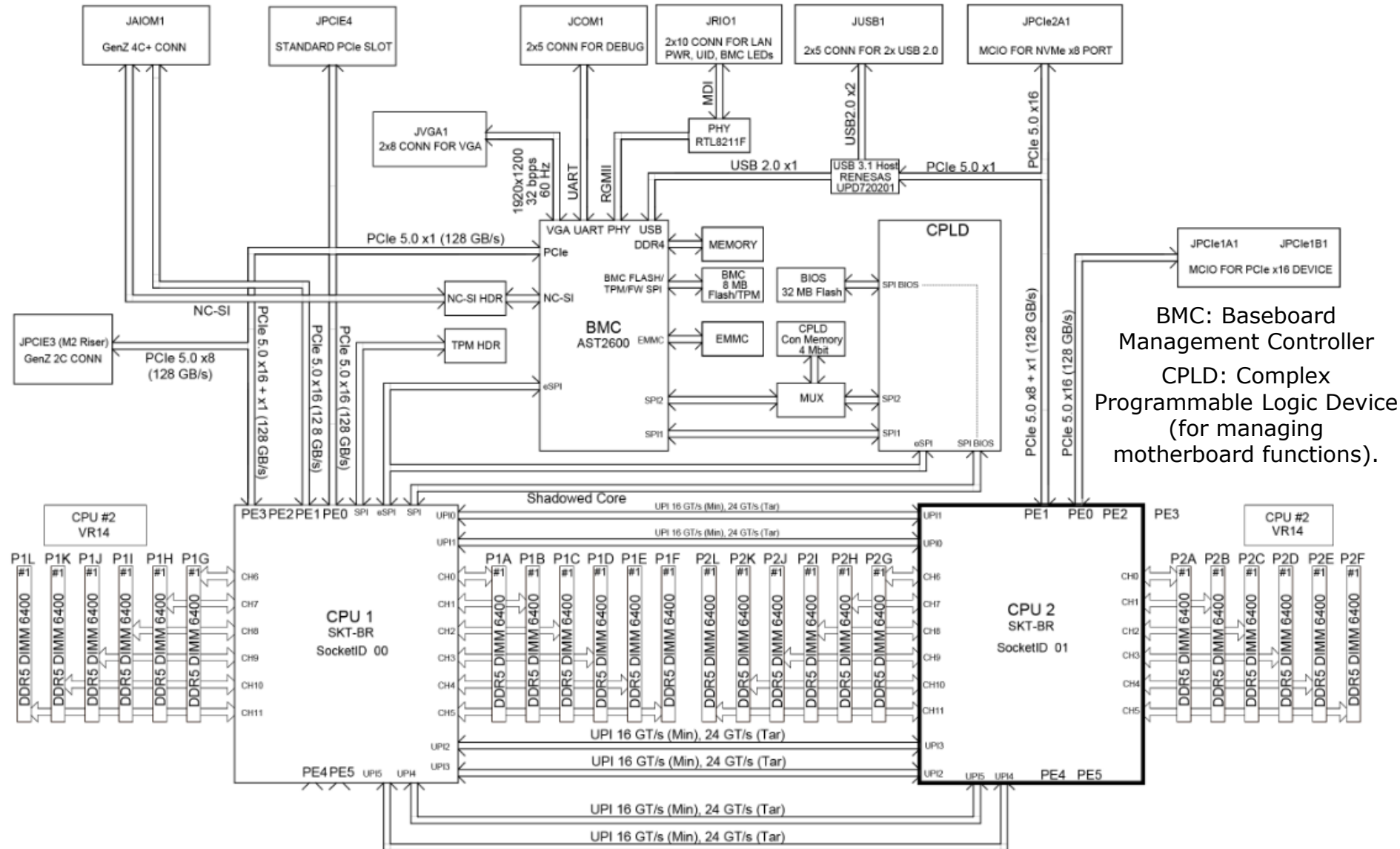


Figure: Principle of operation of MRDIMMs [184]

- MRDIMMs can double memory bandwidth, however, they require memory controllers and connections that operate at double speed.

3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (11)

Block diagram of Supermicro's X14DBT motherboard for 6th gen. scalable 2S servers with Granite Rapids-AP processors (2025) (not detailed) [185]



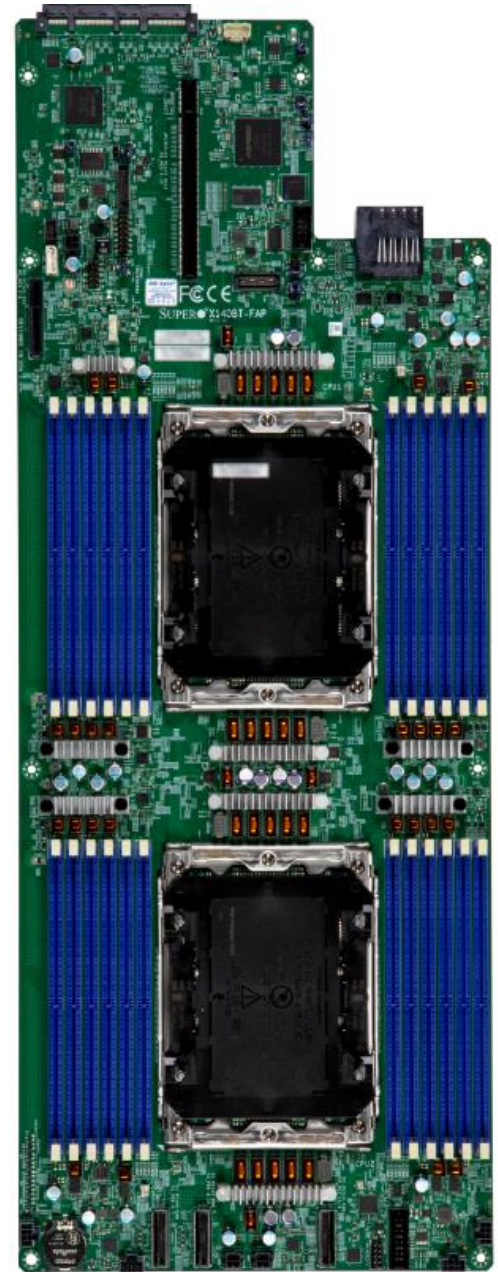
3.2 Case example 2: 6th gen. Granite Rapids-AP based scalable 2S server boards (12)

Picture of Supermicro's X14DBT motherboard for 6th gen. scalable 2S servers with Granite Rapids-AP processors (2025) [185]

- This motherboard supports up to 6144 GB of DDR5-6400 memory in 2x12 DIMM slots for 2x12 memory channels.
- It supports dual Intel Xeon 6900-series Granite Rapids- AP processors with up to 128 P cores, in LGA 7529 socket, and a TDP of up to 500 W.
- There is **no need for a PCH**.

It is Intel's first server processor line that does not require a PCH.

Interestingly, AMD's Zen-based EPYC server processors also do not require PCHs.



3.3 Outlook: The 6+ gen. Clearwater Forest 2S server processor (to be released in 2026)

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (1)

Outlook: Intel's Xeon 6+ series 288 E-core Clearwater Forest server processor line, to be released in the first half of 2026

It is Intel's **second-generation heterogeneous multi-die** server processor line of its scalable processor family.

This E-series includes **efficiency cores** and offers **up to 288 E cores** (known as Darkmont E-cores).

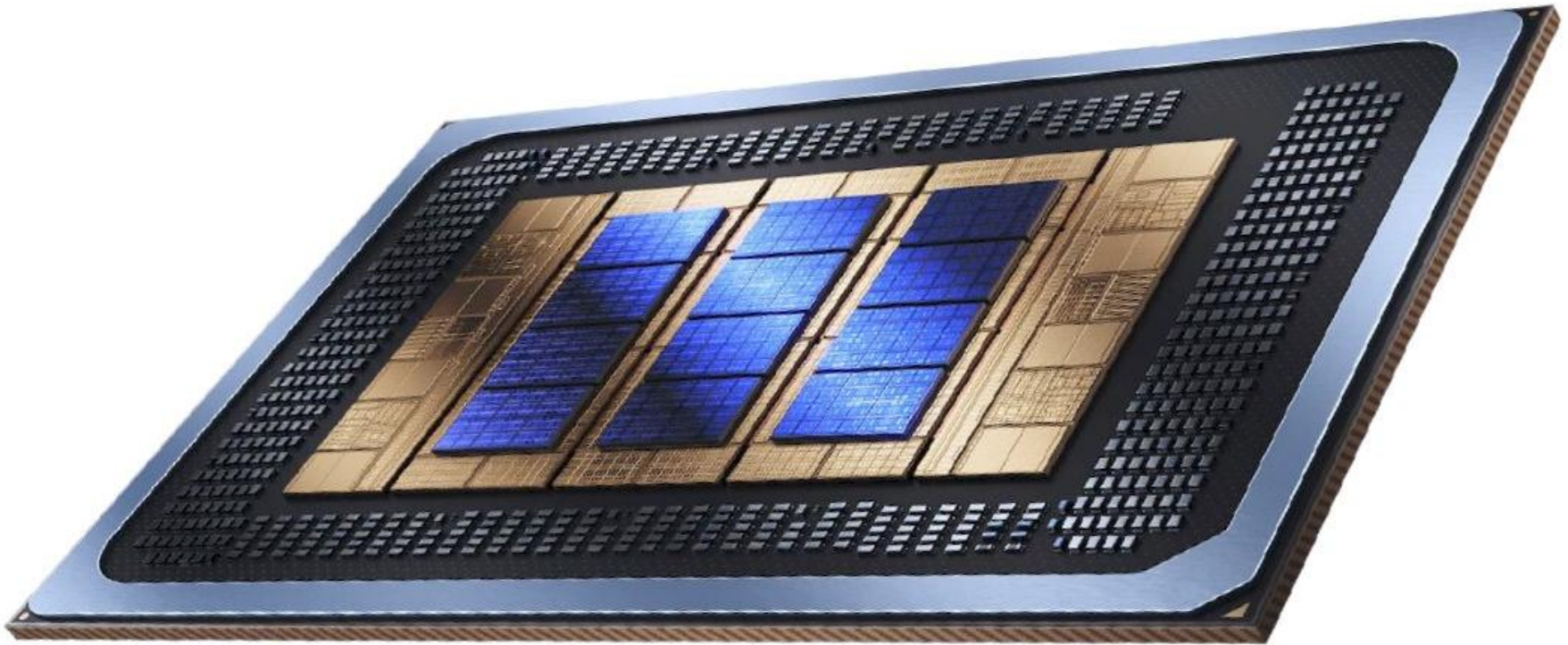
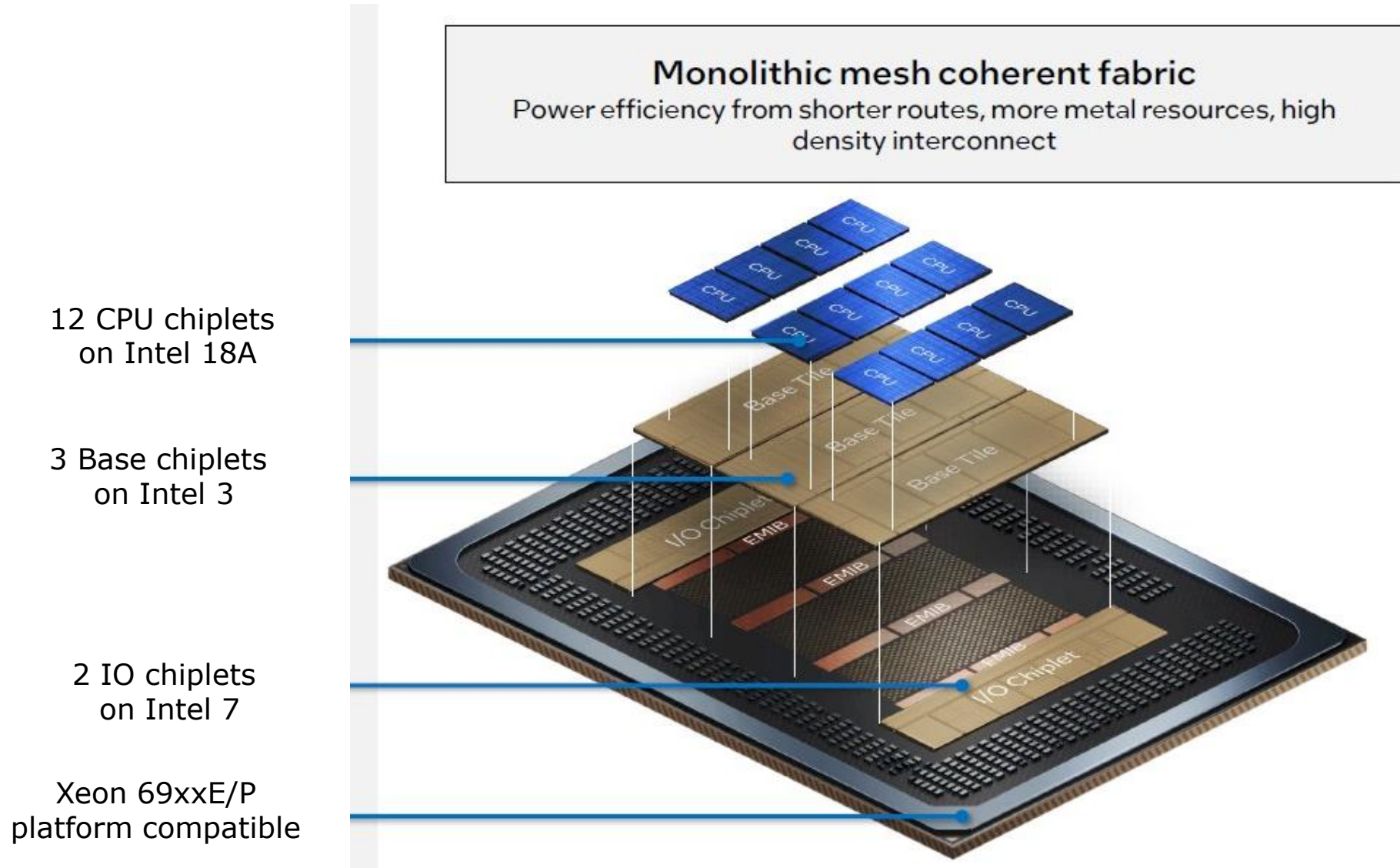


Figure: Layout of a Clearwater Forest server processor [191]

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (2)

Layout of the Clearwater Forest server processor [191] -1

It is built from **3 different die types**, each manufactured by a **different process**, as shown below, to capitalize on reducing fabrication costs.

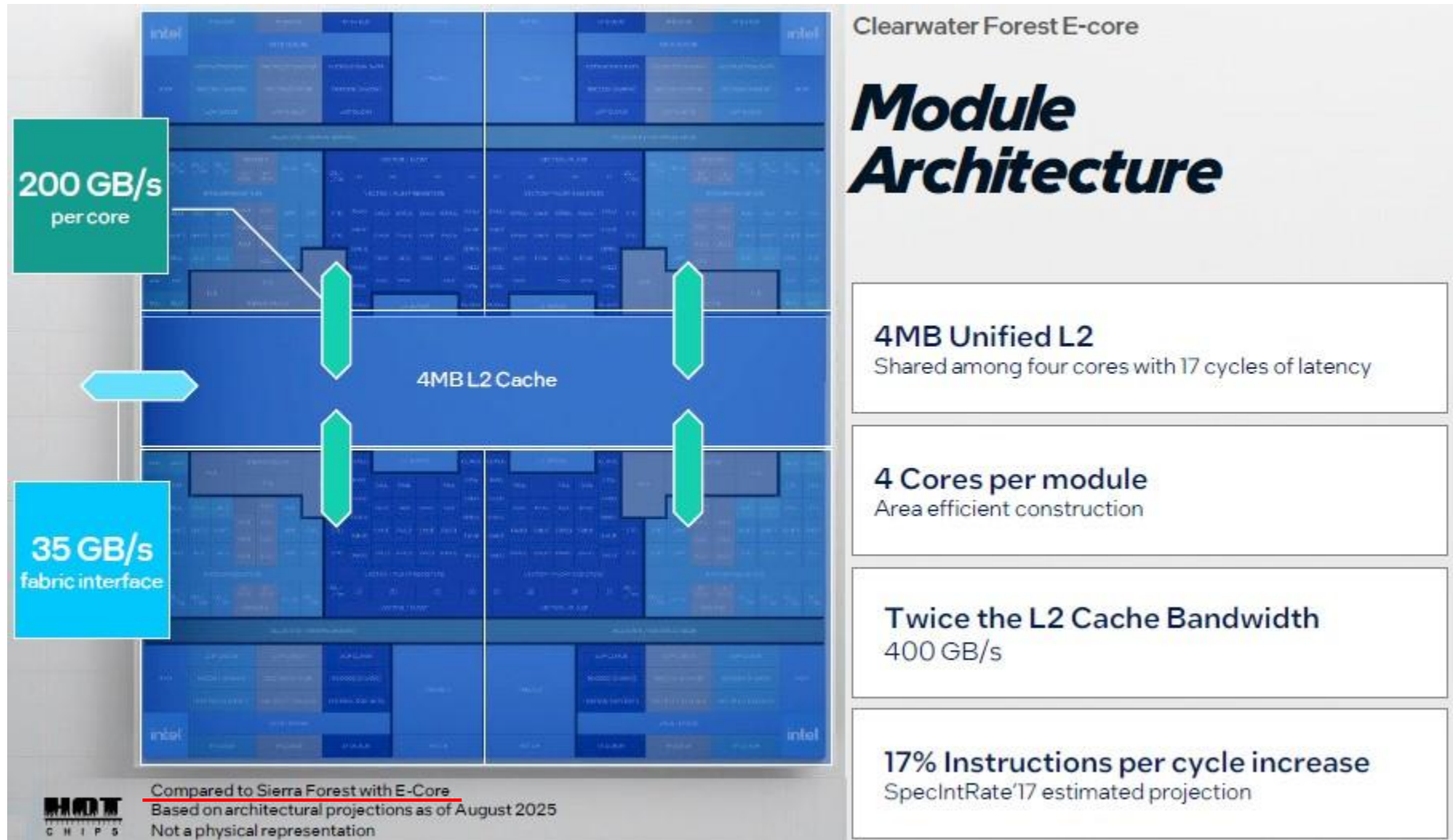


Layout of the Clearwater Forest server processor [191] -2

- We point out that the Compute dies utilize [Intel's cutting-edge 18A process](#), along with the company's Core Ultra Series 3 (Panther Lake) laptop family, discussed in the Chapter "Intel's Core-based client processors".
- Additionally, it is important to note that the Clearwater Forest line is [platform-compatible](#) with the previous Xeon 69xxE/P (Sierra Forest/Granite Rapids) server line.

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (4)

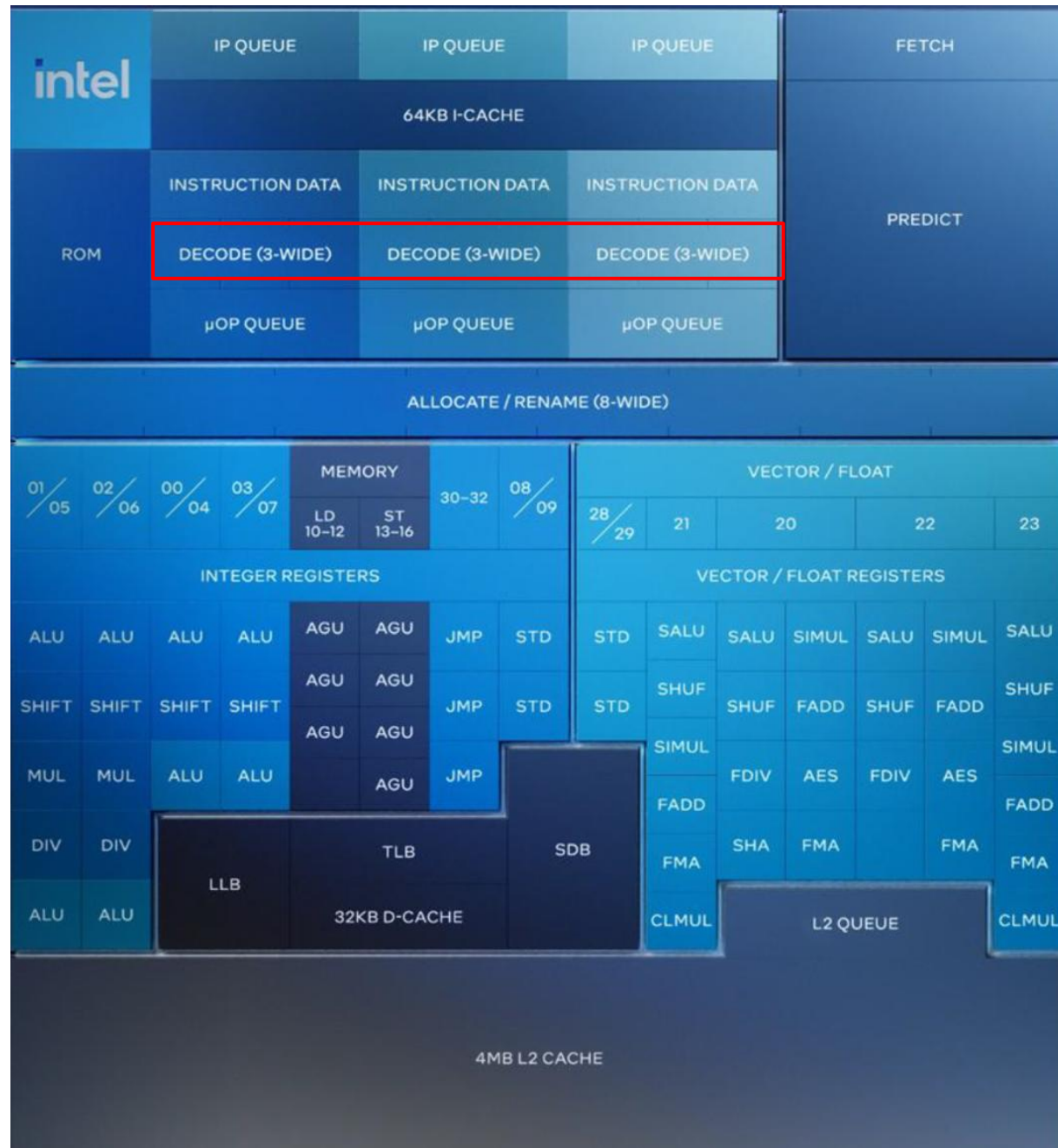
Layout of the 4-core CPU chiplets, referred to as modules [191]



3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (5)

The Darkmont E-core [192]

Note that Darkmont E cores are now 3x3 wide, up from the 2x3 width of the E cores of the previous Sierra Forest line (known as the Sierra Glen cores).



3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (6)

Improvements vs. the previous Cresmont core (of Sierra Forest)

Double Digit IPC Uplift Per Core

Crestmont

on Sierra Forest



-	Branch predictor	Wider & deeper
6-wide	Decode	9-wide
6-wide	Allocate	8-wide
64 entries	uOp queue	96 entries
256 entries	ROB window	416 entries
17 ports	Dispatch	26 ports
4 ALUs	Scalar ALUs	8 ALUs
2x128b	Vector FMAs	4x128b
2 AGUs	Address generation	4 AGUs
64B/cycle	L2 bandwidth	128B/cycle

Darkmont

on Clearwater Forest



As shown above, Darkmont P cores offer significant improvements in many aspects [192].

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (7)

Performance and performance/Watt improvements compared to the previous Cresmont core (of the Sierra Forest)

Darkmont Projections*

μArchitecture Power & Performance

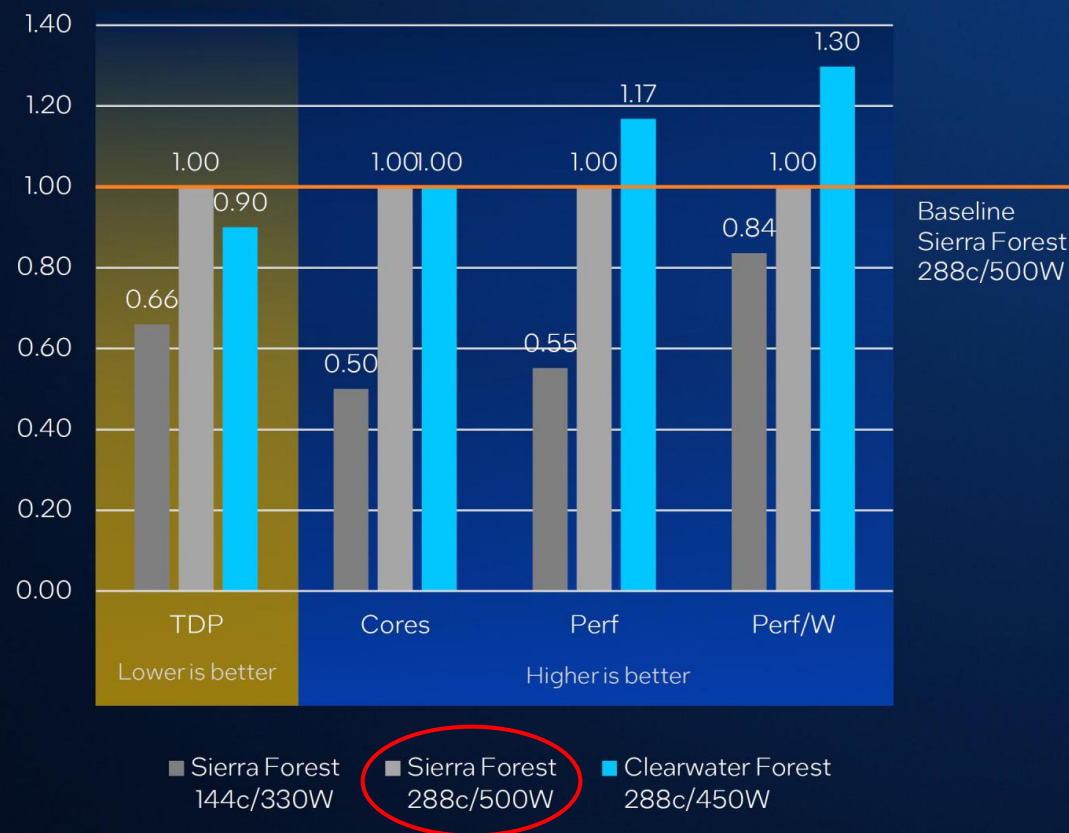
vs. Crestmont

up to **1.17x** higher performance
with Clearwater Forest (288c/450W)
vs Sierra Forest (288c/500W)

up to **1.3x** higher performance per watt
with Clearwater Forest (288c/450W)
vs. Sierra Forest (288c/500W)

Based on 1-socket projections as of 9/19/25. Results may vary.

* Pre-production projections 288c/450W Intel Xeon 6+



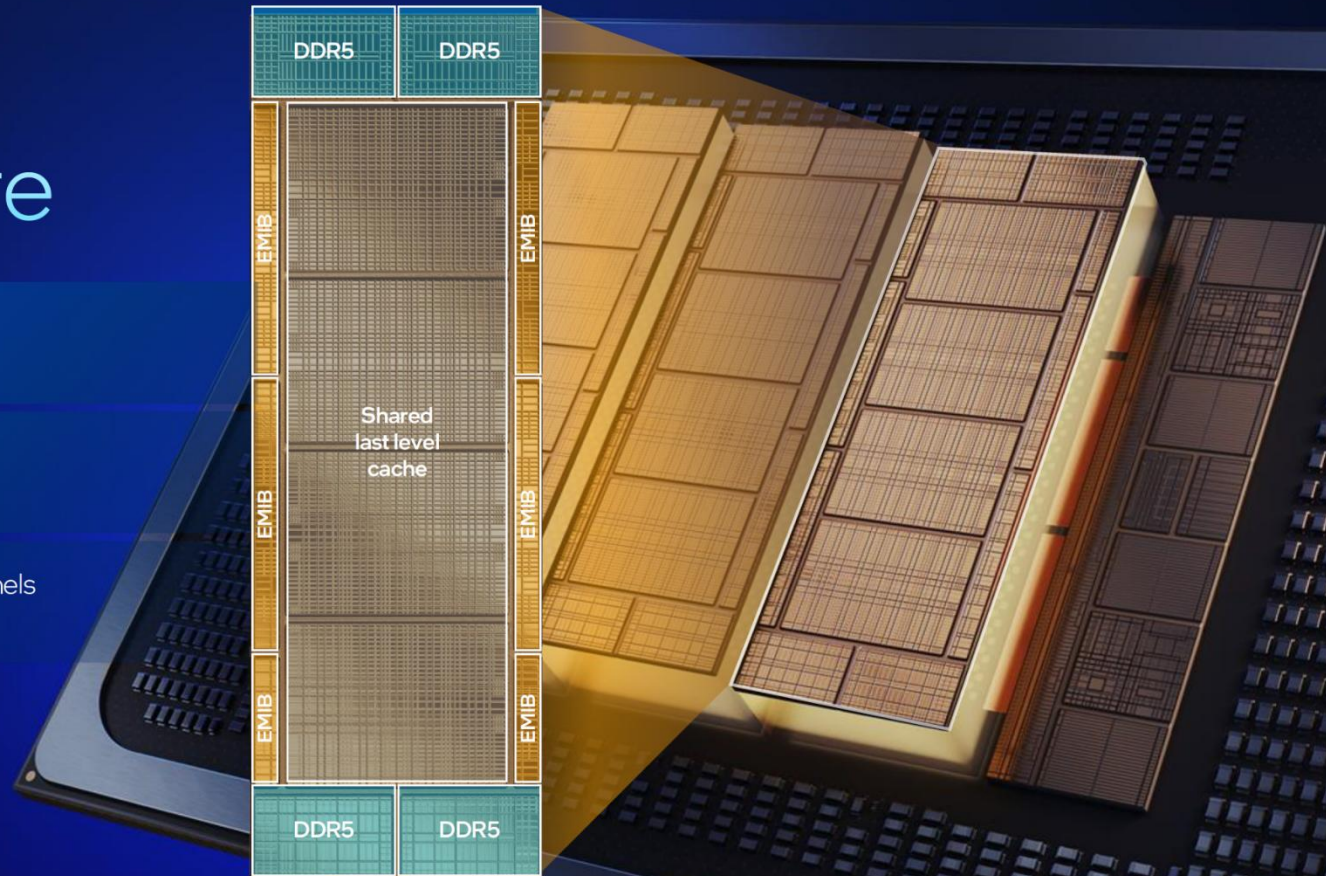
The new P cores demonstrate a 17% increase in performance and a 30% boost in efficiency compared to the previous P cores [192].

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (8)

Base tile architecture [192]

Base Tile Architecture

192 MB	Last level cache per base tile
48 MB	Last level cache per compute tile
4	DDR5 memory channels per base tile



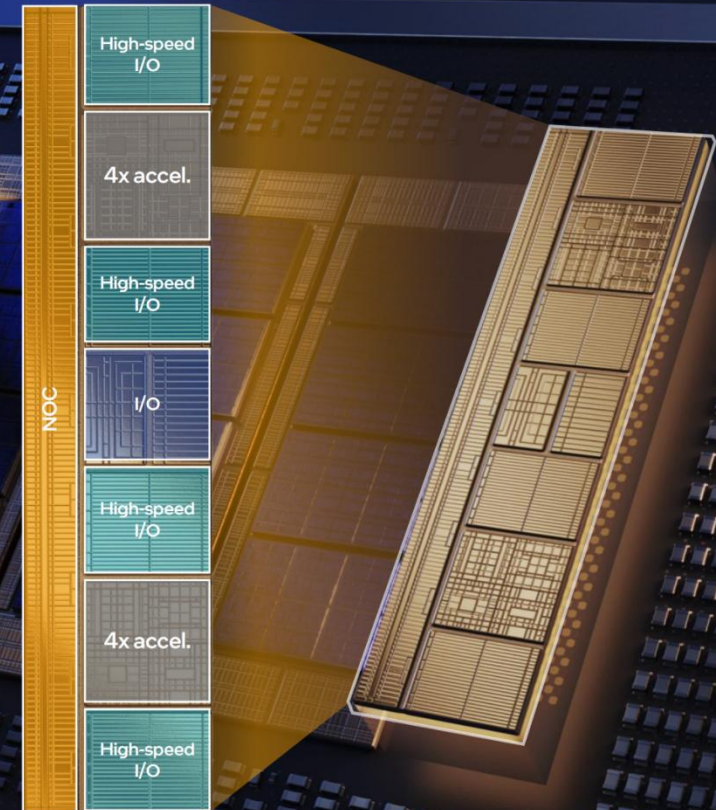
Each Base tile contains 4 CPU chiplets, each with 4 cores, providing 3 MB of LLC per core.

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (9)

I/O tile architecture [192]

I/O Tile Architecture

8	Accelerators per tile Intel QuickAssist Technology, Intel Dynamic Load Balancer, Intel Data Streaming Accelerator, Intel In-Memory Analytics Accelerator
48	Lanes of PCIe gen 5.0 per tile
32	Lanes of CXL 2.0 per tile
96	UPI 2.0 lanes per tile
Tile reuse from Granite Rapids	



3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (10)

The Intel 18A process with back-side power delivery and all-around gates []

Intel 18A Process

Backside metal, combined with gate-all-around, provides numerous additional benefits beyond FET Z scaling

Intel's latest process node, 18A

Best power efficiency for core logic by lowering gate capacitance

Higher cell density

Over 90% cell utilization

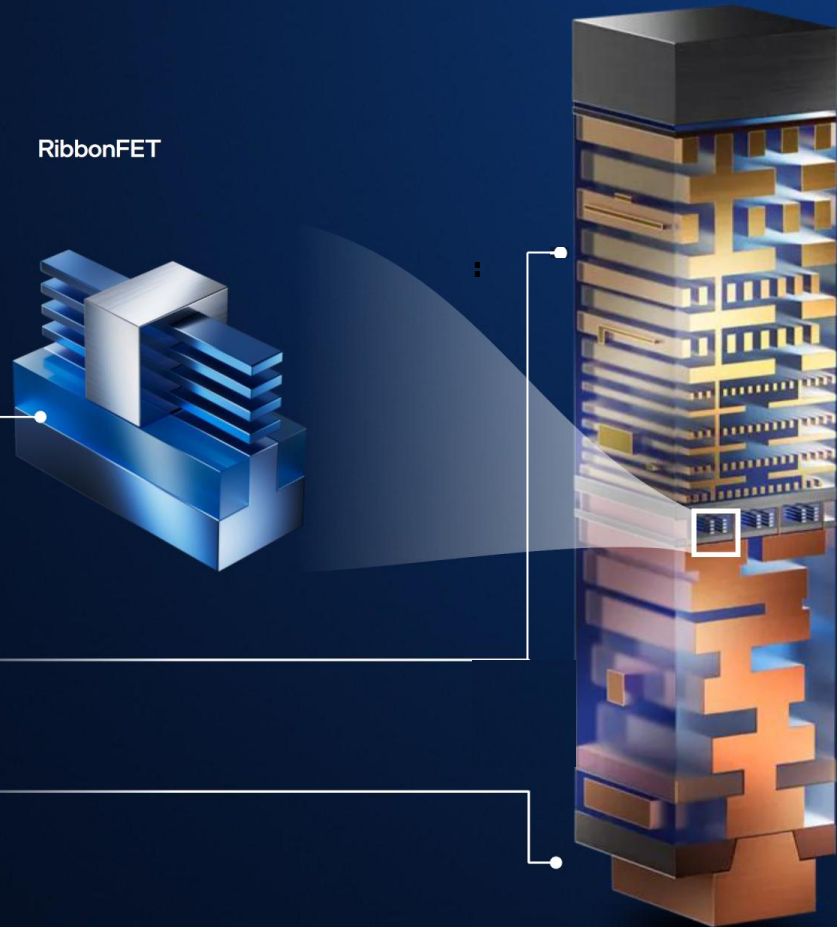
Improved signal routing

Improved signal routing (power rails in backside)

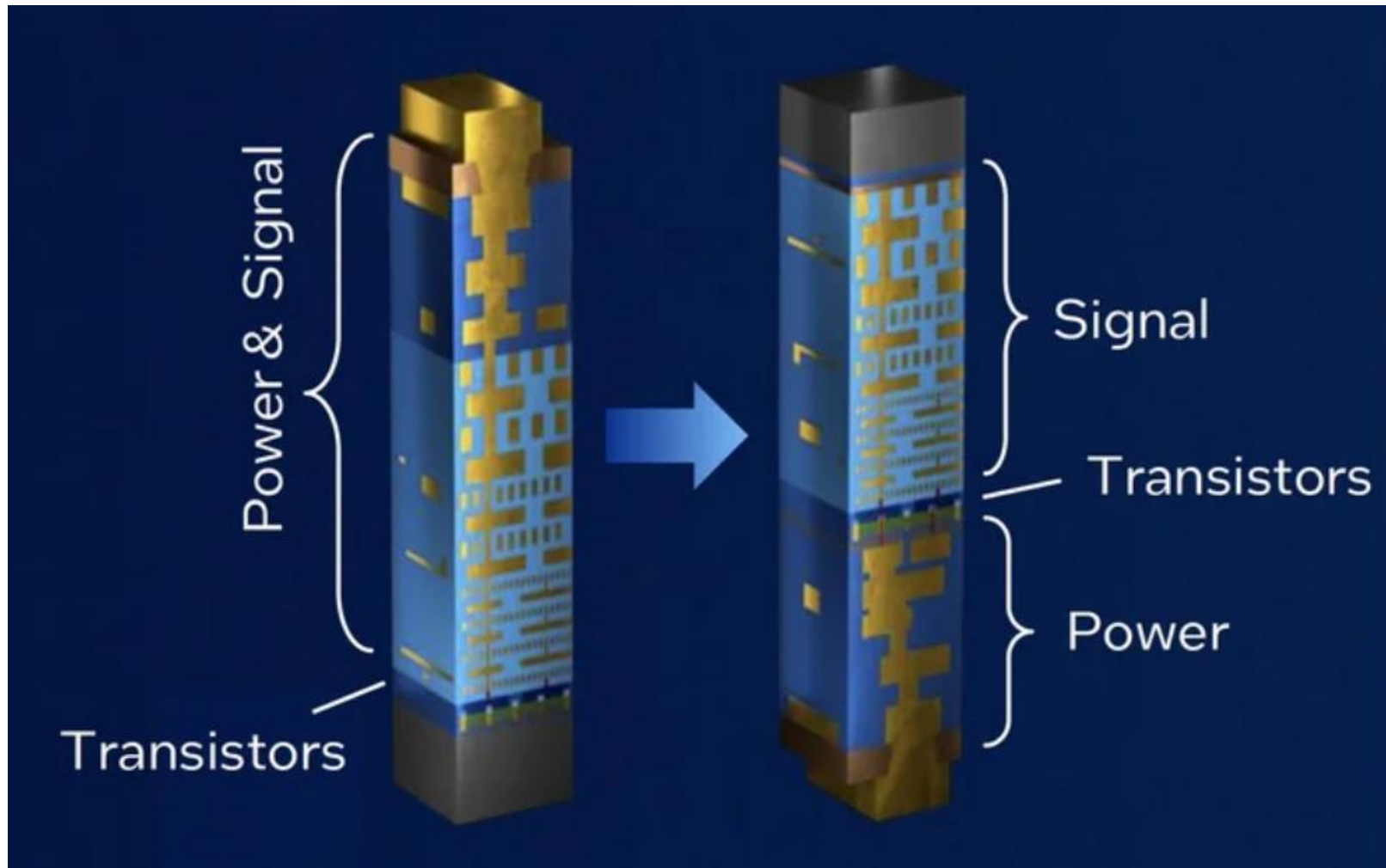
Low-loss power delivery

Reduce losses 4-5%

RibbonFET



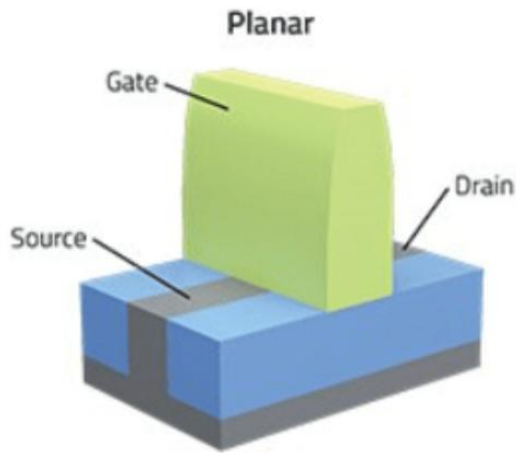
Traditional and new back-side power delivery [198]



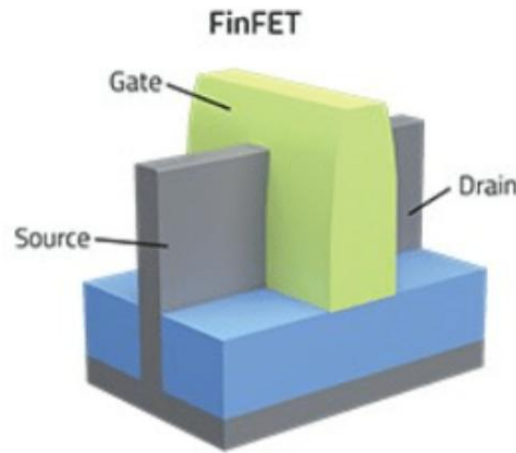
- Traditionally, signal and power interconnects are mixed.
- With **back-side power delivery**, they are **separated**, as seen above.

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor ()

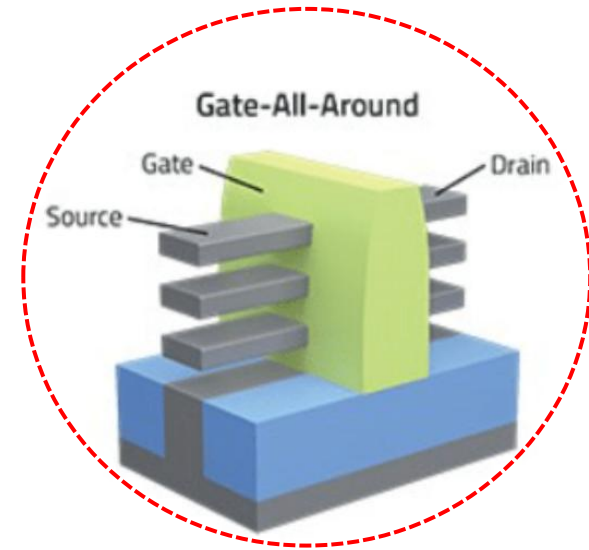
The evolution of transistor technology to the GAA technology [193]



1959



2012 (Intel 22 nm)



(Samsung 3nm, 2022)
(TSMC, Intel 2025, 2 nm)

The **planar topology** of FETs allowed the design and manufacture of **integrated circuits**. Noyce's achievement was to use the planar process (developed by Jean Hoerni) to fabricate transistors, resistors, and interconnects on a single piece of silicon.

FET: Field Effect Transistor (térvezérlésű tranzisztor) vs. bipolar junction transistor.

GAA/RibbonFET technology: The gate **wraps entirely around** each ribbon (channel).

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (12)

Using EMIB 2.5D technology along with the 3D Foveros packaging technology to interconnect adjacent dies [192]

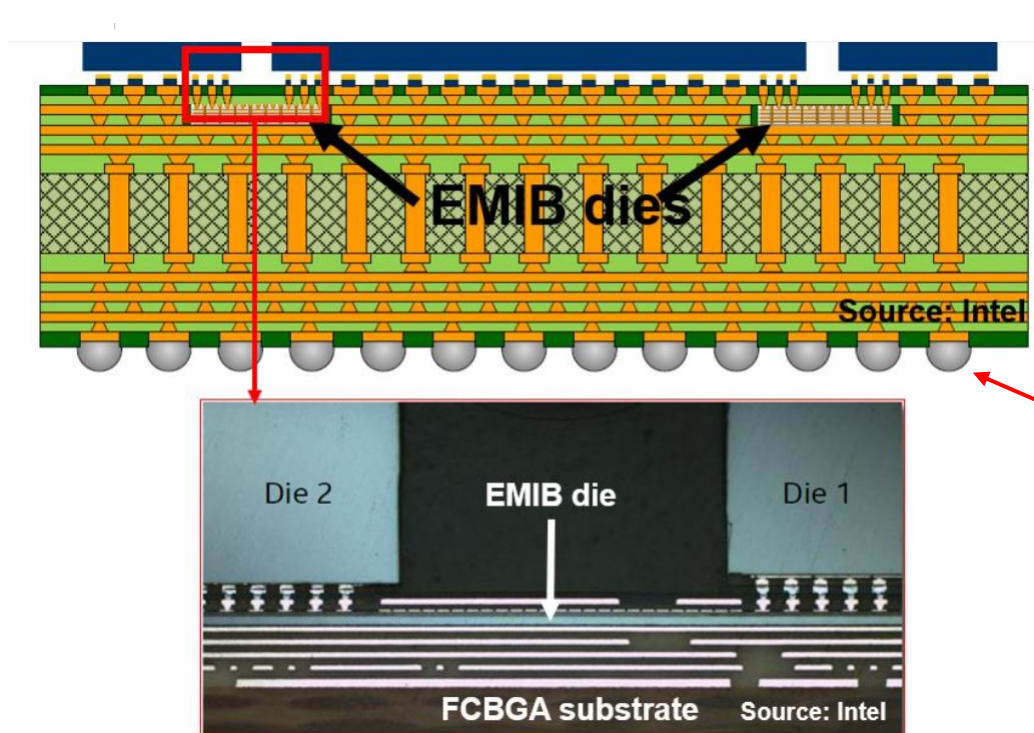
Clearwater Forest 3D Construction

Leveraging advanced packaging
across the entire SOC



3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (13)

The EMIB 2.5D (Embedded Multi-die Interconnect Bridge) packaging technology []



Its main components are:

- Soldered microbumps with a pitch of 45-55 μm .
- Organic package substrate
- Cu routing layers
- Dielectric layers isolating embedded Si bridges
- BGA ball grid

It was **first** implemented in the FPGA Field-Programmable Gate Array) chip (Stratix 10) **in 2017**, referred to as Intel's EMIB technology.

Soldered microbumps: forraszgöbök Pitch: osztás-távolság

Organic package substrate: Polimer-alapú tokozó anyag

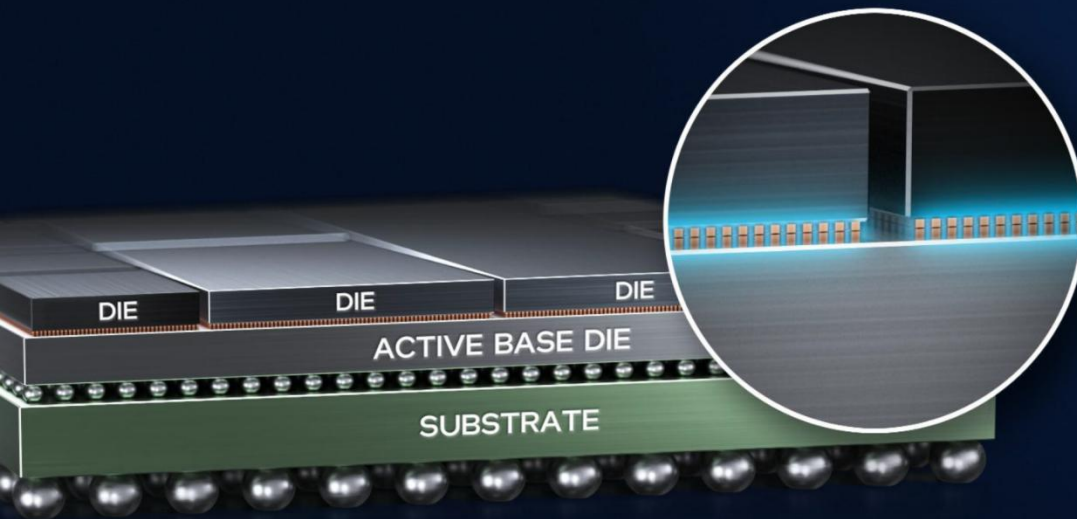
Cu routing layers: réz vezetősávok BGA ball grid: BGA forraszgolyó-rács

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (14)

3D Foveros packaging technology to interconnect stacked dies [192]

Foveros Direct 3D

Clearwater Forest first production in HVM

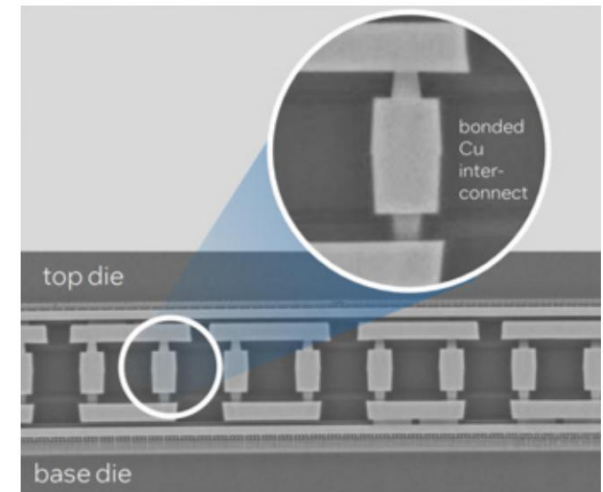


A microscopic view of the Cu-to-Cu bonding process, showing the 'bonded Cu interconnect' structure.

- 9 μ m bump pitch**
- Cu-to-Cu bonding**
- Active Si interposer**
- High-density & low-resistance die-to-die interconnects**
- Superior power/bit performance (~0.05pJ/bit)**

Please refer to the next slide for a detailed explanation.

The Foveros Direct 3D packaging technology [220]



- It **eliminates traditional solder microbumps** by utilizing direct **Cu-to-Cu hybrid bonding** between the chiplets and the base die with a pitch of less than 10 μm (instead of $\sim 50 \mu\text{m}$). To achieve direct Cu-to-Cu hybrid bonding, **Cu pads** (réz érintkezők) are **pressed together under heat and pressure**, allowing Cu atoms to diffuse across the interface and create a hybrid metallic interconnect.
- This method **removes the resistance and parasitic effects of solder bumps**, while enabling a **much tighter pitch**.
- The roles of the base die and the package substrate remain largely the same as in the Foveros 2.5D technology.
- The Foveros Direct 3D packaging will **first** be implemented in Intel's 6+ gen. Clearwater server processor series, which is set to be launched in **H1 2026**.

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (16)

Clearwater Forest – platform overview [192]

Clearwater Forest Feature Overview

Platform

Sockets 1S - 2S (Xeon 6900P compatible)

Max TDP 300 to 500W per CPU

Compute & memory

Cores up to **288** efficient cores

L2 cache up to **288MB** (up to 4MB / cluster)

Last level cache **576MB**

Memory **12ch DDR5 8000MT/s**

Compute & memory

Intel® UPI up to **6 UPI 2.0** (up to 24 GT/s per lane)

PCI Express up to **96 lanes PCIe 5.0** (x16, x8, x4, x2)

Compute Express Link up to **64 lanes CXL 2.0**

Security & efficiency

Security Intel Software Guard Extensions (Intel SGX)
Intel Trust Domain Extensions (Intel TDX)

Power management Intel Application Energy Telemetry (Intel AET)
Intel Turbo Rate Limiter

Acceleration

Acceleration Intel Advanced Vector Extensions 2 (VNNI/INT8)

Integrated accelerators

up to **16 accelerators**

4x Intel QuickAssist Technology
4x Intel Dynamic Load Balancer
4x Intel Data Streaming Accelerator
4x Intel In-memory Analytics Accelerator



The 6 UPI 2.0 links (enhanced QPI links) can support 4–8-socket topologies.

3.3 Case example 3: The 6+ gen. Clearwater Forest 2S server processor (17)

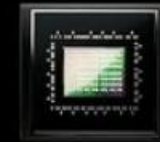
Historical trends of main features of processors (Source: GTC 2025)

CPU General Purpose to GPU Accelerated Computing Platform Shift

Microprocessor Trend Data



GPU



CPU